Amaro

de Matos, João Amaro - Theoretical Foundations of Corporate

- Capital markets, Fisher separation, NPV rule under certainty: Ch. 1, §1.1.4–1.1.6 (p. 9–15)
- One-period model under uncertainty, no arbitrage, state prices, martingale measure: Ch. 1, §1.2.1–1.2.4 (p

1.1.4 The Role of Capital Markets

Ignore for now the existence of an investment schedule such as the one just described. But now consider that the island has been invaded and that Robinson still lives in a one-period economy, but with the difference that there are now many agents.

The existence of different people trading means that there is a market for trading time preferences. Some people may want to consume at time $t = 0$ more than their endowment allows them to do. Some other agents, however, may initially have more than they are willing to consume at that point in time. The former agents will be willing to exchange some of their future endowment for part of the latter's current endowment.

In equilibrium the interaction of demand and supply for endowments at different times should determine a price for transferring consumption from time $t = 0$ to $t = 1$. This price represents a premium for postponing consumption. Clearly, Robinson may leave an amount ϕ of his current consumption to get it back in the future with a premium of ϕr, proportional to ϕ.

Recall that it has been assumed that the utility function is increasing in current and future consumption (i.e., Robinson prefers more wealth to less). Therefore, with this premium ϕr, Robinson will be better off lending his current consumption than keeping it for his own use in the future if and only if

$$U(X - \phi, Y + \phi(1 + r)) \geq U(X, Y).$$

The positiveness of the proportionality constant r, termed the *interest rate*, provides the right incentive for such exchanges to happen given the decrease in the first argument of the utility function.

The maximum amount that Robinson could lend would be X, in which case his current consumption would be null. In the future he would have Y plus the initial endowment X back with the premium rX, a total amount $Y + X(1 + r)$. The amount $X(1 + r)$ is denominated the *future value* of X.

Alternatively, he may wish to consume in the present more than his current endowment X by an additional amount ϕ, say, that someone in the market is willing to lend him. This may happen provided that he pays it back from his future endowment Y plus a premium of ϕr. Under these circumstances, the maximum possible ϕ in this case would be $Y/(1 + r)$. This amount is called the *present value* of Y.

The issue here is to determine how this possibility affects Robinson's consumption decisions. Given the endowment pair (X, Y), the current consumption is not bounded anymore by X. This bound can be augmented to $0 \leq x \leq X + Y/(1 + r)$. It is then said that the pair of

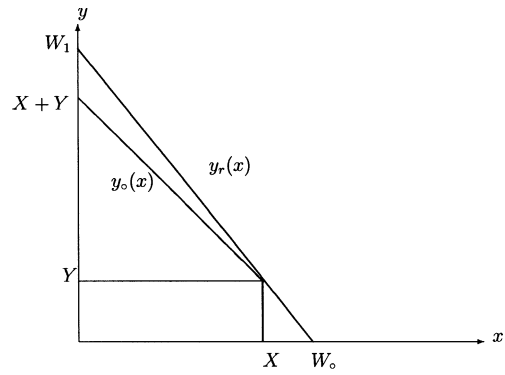


Figure 1.3. *The Region of Admissible Consumptions in the Presence of Capital Markets*

endowments (X, Y) generates a *current wealth* of $W_\circ = X + Y/(1 + r)$, which reflects the maximum possible current consumption. In the same fashion, future consumption is bounded by $0 \leq y \leq Y + X(1 + r)$ as illustrated in Figure 1.3, and one defines the *future wealth* of endowments (X, Y) as $W_1 = X(1 + r) + Y$, which reflects the maximum possible future consumption.

This situation may be compared with that of the previous section, where there was an investment schedule with a positive (decreasing) marginal rate of return $\varphi(I)$ for the level of investment I. Here, it follows that the marginal rate of return is a positive constant r. This leads to a total amount y_r for consumption at time 1 given by

$$\begin{aligned} y_r(x) &= Y + \int_0^{X-x} [1 + r] dI \\ &= Y + (X - x)(1 + r) \\ &> Y + X - x = y_\circ(x). \end{aligned}$$

Figure 1.3 illustrates this inequality. Notice that $W_\circ$ can be easily identified with the intercept of the straight line $y_r(x)$ with the x axis, whereas W_1 is the intercept with the y axis. Note also that the existence of markets allows the choice of any point on the straight line $(x, y_r(x))$. Any chosen point on this line has the following important property: The current consumption plus the present value of the future consumption add up to the current wealth generated by the endowment pair (X, Y). In other words,

$$x + \frac{y_r(x)}{1 + r} = X + \frac{Y}{1 + r}.$$

Thus, all points on the straight line $(x, y_r(x))$ are associated with the same level of wealth, $W_\circ$. This straight line is called *market line*. Then, as before, it is simple to show that $x_r^* = \arg\max U[x, y_r(x)]$ is given by

$$x_r^* = \begin{cases} 0 & U_1[0, Y + X(1+r)] \le (1+r)U_2[0, Y + X(1+r)] \\ X + \frac{Y}{1+r} & U_1[X + Y/(1+r), 0] \ge (1+r)U_2[X + Y/(1+r), 0] \\ x^* & \text{otherwise,} \end{cases}$$

the interpretation being identical as for $x_\circ^*$, and x^* now being the solution of $U_1(x^*, Y + X - x^*) = (1+r)U_2(x^*, Y + X - x^*)$. In other words, now the marginal ratio of substitution must be $-(1+r)$ at the optimal point of consumption, thus leading to $r = \rho$, the parameter ρ being defined in equation (1.4).

The effect of capital markets may be read from the expression of $y_r(x)$. For any level of current consumption x, the fact that $y_r(x) \ge y_\circ(x)$ implies that consumption in the future can be greater with than without the capital markets. Since the presence of capital markets implies that future and current consumption can be exchanged, this means that current consumption can also increase from its original maximum X up to the level of current wealth. Thus, the optimal choice has a greater level of utility with financial markets than without them.

1.1.5 Consumption and Investment with Capital Markets

Now consider the same one-period economy with many agents, again including an investment schedule. Assume that there are different time preferences. Also, demand and supply are assumed to exist for both current and future consumption. This leads to an equilibrium value for the discount factor r reflecting the premium that one gets for postponing consumption. Being determined in equilibrium, this premium is exactly how much one must pay to consume earlier. In the presence of an investment schedule, it is also reasonable to make the following assumption:

Assumption 1.1.5. *The infimum of the marginal decreasing rate of return $\varphi(I)$, which is always positive, is smaller than the interest rate r.*

In other words, after a certain amount of investment, it is more interesting to invest in the financial market than in the investment schedule.

Up to now, only two types of investment have been considered: an investment schedule as analyzed in section 1.1.3 and the capital markets as just described. Now a more general situation is considered since there are *two* different places to invest: the production schedule and the financial markets.

Given the endowment pair (X, Y), the existence of financial markets allows us to borrow money for investment. Let ϵ be the amount to be

invested in the investment schedule. Thus, consuming x at the initial time $t = 0$, the amount available for the investment is $X - x$. If the required investment ϵ is greater than $X - x$, the amount $\epsilon - (X - x)$ must be borrowed from the bank. This implies that at time $t = 1$ the investor must pay back $[\epsilon - (X - x)](1 + r)$ to the bank. If, on the other hand, the required investment ϵ is smaller than $X - x$, the amount $(X - x) - \epsilon$ must be lent to the bank. Hence, in any case, the future consumption will be given by

$$y(x) = Y + \int_0^{\epsilon} [1 + \varphi(I)]dI - [\epsilon - (X - x)](1 + r) \tag{1.5}$$

$$= X + Y - x + \int_0^{\epsilon} \varphi(I)dI - [\epsilon - (X - x)]r. \tag{1.6}$$

Then the problem is to maximize the utility $U(x, y)$ subject to the possibilities allowed by the production schedule or, equivalently, maximizing $U(x, y(x))$ with $y(x)$ given by equation (1.5), subject to $\epsilon \geq 0$. Maximization must be performed with respect to x and ϵ. It is simple to check that the first-order condition with respect to ϵ does not depend on x and y, leading to an optimal value ϵ^* of ϵ given by

$$\epsilon^* = \begin{cases} 0 & r \geq \varphi(0) \\ \varphi^{-1}(r) & \varphi(0) \geq r. \end{cases}$$

This result may be interpreted as follows: The optimal level of investment ϵ^* does not depend on the preferences of the agents over the levels of consumption in both periods. A rational agent maximizing his/her utility will invest in all available investment projects up to the level where the marginal rate of return equals the market interest rate, r. From that point on, the agent prefers to lend money in the market at that interest rate.

The fact that the market rate r is the threshold that determines up to what point an investment project is interesting establishes a very important role for the capital markets. Markets actually become the term of comparison for analyzing investments in the sense that if one's nonconsumed current wealth has not been invested, it may be invested in the market at the rate r. For that reason, the market rate is often referred to as the *cost of opportunity of capital* or simply *cost of capital*.

The fact that the preceding optimal solution for the investment level depends only on the market rate r and is completely independent of consumption levels leads to the following proposition:

Proposition 1.1.1 (Fisher separation). *The optimal production decision is determined by an objective market criterion independently of individuals' subjective preferences that define their consumption decisions.*

This result has strong implications in that the structure of any investment policy can be seen as a sequential decision in two phases. In the first phase, one chooses the level of investment. In the second, a decision is made about the intertemporal consumption path, or how much to consume at each point in time. Therefore, the level of investments is decided independently of the consumption preferences, but the level of possible consumption depends on the investment decision.

The consumption problem should actually be solved as follows: First, define the function

$$f(x, y) = x + y - X - Y - \int_0^{\epsilon} \varphi(I)dI + r[\epsilon - (X - x)]. \tag{1.7}$$

Then the problem is to maximize the utility $U(x, y)$ subject to the possibilities allowed by the production schedule, described from equation (1.5) as $f(x, y) = 0$. One may find the optimal solution either by constructing the Lagrangean

$$\mathcal{L}(x, y) = U(x, y) - \lambda f(x, y)$$

and then working out the first-order conditions $U_1 = \lambda f_1$ and $U_2 = \lambda f_2$ or, equivalently, by maximizing $U[x, y(x)]$ with $y(x)$ given by equation (1.5). This last formulation reduces to a one-dimensional optimization in x. The first-order condition is $U_1 + U_2 y' = 0$, implying once again that the marginal rate of substitution at the point of optimal consumption should be $-(1+r)$. This leads once more to $r = \rho$, the parameter ρ being defined in equation (1.4).

Once the optimal level x of initial consumption is determined, the future consumption is maximized for an investment level ϵ^*. The optimal value ϵ^* of investment leads to a total amount $y_f(x)$ for consumption at time 1 given by

$$y_f(x) = \begin{cases} Y + (X - x)(1 + r) & r \geq \varphi(0) \\ Y + (X - x)(1 + r) + \int_0^{\varphi^{-1}(r)} [\varphi(I) - r]dI & r \leq \varphi(0). \end{cases}$$

It is a very important and intuitive result that for a given level of current consumption x,

$$y_f(x) \geq y_p(x) \geq y_\circ(x), \tag{1.8}$$

which can be checked by simple inspection. This means that the introduction of financial markets allows a larger range of current and future consumption alternatives for the economic agents, as illustrated in Figure 1.4. Thus, the existence of financial markets leads to a choice of optimal solutions of investment and consumption choices with higher levels of utilities for the agents as compared with the case where no such markets exist.

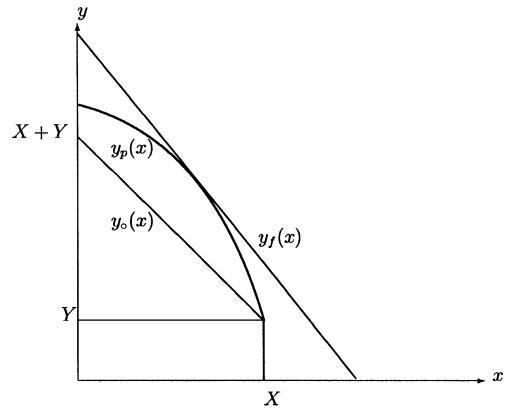


Figure 1.4. *The Region of Admissible Consumptions in a Robinson Crusoe Economy with an Investment Schedule and Capital Markets*

1.1.6 The Value of an Investment Project

In an earlier subsection the consumption problem was analyzed in the absence of an investment schedule, leading to an optimal consumption at $t = 1$ expressed by $y_r(x)$. One may wonder where would $y_r(x)$ be placed in the set inequalities (1.8).

Formally the problem may be solved very simply by noticing that the consumption decision is based on the optimization of the utility $U(x, y_f(x))$, in the case of determining $y_f(x)$, and on maximizing $U(x, y_r(x))$ in the case of $y_r(x)$. Comparing $y_r(x)$ with $y_f(x)$, one notices that the difference between them is expressed as

$$y_f(x) - y_r(x) = \int_0^{\varphi^{-1}(r)} [\varphi(I) - r]dI$$

if there are projects more interesting than the financial markets, or as zero otherwise. The important thing is that this difference is independent of the optimal choice of x and y. Thus, the introduction of an investment works as if the future endowment Y were shifted by

$$\Delta = \begin{cases} 0 & r \geq \varphi(0) \\ \int_0^{\varphi^{-1}(r)} [\varphi(I) - r]dI & r \leq \varphi(0) \end{cases}$$

This means that the introduction of the investment schedule is equivalent to increasing the future wealth by Δ. In other words, the current wealth is increased by $\Delta/(1 + r)$.

Given an investment schedule, its *net present value* (NPV) is defined as the increase in the current wealth generated by the investment. In this case, for the given investment schedule φ,

$$\text{NPV}(\varphi) = \frac{\Delta}{1+r}.$$

Notice that if the investment schedule does not allow return rates greater than r, its NPV is null. In other words, the corresponding investments are rationally ignored. It follows that

Proposition 1.1.2. *It is rational to invest in a project if its net present value is positive.*

This is simply another way of saying that the optimal level of investment is ϵ^*. By construction, this is also equivalent to saying that one should invest in a project if its marginal rate of return is at least equal to r.

A corollary of this proposition is that the optimal level of investment is the one that maximizes the present wealth of the agents or, in other words, is the level of investments that maximizes their capacity to generate higher levels of consumption from the pair (X, Y) for both periods.

Corollary 1.1.1. *The optimal investment decision is the one that maximizes the current wealth of the agents.*

1.1.7 Multiperiod Economy with Capital Markets

There is no mystery in how to generalize these ideas to a multiperiod economy. Suppose now that there is a two-period economy with capital markets together with production arising from a given investment schedule. Introducing z as the consumption at time $t = 2$ and Z as the endowment at that point in time, the formalism follows the earlier sections. Let ϵ_0 and ϵ_1 be the amounts invested at times $t = 0$ and $t = 1$, respectively. The consumption problem should actually be solved as follows: Analogously to equation (1.7), one may define the pair of functions

$$f(x, y) = x + y - X - Y - \int_0^{\epsilon_0} \varphi(I)dI + r[\epsilon_0 - (X - x)]$$

and

$$g(y, z) = y + z - Y - Z - \int_0^{\epsilon_1} \varphi(I)dI + r[\epsilon_1 - (Y - y)].$$

Then the problem is to maximize the utility $U(x, y, z)$ subject to the possibilities allowed by the production schedule, described as $f(x, y) = 0$ and $g(y, z) = 0$. The optimal solution is found by constructing the Lagrangean

$$\mathcal{L}(x, y, z) = U(x, y, z) - \lambda f(x, y) - \gamma g(y, z)$$

1.2 Valuation under Uncertainty

The principles already developed may well be extended to situations in which the future outcome is not certain. Suppose that Robinson Crusoe now has a bundle of investment projects available to him, but he is not sure of the outcome of any of them. This section addresses how this uncertainty affects his optimal decision of consumption and investment.

In other words, considering that there are several available projects, and that each of them requires an initial investment I_i and has an intrinsic present value of V_i, this section will describe conditions under which the decision about whether or not to invest depends only on whether or not V_i is greater than I_i. In order to do that one must know how to properly characterize the value V_i of each project, given the uncertainty underlying their payoffs. This is the main purpose of this section.

It will be assumed that financial markets are available in the context of an exchange economy. In that setting conditions will be established so that the analysis under uncertainty may be focused strictly on the valuation process of the projects and not on the consumption path.

1.2.1 One-Period Model

The expression *future uncertainty in a one-period economy* means that someone investing at time $t = 0$ is not sure of the market conditions at $t = 1$. Both the endowment and the investment's payoff at that point in time depend on the particular conditions of nature at $t = 1$. Uncertainty is thus described by a set of states belonging to a finite sample space $\Omega = \{w_1, w_2, \ldots, w_K\}$, where each state w_j occurs with probability $p(w_j)$, denoted for simplicity as p_j. Also, if $x \in R^K$ is a random variable in Ω, let

$$E(x) \equiv \sum_{j=1}^{K} x_j p_j$$

denote the expectation of x under the probability measure p.

The notion of utility must be refined under uncertainty. In Section 1.1, utility was introduced as a real function $U(x, y)$, increasing and differentiable in the present consumption x and in the future consumption y. Here, however, future consumption is a random variable. Let $y \in R^K$ denote a *consumption plan* (i.e., the vector of possible consumptions at time $t = 1$), with the j-th component $y_j \equiv y(w_j)$ denoting consumption in state j. In this setting utility must be a real function $U^*(x, y)$ defined on R^{K+1}, allowing us to define preferences over different plans of consumption. Comparing two different consumption plans, y and z, the utility function U^* of an individual must be constructed in a way such that

$$U^*(x, y) > U^*(x, z)$$

if this individual prefers plan y to plan z. In more formal terms, the individual's preferences are said to have a utility representation if there is a function U^* with the preceding property. A commonly accepted utility representation is the *von Neumann-Morgenstern utility function*,

$$U^*(x, y) = EU(x, y) = \sum_{j=1}^{K} U(x, y_j)p_j,$$

also known as the *expected utility function*. Notice that under a *certain consumption plan*, where consumption is a sure thing or, in other words, y_j is a constant k independent of j, $U^*(x, y) = U(x, k)$ and the utility function under certainty is recovered.

The *certainty equivalent* of an uncertain consumption plan y can be defined as the certain consumption plan γ with the same expected utility as y. In other words, $U^*(x, y) = U(x, \gamma)$.

An agent is said to be *risk-averse* if, for any consumption plan y, the certainty equivalent γ is smaller than $E(y)$ or

$$EU(x, y) < U[x, E(y)].$$

It follows that the utility function U must be concave. If, on the other hand, an agent has a utility such that the certainty equivalent of a risky consumption plan is greater than the expected future consumption, the agent is said to be *risk-seeking* and U must be convex. Finally, risk-neutrality characterizes an agent that is indifferent between a certain consumption plan and a risky plan, provided that the expected consumption coincides with the certain consumption. In this case, the utility function U must be linear.

Now, consider the consumption problem in a one-period setting under uncertainty. Regarding the endowments, define $Y_j \equiv Y(w_j)$ as the endowment received at $t = 1$ in state $j = 1, 2, \ldots, K$. Also, instead of having one single project to consider for investment, assume that the investor faces a bundle of N different one-period projects. Each of these projects has an initial value V_i, $i = 1, \ldots, N$, and at time $t = 1$ pays $\Gamma_i(w_j)$ in state $j = 1, 2, \ldots, K$. For simplicity of notation, let $\Gamma_i(w_j)$ be denoted by Γ_{ij}. These values define a value vector $V \in R^N$ and a payoff $N \times K$ matrix Γ, given by

$$\Gamma = (\Gamma_1, \Gamma_2, \ldots, \Gamma_K) = \begin{bmatrix} \Gamma_{11} & \Gamma_{12} & \cdots & \Gamma_{1K} \\ \Gamma_{21} & \Gamma_{22} & \cdots & \Gamma_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_{N1} & \Gamma_{N2} & \cdots & \Gamma_{NK} \end{bmatrix}$$

where Γ_j denotes the N-dimensional column vector of payoffs in state j.

An investment decision amounts to choosing a portfolio of projects, namely a vector $\theta \in R^N$ with one component for each project. It follows that a portfolio has market value $V^\top\theta = \sum_i V_i\theta_i$ and payoff vector $\Gamma^\top\theta \in R^K$. In other words, for a given choice of θ, consumption at $t = 0$ is

$$x(\theta) = X - V^\top\theta \tag{1.9}$$

and consumption at $t = 1$ will be

$$y(\theta) = Y + \Gamma^\top\theta, \tag{1.10}$$

where y is a K-dimensional vector, with the j-th component y_j denoting consumption in state j.

From the consumption point of view, this setting of uncertain outcomes poses new problems. If we suppose that an agent has a specific consumption target y for $t = 1$, the choice of θ should be made in such way as to satisfy

$$\Gamma^\top\theta = y - Y$$

or, more explicitly,

$$\begin{bmatrix} \Gamma_{11} & \Gamma_{21} & \cdots & \Gamma_{N1} \\ \Gamma_{12} & \Gamma_{22} & \cdots & \Gamma_{N2} \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_{1K} & \Gamma_{2K} & \cdots & \Gamma_{NK} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_N \end{bmatrix} = \begin{bmatrix} y_1 - Y_1 \\ y_2 - Y_2 \\ \vdots \\ y_K - Y_K \end{bmatrix}. \tag{1.11}$$

It has been assumed that N projects are available, but when analyzing the consumption problem it is reasonable to consider only projects with linearly independent payoffs. In other words, suppose that there is a project k such that its payoff in any state of nature can be written as a combination of the payoffs of the other available projects. Then, holding an investment in project k is not relevant since such a position can be substituted for by holding a portfolio of the other projects, as defined by the linear combination of payoffs.

Formally, consider a portfolio choice θ. If there exists a real vector $\alpha \in R^N$ such that $\alpha_k = 0$ and $\Gamma_{kj} = \sum_{i\neq k} \alpha_i\Gamma_{ij}$, the $t = 1$ consumption is the same as if the chosen portfolio were θ^* with $\theta^*_k = 0$ and $\theta^*_i = \theta_i + \alpha_i\theta_k$. In that sense we say that project k is *redundant*. Hence, let N be the number of projects with linearly independent payoffs—that is, such that there is no real vector $\alpha \in R^N$ such that $\sum_i \alpha_i\Gamma_{ij} = 0$.

The existence of a solution to the system of K equations on N independent variables θ as described in equation (1.11) depends only on the number of linearly independent lines of the matrix $\Gamma^\top$ (or the number

of linearly independent columns of Γ). It is a standard result from linear algebra that a unique solution θ to the preceding system of equations exists for any vector $(y - Y)$ if and only if $K = N$. When a unique solution is possible, the market is said to be *complete* at time $t = 0$. Otherwise, is said to be *incomplete* at that point in time.

In other words, completeness of the market is equivalent to having the number of projects with linearly independent payoffs N equal to the number of states K.

Using this notation and assuming as before that the investors maximize strictly increasing, concave, differentiable utility functions, the optimal portfolio choice can be characterized as follows: Let $U(x, y)$ be one such utility function. Then an optimal portfolio must maximize the expected utility, satisfying

$$\max_{\theta} E[U(x, y)] \equiv \max_{\theta} E\{U[x(\theta), y(\theta)]\}$$

subject to the constraints (1.9) and (1.10). First-order conditions imply that at the optimal portfolio choice

$$dE(U) = 0 \Rightarrow -U_x V^{\top} + E(U_y \Gamma^{\top}) = 0.$$

The interpretation of this condition is simple. The amount of utility decrease because consumption was diminished at $t = 0$ must be compensated by the increased expected utility due to the larger future consumption.

In other words, an optimal portfolio choice links the N-dimensional value vector of the projects V to the $N \times K$ matrix of future payoffs by the relationship

$$V = \sum_{j=1}^{K} p_j \Gamma_j \frac{\partial U / \partial y_j}{\partial U / \partial x} = \sum_{j=1}^{K} \Gamma_j \psi_j,$$

where

$$p_j \frac{\partial U / \partial y_j}{\partial U / \partial x} \equiv \psi_j > 0 \tag{1.12}$$

given the assumption that utilities are strictly increasing. The interpretation for the components of the K-dimensional vector ψ is as follows: We defined Γ_{ij} as the payoff of project i in state j. Suppose that such payoff increases to $\Gamma_{ij} + 1$. This implies that the value of the project would change from V_i to $V_i + \psi_j$. Therefore, ψ_j measures the change in value of any project for a unitary increase of the payoff in state j. For that reason the K-dimensional vector ψ is termed the vector of *state-prices* and ψ_j is called the price of state j.

1.2.2 Value and the Absence of Arbitrage

Under the assumption that all agents have increasing utility functions in their wealth, simple economic considerations can be developed.

An *arbitrage opportunity* is a portfolio θ such that initial consumption is not reduced below X and, still, future consumption may be greater than Y. Hence, if all agents have increasing utility functions in their wealth and choose portfolios so as to optimize their utilities, an existing arbitrage opportunity would be the optimal portfolio choice for all such agents. In this way, an unbounded demand for that portfolio would be generated, allowing unbounded future consumption with no sacrifice of present consumption.

However, by construction an arbitrage opportunity is a portfolio θ such that $V^\intercal\theta \leq 0$ and, therefore, at least one of the components of θ is negative. This means that any holder of such a portfolio would be selling at $t = 0$ a project without holding the claim to its payoff at $t = 1$. If all the agents were optimally holding the arbitrage opportunity, they would all be selling that project, and there would be no one to whom they could sell it. It follows that in an exchange economy with several agents optimizing their utilities under the preceding assumptions, arbitrage opportunities cannot exist for they generate an inconsistent collective behavior.

The absence of arbitrage opportunities implies that any portfolio θ such that the K-dimensional vector of payoffs $\Gamma^\intercal\theta$ is positive in all states of nature must also satisfy $V^\intercal\theta \geq 0$. There is a mathematical result known as Farka's lemma[1] that states that this is true if and only if there exists a vector $\lambda \in R^K_+$ of strictly positive components such that

$$V = \Gamma\lambda.$$

Thus, one easily identifies each component λ_j with the state-prices ψ_j, in the case where utilities are differentiable. Notice that the reasoning to get to state-prices here is independent of any differentiability assumption involving the utility functions, as it is the interpretation of the components of ψ as state-prices. This result is summarized in the following:

Proposition 1.2.1. *The absence of arbitrage opportunities is equivalent to the existence of a strictly positive vector of state-prices $\psi \in R^K_+$ such that $V = \Gamma\psi$.*

Under the absence of arbitrage opportunities, it is clear that a redundant project such that $\Gamma_{kj} = \sum_{i\neq k} \alpha_i\Gamma_{ij}$ has value

$$V_k = \sum_{j=1}^{K} \sum_{i\neq k} \alpha_i\Gamma_{ij}\psi_j = \sum_{i\neq k} \alpha_i V_i.$$

It follows that restraining the number of projects to those with linearly independent payoffs does not affect the consumption possibilities of the

investors at times $t = 0$ and $t = 1$. Otherwise stated, the only reason for considering redundant projects, given that they provide a payoff similar to a portfolio of the other projects, would be if they were cheaper than the equivalent portfolio, allowing a larger initial consumption. Under absence of arbitrage opportunities, this is not possible.

Whether or not the vector ψ of state-prices is uniquely defined depends once more on the number of independent lines of the matrix Γ, defined as the number of projects with linearly independent payoffs, N. Again, if the market is complete at $t = 0$, the K components of ψ solve a system of $K = N$ independent linear equations. In other words, the vector of state-prices is clearly uniquely defined as

$$\psi = \Gamma^{-1} V.$$

1.2.3 Arbitrage Opportunities and Investment

Markets being complete or not may have implications regarding the separation between the investment and consumption decision. In fact, under the absence of arbitrage opportunities, the required investment $I^\top \theta$ in a portfolio θ must equal its value $V^\top \theta$. Otherwise an arbitrage opportunity exists, allowing an increase of current consumption by

$$(V - I)^\top \theta,$$

for fixed future consumption or, equivalently, increasing future consumption by

$$(V - I)^\top \theta (1 + r),$$

for fixed current consumption. In the language of the first section of this chapter, $(V - I)^\top \theta$ would be the net present value of such an arbitrage opportunity.

Hence, market *completeness* together with the absence of arbitrage opportunities implies that all investments have zero net present value. In other words, if markets are complete, then any portfolio θ is a redundant security, and there is no special θ^* that improves the set of intertemporal consumption possibilities.

However, under *incomplete* markets V is not uniquely defined, since there is a family of possible vectors of state-prices. Consider that markets are incomplete and that both

$$\psi^1 \equiv \{\psi_j^1\} \quad \text{and} \quad \psi^2 \equiv \{\psi_j^2\}$$

are possible vectors of state-prices. Then, two possibly different vectors of values would exist—namely, $V^1 = \Gamma^\top \psi^1$ and $V^2 = \Gamma \psi^2$.

In this case, if $(V^1)^\top\theta \neq (V^2)^\top\theta$ for a given portfolio θ, then there is a positive net present value at least with respect to one of the state-price vectors. Suppose, without loss of generality, that $(V^1 - I)^\top\theta = 0$. In that case, under ψ^2

$$(V^2 - I)^\top\theta \neq 0.$$

If $(V^2 - I)^\top\theta > 0$, the investment in the portfolio θ has a positive net present value of $(V^2 - I)^\top\theta$. If $(V^2 - I)^\top\theta < 0$, a positive net present value strategy would be to sell at time $t = 0$ the portfolio θ by $I^\top\theta$, paying the respective payoff $\Gamma^\intercal\theta$ at time $t = 1$.

The issue here is that, given the nonuniqueness of the state-prices under market incompleteness, the value of a portfolio may be not uniquely defined, allowing for positive net present value investments.

1.2.4 Value and the Martingale Measure

Assume the existence of an investment such that its payoff is a constant and therefore does not depend on state of nature to be realized in the future. Such an investment is said to be *risk-free*. As in section 1.1, assume that for each unit invested at time 0, a number $(1 + r)$ of consumption units are received at time 1. It then follows from Proposition 1.2.1 that

$$1 = (1 + r)\sum_{j=1}^{K}\psi_j$$

and the value vector may be rewritten as

$$V = \frac{1}{1 + r}\Gamma\pi, \tag{1.13}$$

where π is an K-dimensional vector with components

$$\pi_i = \psi_i \Big/ \sum_{j=1}^{K}\psi_j.$$

By construction it follows that $\pi_i \in [0, 1]$ and $\sum_{i=1}^{K}\pi_i = 1$. Therefore, π may be interpreted as a probability vector in the space of the states of nature. In that case, the value of each project reads from the preceding expression as its expected discounted payoff, the expectation being defined under the probability measure π and the discounting using the risk-free interest rate.

When the expectation of future values coincides with the present value, the value process is said to follow a *martingale*. From equation (1.13) the payoffs discounted at the risk-free rate follow a martingale under the π-measure. For that reason, the π-measure is termed a *martingale measure*.

Also by construction, uniqueness of the probability measure π depends on the market being complete. If markets are not complete, different probability measures may arise leading to different value vectors V.

Given equation (1.13), the value of an arbitrary portfolio θ is given by

$$V^{\intercal}\theta = \frac{1}{1+r}\sum_{j=1}^{K}\pi_j(\Gamma^{\intercal}\theta)_j. \tag{1.14}$$

The vector of probabilities π does not necessarily correspond to the vector of real probabilities of occurrence of the different states of nature. Notice that the vector π is a theoretical construct that appears from the absence of arbitrage possibilities.

A natural point to raise is whether it would be possible to write the present value of an uncertain future cash flow as its discounted expected payoff, but now using the real probabilities of occurrence of the different states. The answer is yes, this is possible to do. However, if one changes the probabilities into the real ones and if the value is assumed to be the same as that which does not allow for arbitrage opportunities, the discount factor will have to be different from the preceding risk-free rate. This section shows how to make such an adjustment.

Given the original set of states of nature $\Omega = \{w_1, w_2, \ldots, w_K\}$, where each state w_i occurs with probability p_i, one may write

$$V_i = \Gamma_i\psi = \sum_{j=1}^{K}\Gamma_{ij}\psi_j = \sum_{j=1}^{K}\Gamma_{ij}\delta_j p_j,$$

where the factor $\delta_j = \frac{\psi_j}{p_j}$ is called the *state-price deflator*. Under complete markets, both ψ and p being unique, the resulting state-price deflator vector δ is also unique. It follows from the definition of the state-price vector in (1.12) that the state-price deflator is simply the ratio of marginal utilities

$$\delta_j = \frac{\partial U/\partial y_j}{\partial U/\partial x},$$

measuring the ratio of increased future utility in state j per decreased current utility.

A portfolio on the available investment projects may then be valued as the weighted sum of the corresponding expected deflated payoffs

$$V^{\intercal}\theta = \sum_{i=1}^{N}\sum_{j=1}^{K}\delta_j p_j\Gamma_{ij}\theta_i = \sum_{j=1}^{K}\delta_j p_j(\Gamma^{\intercal}\theta)_j = E(\delta^{\top}\Gamma^{\intercal}\theta).$$

Notice that this expression compares with equation (1.14) in the sense that it gives the same value. The difference, however, is that here *real* probabilities are used instead of the martingale measure π.

1.2.5 Beta Values

The present value of the portfolio has to be calculated as an expectation because it is not known ex-ante which state of nature will be realized. Thus, a weighted average over all possible outcomes is required. Ex-post, however, one knows exactly which state has been realized (say, state j) and may compare the resulting payoff $(\Gamma^\intercal\theta)_j$ of holding that portfolio with the value initially paid, $V^\intercal\theta$. The measure of such comparison is known as the total return on the portfolio, defined as

$$R_j^\theta = \frac{(\Gamma^\intercal\theta)_j}{V^\intercal\theta}, \tag{1.15}$$

which is clearly a random variable. Notice that if it is possible to choose a portfolio θ such that its payoff is independent of the future state of nature or, in other words, such that $(\Gamma^\intercal\theta)_j$ is independent of j, then it follows that

$$V^\intercal\theta = \sum_{j=1}^{K} \delta_j p_j (\Gamma^\intercal\theta)_j = (\Gamma^\intercal\theta)_1 \sum_{j=1}^{K} \delta_j p_j = \frac{(\Gamma^\intercal\theta)_1}{1+r}$$

where use has been made of the equality

$$E(\delta) = \sum_{j=1}^{K} \delta_j p_j = \sum_{j=1}^{K} \psi_j = \frac{1}{1+r},$$

which means that the expected deflator is the risk-free discount rate. The return on the preceding portfolio will be denoted by R and is given by $R = 1 + r$. Such a portfolio is risk-free by construction and, as one would expect, for each unit invested at time 0 it gives back $(1 + r)$ units at $t = 1$. This is what the total return measure tells.

In the case of a risky portfolio, the return is a random variable, whose expected value may be obtained from the expression

$$\operatorname{cov}(R^\theta, \delta) = E(R^\theta\delta) - E(R^\theta)E(\delta)$$

or, equivalently,

$$E(R^\theta) - \frac{E(R^\theta\delta)}{E(\delta)} = -\frac{\operatorname{cov}(R^\theta, \delta)}{E(\delta)}.$$

Since

$$E(R^\theta\delta) = \sum_{j=1}^{K} \frac{(\Gamma^\intercal\theta)_j}{V^\intercal\theta} \delta_j p_j = \frac{\sum_{j=1}^{K} (\Gamma^\intercal\theta)_j \delta_j p_j}{V^\intercal\theta} = 1,$$

it follows that

$$E(R^\theta) - R = -\frac{\operatorname{cov}(R^\theta, \delta)}{E(\delta)}. \tag{1.16}$$

This equality gives the expression for how much more return is expected from a portfolio of risky investments when compared to a risk-free investment. It may simplify into a more familiar relationship by taking a particular portfolio θ^* such that $\delta = \Gamma^\top \theta^*$. In that case,

$$\text{var}(\Gamma^\top \theta^*) = \text{cov}(\Gamma^\top \theta^*, \Gamma^\top \theta^*).$$

Now denote by $R^* = (\Gamma^\top \theta^*)/V^\top \theta^*$ the return on this particular portfolio. Multiplying both sides of the preceding equality by δ, dividing by $(V^\top \theta^*)^2$, and recalling that $\delta = \Gamma^\top \theta^*$, one gets

$$\delta = \frac{\text{cov}(R^*, \delta)}{E(\delta)} \frac{R^*}{\text{var}(R^*)} E(\delta)$$

or still

$$\delta = -[E(R^*) - R] \frac{R^*}{\text{var}(R^*)} E(\delta).$$

Using the linearity of the covariance operator, it then follows that

$$-\frac{\text{cov}(R^\theta, \delta)}{E(\delta)} = \frac{\text{cov}(R^\theta, R^*)}{\text{var}(R^*)} [E(R^*) - R],$$

which, together with equation (1.16), leads to the state-price model

$$E(R^\theta) - R = \beta_\theta [E(R^*) - R], \tag{1.17}$$

with

$$\beta_\theta = \frac{\text{cov}(R^\theta, R^*)}{\text{var}(R^*)}. \tag{1.18}$$

This so-called beta model states that the excess expected return on any arbitrary portfolio is proportional to the excess expected return on the portfolio θ^*. Portfolios with the same proportionality coefficient β are said to be in the same *class of risk* and β is said to be a measure of the systematic risk of portfolios with respect to the given portfolio θ^*. Notice in particular that the existence of a portfolio satisfying $\delta = \Gamma^\top \theta^*$ relies on the assumption that markets are complete.

Going back to the original problem of evaluating a project, recall from equation (1.15) that

$$V^\top \theta E(R^\theta) = E(\Gamma^\top \theta)$$

and, hence, the value of any portfolio of projects is simply written as

$$V^\top \theta = \frac{1}{E(R^\theta)} E(\Gamma^\top \theta),$$

the expected discounted payoff of the portfolio.

There are two main differences between the interpretation of this equation and the interpretation of equation (1.14). Although they provide the same vector, this last equation uses the *real* probabilities for the occurrence of the different states of nature. Also, the expected cash flow generated by the portfolio of projects is discounted not at the risk-free rate but at the expected return that is adjusted for the class of risk of the considered portfolio.

1.2.6 Exercises

1. Prove that if there is a strictly positive vector of state-prices $\psi \in R^K$ such that $V = \Gamma\psi$, then there are no arbitrage opportunities.
2. Derive the expression

$$-\frac{\text{cov}(R^\theta, \delta)}{E(\delta)} = \frac{\text{cov}(R^\theta, R^*)}{\text{var}(R^*)}[E(R^*) - R].$$

3. When developing the beta-valuation model, it is stated that the existence of a portfolio satisfying $\delta = \Gamma^\intercal\theta^*$ relies on the assumption that markets are complete. Explain why.
4. Consider two agents, A and B, who live in a one-period economy defined by two points in time. They are given endowments at both points in time. Both agents have the possibility of making productive investments. However, there is uncertainty with respect to the return of this investment in the future. With probability p it is $F_1(\epsilon)$ and with probability $1 - p$ it is $F_2(\epsilon)$. As a consequence, an agent who makes a productive investment cannot be certain about the future consumption. With probability p agent i can consume y_i^1 and with probability $1 - p$ can consume y_i^2. Their preferences are described by a utility function $V_i = EU_i(x, y)$ where $U_i(x, y)$ has the properties described in section 1.1. Find sufficient conditions for the level of investment to be independent of the preferences and of the endowments of the agents.
5. Consider a one-period economy with two possible states of nature at time $t = 1$. The risk-free rate in this economy is 1%. There is a project (call it project 1) in this economy with value today of 6.7 and payoffs $\Gamma_{11} = 10$ and $\Gamma_{12} = 5$.
   (a) Compute the vector of state-prices when there are no arbitrage opportunities.
   (b) Consider another project (project 2) with payoffs $\Gamma_{21} = -2$ and $\Gamma_{22} = 4$. The investment necessary to do project 1 is 5; to do project 2, we must invest 2. Will an agent invest in both projects?
6. Consider a two-period model with two assets: a risk-free asset that pays R in each state of nature and a risky asset that pays D in the first state

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- NPV decision rule: Ch. 8, §8.1–8.2 (p. 212–216)
- IRR and multiple IRRs: Ch. 8, §8.3 (p. 217–226)
- Choosing between projects; crossover rate: Ch. 8, §8.4 (p. 227–232)
- Incremental earnings and free cash flow; NWC, opportunity and sunk costs: Ch. 9, §9.2–9.4 (p. 250–265)
- CAPM and portfolio beta: Ch. 12, §12.4 (p. 366–372)
- WACC and the return on assets: Ch. 13, §13.1 (p. 383–386)
- Project-based costs of capital: unlevering and relevering: Ch. 13, §13.5 (p. 396–399)

the alternative rules is likely to lead to bad investment decisions. After establishing these rules in the context of a single, stand-alone project, we broaden our perspective to include evaluating multiple opportunities to select the best one. We conclude with a look at project selection when the firm faces limits on capital or managers' time.

8.1 The NPV Decision Rule

In the previous chapters, we learned how to convert between cash today and cash in the future using the discount rate. As long as we convert costs and benefits to the same point in time, we can use the Valuation Principle to make a decision. In practice, most corporations prefer to measure values in terms of their present value—that is, in terms of cash today. In this section, we apply the Valuation Principle to derive the concept of the *net present value* or *NPV*, which we can use to define the "golden rule" of financial decision making, the *NPV rule*.

Net Present Value

net present value (NPV) The difference between the present value of a project's or investment's benefits and the present value of its costs.

When the value of a cost or benefit is computed in terms of cash today, we refer to it as the present value (PV). Similarly, we define the **net present value (NPV)** of a project or investment as the difference between the present value of its benefits and the present value of its costs:

Net Present Value

$$NPV = PV(\text{Benefits}) - PV(\text{Costs}) \tag{8.1}$$

Let's consider a simple example. Suppose your firm is offered the following investment opportunity: In exchange for \$500 today, you will receive \$550 in one year. If the interest rate is 8% per year, then:

$$\begin{aligned} PV(\text{Benefit}) &= (\$550 \text{ in one year}) \div (\$1.08 \text{ in one year}/\$1 \text{ today}) \\ &= \$509.26 \text{ today} \end{aligned}$$

This PV is the amount you would need to put in the bank today to generate \$550 in one year ($\$509.26 \times 1.08 = \$550$). In other words, *the present value is the amount you need to invest at the current interest rate to recreate the cash flow.* We can think of this as the cash cost today of generating the cash flow ourselves.

Once the costs and benefits are in present value terms, we can compare them by computing the investment's NPV:

$$NPV = \$509.26 - \$500 = \$9.26 \text{ today}$$

The NPV is positive, so the benefits outweigh the costs, which means the firm should undertake this investment opportunity.

But what if you don't have the \$500 needed to cover the initial cost of the project? Does the project still have the same value? Because we computed the value using competitive market prices, it should not depend on your tastes or the amount of cash you have in the bank. If you don't have the \$500, suppose you borrow \$509.26 from the bank at the 8% interest rate and then take the project. What are your cash flows in this case?

$$\begin{aligned} &\text{Today: } \$509.26 \text{ (loan)} - \$500 \text{ (invested in the project)} = \$9.26 \\ &\text{In one year: } \$550 \text{ (from project)} - \$509.26 \times 1.08 \text{ (loan balance)} = \$0 \end{aligned}$$

This transaction leaves you with exactly $9.26 extra cash in your pocket today and no future net obligations. So taking the project is similar to having an extra $9.26 in cash up front. Thus, the NPV expresses the value of an investment decision as an amount of cash received today. *As long as the NPV is positive, the decision increases the value of the firm and is a good decision regardless of your current cash needs or preferences regarding when to spend the money.*

The NPV Decision Rule

As shown in the last example, the Valuation Principle implies that we should undertake projects with a positive NPV because undertaking those projects increase wealth. That is, good projects are those for which the present value of the benefits exceeds the present value of the costs. As a result, the value of the firm increases and investors are wealthier. Projects with negative NPVs have costs that exceed their benefits. Accepting them is equivalent to losing money today.

NPV decision rule When choosing among investment alternatives, take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today.

We capture this logic in the **NPV Decision Rule**:

When making an investment decision, take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today.

Because NPV is expressed in terms of cash today, using the NPV decision rule is a simple way to apply the Valuation Principle. Decisions that increase wealth are superior to those that decrease wealth. We don't need to know anything about the investor's preferences to reach this conclusion. As long as we have correctly captured all the cash flows of a project and applied the appropriate discount rate, we can determine whether the project makes us wealthier. Being wealthier increases our options and makes us better off, whatever our preferences are.

A common way companies apply the NPV rule in financial practices is when deciding whether to accept or reject a project. Because rejecting the project generally has $NPV = 0$ (there are no new costs or benefits from not doing the project), the NPV decision rule implies that we should

- Accept positive-NPV projects; accepting them is equivalent to receiving their NPV in cash today, and
- Reject negative-NPV projects; accepting them would reduce the value of the firm, whereas rejecting them has no cost $(\text{NPV} = 0)$.

If the NPV is exactly zero, then you will neither gain nor lose by accepting the project instead of rejecting it, which also has an NPV of zero. It is not a bad project because it does not reduce the firm's value, but it does not add value to the firm either.

EXAMPLE 8.1 The NPV Is Equivalent to Cash Today

Problem

After saving $1500 by waiting tables, you are about to buy a 50-inch LCD TV. You notice that the store is offering a "one year same as cash" deal. You can take the TV home today and pay nothing until one year from now, when you will owe the store the $1500 purchase price. If your savings account earns 5% per year, what is the NPV of this offer? Show that its NPV represents cash in your pocket.

Solution

Plan

You are getting something worth $1500 today (the TV) and in exchange will need to pay $1500 in one year. Think of it as getting back the $1500 you thought you would have to spend today to get the TV. We treat it as a positive cash flow.

Cash flows:

Today	In one year
+$1500	−$1500

The discount rate for calculating the present value of the payment in one year is your interest rate of 5%. You need to compare the present value of the cost ($1500 in one year) to the benefit today (a $1500 TV).

Execute

$$NPV = +1500 - \frac{1500}{(1.05)} = 1500 - 1428.57 = \$71.43$$

You could take $1428.57 of the $1500 you had saved for the TV and put it in your savings account. With interest, in one year it would grow to $1428.57 × (1.05) = $1500, enough to pay the store. The extra $71.43 is money in your pocket to spend as you like (or put toward the speaker system for your new media room).

Evaluate

By taking the delayed payment offer, we have extra net cash flows of $71.43 today. If we put $1428.57 in the bank, it will be just enough to offset our $1500 obligation in the future. Therefore, this offer is equivalent to receiving $71.43 today, without any future net obligations.

Concept Check

1. What is the NPV decision rule? How is it related to the Valuation Principle?
2. Why doesn't the NPV decision rule depend on the investor's preferences?

8.2 Using the NPV Rule

We continue our discussion of investment decision rules by considering a take-it-or-leave-it decision about a single, stand-alone project. By undertaking this project, the firm does not constrain its ability to take other projects. In the case of a stand-alone project, the alternatives we are considering are to accept or reject a project. The NPV rule then implies that we should compare the project's NPV to zero (the NPV of rejecting the project and doing nothing). Thus, we should accept the project if its NPV is positive.[1]

Organizing the Cash Flows and Computing the NPV

Researchers at Fredrick's Feed and Farm have made a breakthrough. They believe they can produce a new, environmentally friendly fertilizer at a substantial cost saving over the company's existing line of fertilizer. The fertilizer will require a new plant that can be built immediately at a cost of $81.6 million. Financial managers estimate that the benefits of the new fertilizer will be $28 million per year, starting at the end of the first year and lasting for four years, as shown by the following timeline:



Thus, the cash flows are an immediate $81.6 million outflow followed by an annuity inflow of $28 million per year for four years. Therefore, given a discount rate r, the NPV of this project is:

[1]Some prefer to state the rule that the project should be accepted if its NPV is non-negative. Since the firm should be indifferent at an NPV of exactly zero (you will neither gain nor lose by accepting the project instead of rejecting it), we prefer to state it this way. A zero NPV project is not a bad project because it does not reduce the firm's value, but it does not add value to the firm either.

$$NPV = -81.6 + \frac{28}{1+r} + \frac{28}{(1+r)^2} + \frac{28}{(1+r)^3} + \frac{28}{(1+r)^4} \tag{8.2}$$

We can also use the annuity formula from Chapter 4 to write the NPV as:

$$NPV = -81.6 + \frac{28}{r}\left(1 - \frac{1}{(1+r)^4}\right) \tag{8.3}$$

To apply the NPV rule, we need to know the cost of capital. The financial managers responsible for this project estimate a cost of capital of 10% per year. If we replace r in Eq. 8.2 or 8.3 with the project's cost of capital of 10%, we get an NPV of $7.2 million, which is positive. Recall that a net present value tells us the present value of the benefits (positive cash flows) net of the costs (negative cash flows) of the project. By putting everything into present values, it puts all the costs and benefits on an equal footing for comparison. In this case, the benefits outweigh the costs by $7.2 million in present value. The NPV investment rule indicates that by making the investment, Fredrick's will increase the value of the firm today by $7.2 million, so Fredrick's should undertake this project.

The NPV Profile

NPV profile A graph of a project's NPV over a range of discount rates.

The NPV of the project depends on its appropriate cost of capital. Often, there may be some uncertainty regarding the project's cost of capital. In that case, it is helpful to compute an **NPV profile**, which graphs the project's NPV over a range of discount rates. It is easiest to prepare the NPV profile using a spreadsheet such as Excel. We simply repeat our calculation of the NPV above using a range of different discount rates instead of only 10%. Figure 8.1 presents the NPV profile for Fredrick's project by plotting the NPV as a function of the discount rate, r.[2]

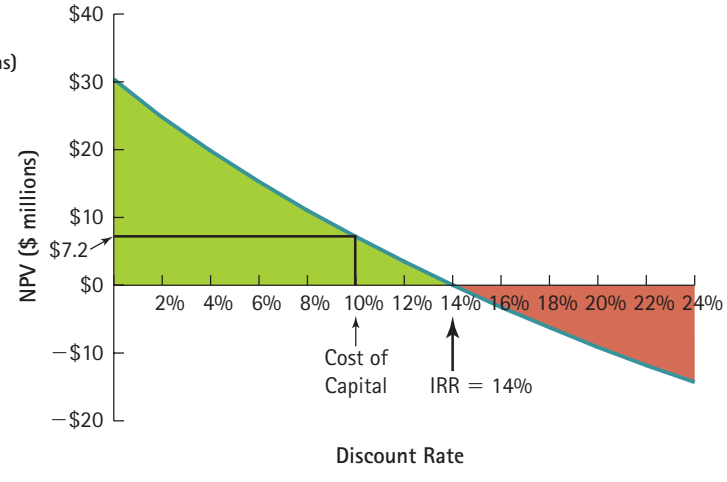
FIGURE 8.1
NPV of Fredrick's New Project

The graph in panel (b) shows the NPV as a function of the discount rate based on the data in panel (a). The NPV is positive, represented by the green-shaded area, only for discount rates that are less than 14%, the *internal rate of return (IRR).* Given the cost of capital of 10%, the project has a positive NPV of $7.2 million. The red-shaded area indicates discount rates above the 14% IRR with negative NPVs.

Panel (a)

Discount Rate	NPV ($ millions)
0%	$30.4
2%	$25.0
4%	$20.0
6%	$15.4
8%	$11.1
10%	$7.2
12%	$3.4
14%	$0.0
16%	−$3.3
18%	−$6.3
20%	−$9.1
22%	−$11.8
24%	−$14.3

Panel (b)



[2]In the online appendix to this chapter, we show you how to create an NPV profile in Excel.

internal rate of return (IRR) The interest rate that sets the net present value of the cash flows equal to zero.

Notice that the NPV is positive only for discount rates that are less than 14% (the green shaded area on the graph). Referring to the graph and the accompanying data table, we see that at 14%, the NPV is zero. The discount rate that sets the net present value of the cash flows equal to zero is an investment's **internal rate of return (IRR)**. Thus, by constructing the NPV profile, we have determined that Fredrick's project has an IRR of 14%. We can also compute the IRR without graphing the NPV by using a financial calculator or a spreadsheet's IRR function (see the appendix to Chapter 4. The calculator instructions for solving for the rate of return can be used to solve for the IRR).

	N	I/Y	PV	PMT	FV
Given:	4		−81.6	28	0
Solve for:		14			

Excel Formula: =RATE(NPER,PMT,PV,FV)=RATE(4,28,−81.6,0)

Measuring Sensitivity with IRR

In our Fredrick's example, the firm's managers provided the cost of capital. If you are unsure of your cost of capital estimate, it is important to determine how sensitive your analysis is to errors in this estimate. The IRR can provide this information. For Fredrick's, if the cost of capital estimate is more than the 14% IRR, the NPV will be negative (see the red shaded area in Figure 8.1). Therefore, as long as our estimate of the cost of capital of 10% is within 4% of the true cost of capital, our decision to accept the project is correct. In general, the difference between the cost of capital and the IRR tells us the amount of estimation error in the cost of capital estimate that can exist without altering the original decision.

Alternative Rules Versus the NPV Rule

The NPV rule indicates that Fredrick's should undertake the investment in fertilizer technology. As we evaluate alternative rules for project selection in the subsequent sections, keep in mind that sometimes other investment rules may give the same answer as the NPV rule, but at other times they may disagree. When the rules conflict, always base your decision on the NPV rule, which is the most accurate and reliable decision rule.

Concept Check

3. Explain the NPV rule for stand-alone projects.
4. How can you interpret the difference between the cost of capital and the IRR?

8.3 Alternative Decision Rules

Even though the NPV rule is the most accurate and reliable rule, in practice a wide variety of rules are applied, often in tandem with the NPV rule. In a 2001 study, John Graham and Campbell Harvey[3] found that 75% of the firms they surveyed used the NPV rule for making investment decisions. This result is substantially different from that found in a similar study in 1977 by L. J. Gitman and J. R. Forrester,[4] who found that only 10% of

[3]John Graham and Campbell Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001): 187–243.

[4]L. J. Gitman and J. R. Forrester, Jr., "A Survey of Capital Budgeting Techniques Used by Major U.S. Firms," *Financial Management* 6 (1977): 66–71.

USING EXCEL

Computing NPV and IRR

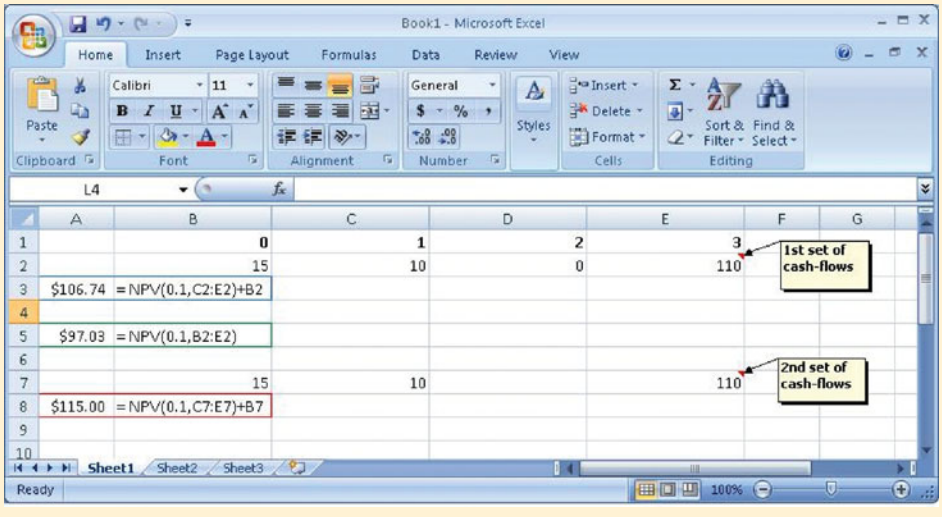
Here we discuss how to use Microsoft Excel to solve for NPV and IRR. We also identify some pitfalls to avoid when using Excel.

NPV Function: Leaving Out Date 0

Excel's NPV function has the format *NPV* (*rate, value1, value2*, . . .), where "rate" is the interest rate per period used to discount the cash flows, and "value1", "value2", etc., are the cash flows (or ranges of cash flows). The NPV function computes the present value of the cash flows *assuming the first cash flow occurs at date 1*. Therefore, if a project's first cash flow occurs at date 0, we cannot use the NPV function by itself to compute the NPV. We can use the NPV function to compute the present value of the cash flows from date 1 onward, and then we must add the date 0 cash flow to that result to calculate the NPV. The screenshot below shows the difference. The first NPV calculation (outlined in blue) is correct: we used the NPV function for all of the cash flows occurring at time 1 and later and then added on the first cash flow occurring at time 0 since it is already in present value. The second calculation (outlined in green) is incorrect: we used the NPV function for all of the cash flows, but the function assumed that the first cash flow occurs in period 1 instead of immediately.

NPV Function: Ignoring Blank Cells

Another pitfall with the NPV function is that cash flows that are left blank are treated differently from cash flows that are equal to zero. If the cash flow is left blank, *both the cash flow and the period are ignored.* For example, the second set of cash flows below is equivalent to the first—we have simply left the cash flow for date 2 blank instead of entering a "0." However, the NPV function ignores the blank cell at date 2 and assumes the cash flow is 10 at date 1 and 110 at date 2, which is clearly not what is intended and produces an incorrect answer (outlined in red).



Because of these idiosyncrasies, we avoid using Excel's NPV function. It is more reliable to compute the present value of each cash flow separately in Excel, and then sum them to determine the NPV.

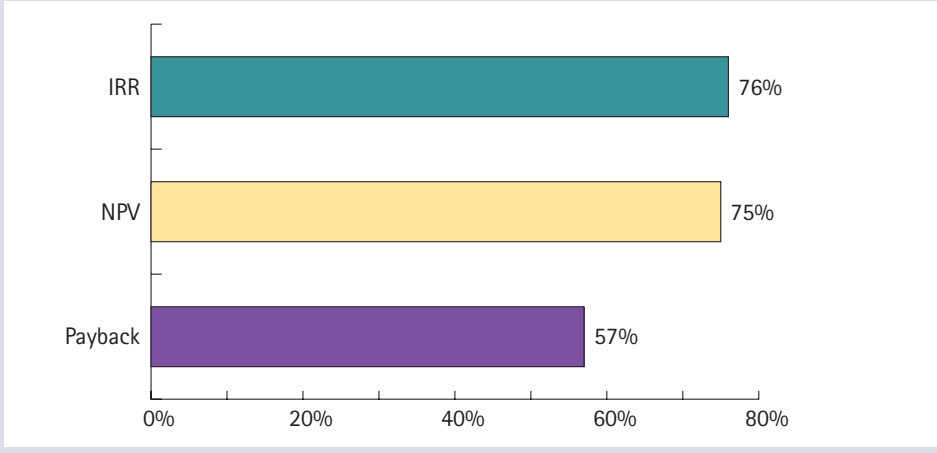
IRR Function

Excel's IRR function has the format *IRR* (*values, guess*), where "values" is the range containing the cash flows, and "guess" is an optional starting guess where Excel begins its search for an IRR. Two things to note about the IRR function:

1. The values given to the IRR function should include all of the cash flows of the project, including the one at date 0. In this sense, the IRR and NPV functions in Excel are inconsistent.
2. Like the NPV function, the IRR ignores the period associated with any blank cells.

FIGURE 8.2
The Most Popular Decision Rules Used by CFOs

The bar graph shows the most popular decision rules used by CFOs in Professors Graham and Harvey's 2001 survey. Many CFOs used more than one method, but no other methods were mentioned by more than half of CFOs.



firms used the NPV rule. Business students in recent years have been listening to their finance professors! Even so, Graham and Harvey's study indicates that one-fourth of U.S. corporations do not use the NPV rule. Exactly why other capital budgeting techniques are used in practice is not always clear. Figure 8.2 summarizes the top three decision rules given in the survey. Because you may encounter these techniques in the business world, you should know what they are, how they are used, and how they compare to NPV. In this section, we examine alternative decision rules for single, stand-alone projects within the firm. The focus here is on *the payback rule* and *the IRR rule*.

The Payback Rule

payback investment rule Only projects that pay back their initial investment within the payback period are undertaken.

payback period The amount of time until the cash flows from a project offset the initial investment. The time it takes to pay back the initial investment.

The simplest investment rule is the **payback investment rule**, which states that you should only accept a project if its cash flows pay back its initial investment within a prespecified period. The rule is based on the notion that an opportunity that pays back its initial investment quickly is a good idea. To apply the payback rule,

1. Calculate the amount of time it takes to pay back the initial investment, called the **payback period**.
2. Accept the project if the payback period is less than a prespecified length of time—usually a few years.
3. Reject the project if the payback period is greater than that prespecified length of time.

For example, a firm might adopt any project with a payback period of less than two years.

EXAMPLE 8.2
Using the Payback Rule

Problem

Assume Fredrick's requires all projects to have a payback period of two years or less. Would the firm undertake the fertilizer project under this rule?

Solution

Plan

In order to implement the payback rule, we need to know whether the sum of the inflows from the project will exceed the initial investment before the end of two years. The project has inflows of $28 million per year and an initial investment of $81.6 million.

Execute

The sum of the cash flows for years 1 and 2 is $\$28 \times 2 = \$56$ million, which will not cover the initial investment of $81.6 million. In fact, it will not be until year 3 that the cash inflows exceed the initial investment ($\$28 \times 3 = \$84$ million). Because the payback period for this project exceeds two years, Fredrick's will reject the project.

Evaluate

While simple to compute, the payback rule requires us to use an arbitrary cutoff period in summing the cash flows. Further, also note that the payback rule does not discount future cash flows. Instead, it simply sums the cash flows and compares them to a cash outflow in the present. In this case, Fredrick's would have rejected a project that would have increased the value of the firm.

Relying on the payback rule analysis in Example 8.2, Fredrick's will reject the project. However, as we saw earlier, with a cost of capital of 10%, the NPV is $7.2 million. Following the payback rule would be a mistake because Fredrick's will pass up a project worth $7.2 million.

The payback rule is not as reliable as the NPV rule because it (1) ignores the time value of money, (2) ignores cash flows after the payback period, and (3) lacks a decision criterion grounded in economics (what is the right number of years to require for a payback period?). Some companies have addressed the first failing by computing the payback period using discounted cash flows, called the **discounted payback rule**. However, this does not solve the fundamental problem because the other two failings remain. Despite these failings, Graham and Harvey found that about 57% of the firms they surveyed reported using the payback rule as part of the decision-making process.

discounted payback rule Only accept projects where the sum of the discounted cash flows within the payback period is greater than or equal to the initial investment.

Why do some companies consider the payback rule? The answer probably relates to its simplicity. This rule is typically used for small investment decisions—for example, whether to purchase a new copy machine or to service the old one. In such cases, the cost of making an incorrect decision might not be large enough to justify the time required to calculate the NPV. The appeal of the payback rule is that it favors short-term projects. Some firms are unwilling to commit capital to long-term investments. Also, if the required payback period is short (one to two years), then most projects that satisfy the payback rule will have a positive NPV.

The Internal Rate of Return Rule

internal rate of return (IRR) investment rule A decision rule that accepts any investment opportunity where the IRR exceeds the opportunity cost of capital and otherwise rejects the opportunity.

Similar to NPV, the **internal rate of return (IRR) investment rule** is based on the concept that if the return on the investment opportunity you are considering is greater than the return on other alternatives in the market with equivalent risk and maturity (i.e., the project's cost of capital), you should undertake the investment opportunity. We state the rule formally as follows:

> ***IRR investment rule**: Take any investment opportunity whose IRR exceeds the opportunity cost of capital. Turn down any opportunity whose IRR is less than the opportunity cost of capital.*

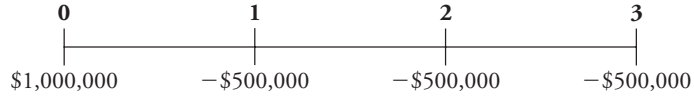
The IRR investment rule will give the correct answer (that is, the same answer as the NPV rule) in many—but not all—situations. For instance, it gives the correct answer for Fredrick's fertilizer opportunity. From Figure 8.1, whenever the cost of capital is in the green area below the IRR (14%), the project has a positive NPV and you should undertake the investment. Table 8.1 summarizes our analysis of Fredrick's new project. The NPV and IRR rules agree, but using the payback rule with a required payback period of two years or less would cause Fredrick's to reject the project.

TABLE 8.1
Summary of NPV, IRR, and Payback for Fredrick's New Project

NPV at 10%	$7.2 million	Accept ($7.2 million > 0)
Payback Period	3 years	Reject (3 years > 2 year required payback)
IRR	14%	Accept (14% > 10% cost of capital)

In general, the IRR rule works for a stand-alone project if all of the project's negative cash flows precede its positive cash flows. But in other cases, the IRR rule may disagree with the NPV rule and thus be incorrect. Let's examine several situations in which the IRR fails.

Delayed Investments. Star basketball player Evan Cole is graduating from college with a degree in finance and preparing for the NBA draft. Several companies have already approached him with endorsement contracts. Two competing sports drink companies are trying to sign him. QuenchIt offers him a single upfront payment of $1 million to exclusively endorse their sports drink for three years. PowerUp offers $500,000 per year, payable at the end of each of the next three years, to endorse their product exclusively. Which offer is better? One direct way to compare the two contracts is to realize that signing with QuenchIt causes Evan to forgo the PowerUp contract, or $500,000 per year. Considering the risk of his alternative income sources and available investment opportunities, Evan estimates his opportunity cost of capital to be 10%. The timeline of Evan's investment opportunity is:



The NPV of Evan's investment opportunity is:

$$NPV = 1{,}000{,}000 - \frac{500{,}000}{1+r} - \frac{500{,}000}{(1+r)^2} - \frac{500{,}000}{(1+r)^3}$$

By setting the NPV equal to zero and solving for r, we find the IRR. We can use either a financial calculator or a spreadsheet to find the IRR:

	N	I/Y	PV	PMT	FV
Given:	3		1,000,000	−500,000	0
Solve for:		23.38			

Excel Formula: =RATE(NPER,PMT,PV,FV)=RATE(3,−500000,1000000,0)

The 23.38% IRR is larger than the 10% opportunity cost of capital. According to the IRR rule, Evan should sign the deal. But what does the NPV rule say?

$$NPV = 1{,}000{,}000 - \frac{500{,}000}{1.1} - \frac{500{,}000}{1.1^2} - \frac{500{,}000}{1.1^3} = -\$243{,}426$$

At a 10% discount rate, the NPV is negative, so signing the deal would reduce Evan's wealth. He should not sign the endorsement deal with QuenchIt, but should sign with PowerUp instead.

To resolve this conflict, we can prepare an NPV profile for the QuenchIt contract. Figure 8.3 plots the NPV of the investment opportunity for a range of discount rates. It shows that, no matter what the cost of capital is, the IRR rule and the NPV rule will give exactly opposite recommendations. That is, the NPV is positive only when the opportunity cost of capital is *above* 23.38% (the IRR). Evan should accept the investment only when the opportunity cost of capital is greater than the IRR, the opposite of what the IRR rule recommends.

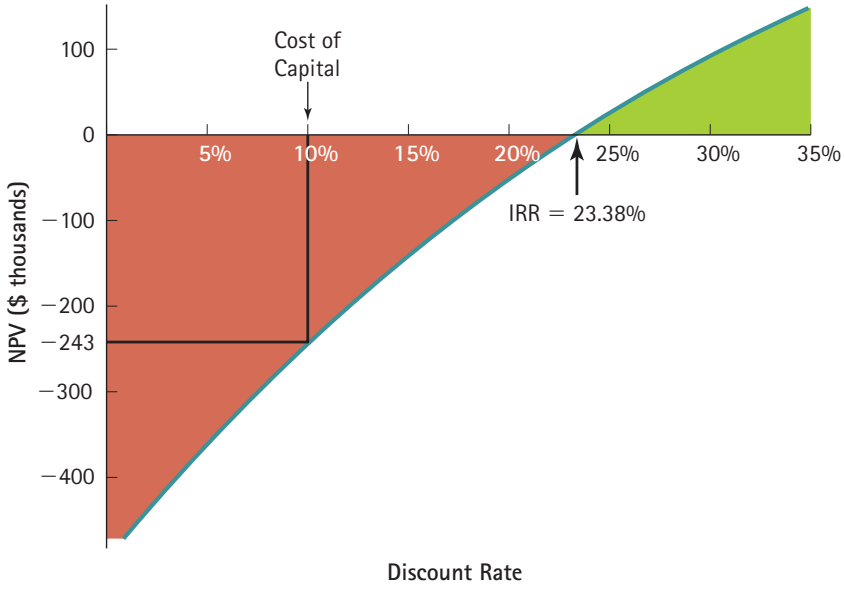
Figure 8.3 also illustrates the problem with using the IRR rule in this case. For most investment opportunities, expenses occur initially and cash is received later. In this case, Evan gets cash *upfront* from QuenchIt but the forgone cash flows from PowerUp occurred later. It is as if Evan borrowed money, and when you borrow money you prefer as *low* a rate as possible. Evan's optimal rule is to borrow money so long as the rate at which he borrows is *less* than the cost of capital.

Even though the IRR rule fails to give the correct answer in this case, the IRR itself still provides useful information *in conjunction* with the NPV rule. As mentioned earlier, the IRR provides information on how sensitive the investment decision is to uncertainty in the cost of capital estimate. In this case, the difference between the cost of capital and the IRR is large—10% versus 23.38%. Evan would have to have underestimated the cost of capital by 13.38% to make the NPV positive.

Multiple IRRs. Evan has informed QuenchIt that it needs to sweeten the deal before he will accept it. In response, the company has agreed to make an additional payment of $600,000 in 10 years as deferred compensation for the long-term increase in sales that

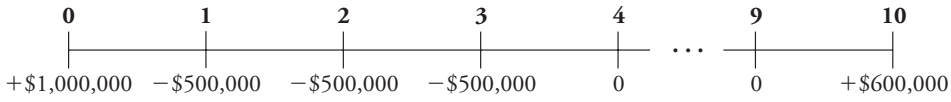
FIGURE 8.3
NPV of Cole's $1 Million QuenchIt Deal

When all the benefits of an investment occur at time zero and before the costs, the NPV is an *increasing* function of the discount rate. The NPV is positive in the green-shaded areas and negative in the red-shaded areas. Notice that the NPV is positive when the cost of capital is above 23.38%, the IRR, so the NPV and IRR rules conflict.



even a short-term endorsement by Evan would bring. Should he accept or reject the new offer?

We begin with the new timeline:



The NPV of Evan's new investment opportunity is:

$$NPV = 1{,}000{,}000 - \frac{500{,}000}{1+r} - \frac{500{,}000}{(1+r)^2} - \frac{500{,}000}{(1+r)^3} + \frac{600{,}000}{(1+r)^{10}}$$

We can find the IRR for this investment opportunity by creating an NPV profile and noting where it crosses zero. Figure 8.4 plots the NPV of the opportunity at different discount rates. In this case, there are *two* IRRs—that is, there are two values of r that set the NPV equal to zero. You can verify this fact by substituting IRRs of 5.79% and 13.80% for r into the equation. Because there is more than one IRR, we cannot apply the IRR rule. It is also worth noting that you should take special care when using a spreadsheet or financial calculator to determine the IRR. Both solve for the IRR through trial and error because you cannot calculate IRR directly. In cases where there is more than one IRR, the spreadsheet or calculator will simply produce the first one that it finds, with no mention that there could be others! Some financial calculators will return an error message if there are multiple IRRs. Thus, it always pays to create the NPV profile.

For guidance, let's turn to the NPV rule. If the cost of capital were *either* below 5.79% or above 13.80%, Evan should undertake the opportunity. But given his cost of capital of 10%, he should still turn it down. Notice that even though the IRR rule fails in this case, the two IRRs are still useful as bounds on the cost of capital estimate. If the cost of

FIGURE 8.4

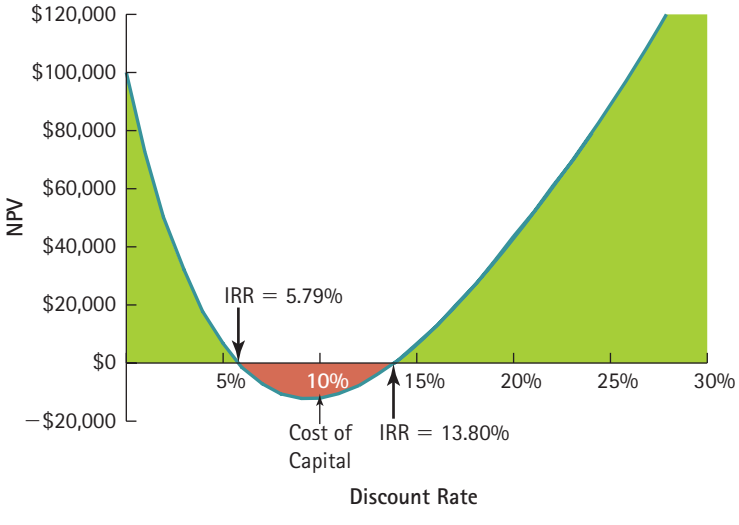
NPV of Evan's Sports Drink Deal with Additional Deferred Payments

The graph in panel (b) shows the NPV of Evan's deal with additional deferred payment based on the data in panel (a). In this case, there are two IRRs, invalidating the IRR rule. If the opportunity cost of capital is *either* below 5.79% or above 13.80%, Evan should accept the deal because the NPV is then positive, as indicated by the green-shaded areas. At any point between the two IRRs, the NPV is negative (see the red-shaded area).

Panel (a)

0%	100,000
1%	72,680
2%	50,267
3%	32,151
4%	17,793
5%	6724
6%	−1469
7%	−7148
8%	−10,632
9%	−12,201
10%	−12,100
11%	−10,547
12%	−7732
13%	−3823
14%	1030
15%	6698
16%	13,065
17%	20,030
18%	27,502
19%	35,403
20%	43,663
21%	52,219
22%	61,019
23%	70,014
24%	79,161

Panel (b)



COMMON MISTAKE

IRR Versus the IRR Rule

Throughout this subsection, we have distinguished between the IRR itself and the IRR rule. While we have pointed out the shortcomings of using the IRR rule to make investment decisions, *the IRR itself remains a very useful tool.* The IRR measures the average return of the investment and indicates the sensitivity of the NPV to estimation error in the cost of capital. Thus, knowing the IRR can be very useful, but relying on it to make investment decisions can be hazardous.

capital estimate is wrong, and it is actually smaller than 5.79% or larger than 13.80%, the decision not to pursue the project will change because it will have a positive NPV.

There is no easy fix for the IRR rule when there are multiple IRRs. Although the NPV is negative between the IRRs in this example, the reverse is also possible (see Figure 8.5). In that case, the project would have a positive NPV for discount rates between the IRRs rather than for discount rates lower or higher than the IRRs. Furthermore, there are situations in which more than two IRRs exist.[5] In such situations, our only choice is to rely on the NPV rule.

modified internal rate of return (MIRR) The discount rate that sets the NPV of modified cash flows of a project equal to zero. Cash flows are modified so there is only one negative cash flow (and one sign-change) to ensure that only one IRR exists.

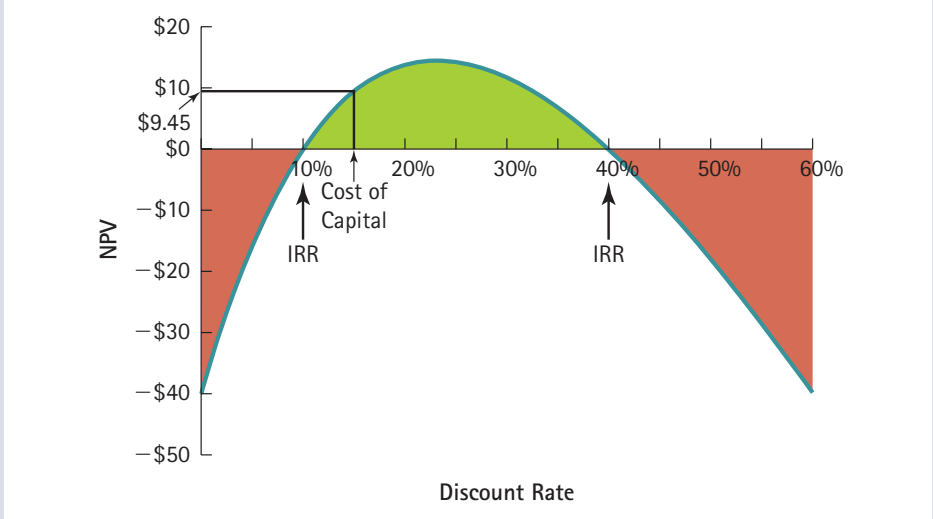
Modified Internal Rate of Return

The fact that there can be multiple IRRs for the cash flows from a project is a clear disadvantage for the IRR. To overcome this, some have proposed various ways of modifying the cash flows before computing the IRR. All these modifications have the common feature that they group the cash flows so that there is only one negative cash flow, occurring at either the beginning or the end. In that case, there is only one sign-change for the cash flows as a whole and hence only one IRR. This new IRR, computed as the discount rate that sets the NPV of the modified cash flows of the project equal to zero, is called the **modified internal rate of return (MIRR)**.

FIGURE 8.5

NPV Profile for a Project with Multiple IRRs

The graph shows the NPV profile for the multiple-IRR project with cash flows of −$1000, $2500 and −$1540 in years 0, 1, and 2, respectively. As the NPV profile shows, the project has two IRRs: 10% and 40%.



[5] In general, there can be as many IRRs as the number of times the project's cash flows change sign over time.

Why Do Rules Other Than the NPV Rule Persist?

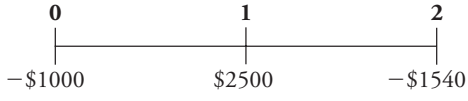
Professors Graham and Harvey found that a sizable minority of firms (25%) in their study do not use the NPV rule at all. In addition, about 50% of firms surveyed used the payback rule. Furthermore, it appears that most firms use *both* the NPV rule and the IRR rule. Why do firms use rules other than NPV if they can lead to erroneous decisions?

One possible explanation for this phenomenon is that Graham and Harvey's survey results might be misleading. CFOs who were using the IRR as a sensitivity measure in conjunction with the NPV rule might have checked both the IRR box and the NPV box on the survey. The question they were asked was, "How frequently does your firm use the following techniques when deciding which projects or acquisitions to pursue?" By computing the IRR and using it in conjunction with the NPV rule to estimate the sensitivity of their results, they might have felt they were using *both* techniques. Nevertheless, a significant minority of managers surveyed replied that they used only the IRR rule, so this explanation cannot be the whole story.

One common reason that managers give for using the IRR rule exclusively is that you do not need to know the opportunity cost of capital to calculate the IRR. On a superficial level, this is true: The IRR does not depend on the cost of capital. You may not need to know the cost of capital to *calculate* the IRR, but you certainly need to know the cost of capital when you *apply* the IRR rule. Consequently, the opportunity cost is as important to the IRR rule as it is to the NPV rule.

In our opinion, some firms use the IRR rule exclusively because the IRR sums up the attractiveness of investment opportunity in a single number without requiring the person running the numbers to make an assumption about the cost of capital. However, if a CFO wants a brief summary of an investment opportunity but does not want her employee to make a cost of capital assumption, she can also request a plot of the NPV as a function of the discount rate. Neither this request nor a request for the IRR requires knowing the cost of capital, but the NPV profile has the distinct advantage of being much more informative and reliable.

MIRR Technique. Let's clarify this with an example. You are considering a project that has the following three cash flows:



The NPV profile for this project, shown in Figure 8.5, identifies the two IRRs for this project as 10% and 40%.

Assume that your discount rate for this project is 15%. As Figure 8.5 shows, the NPV of the project at 15% is $9.45. We could modify the cash flows of the project to eliminate the multiple IRR problem. By discounting all of the negative cash flows to the present and compounding all of the positive cash flows to the end of the project, we have only two cash flows, yielding a single IRR. What discount rate and compounding rate should we use? One natural choice is our cost of capital for this project, which is 15%.

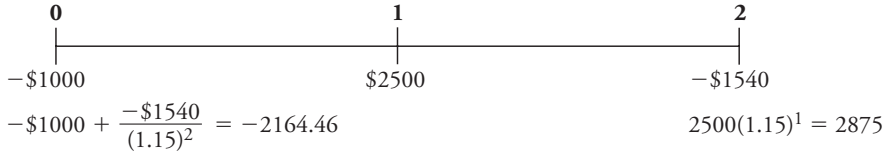
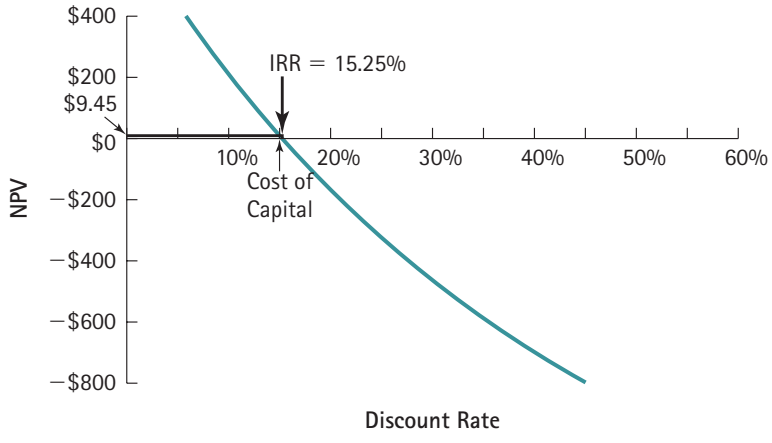


Figure 8.6 presents the NPV profile for our modified cash flows. As Figure 8.6 shows, there is now only a single IRR, at 15.25%. Because our cost of capital is 15%, we would properly accept the project using the IRR rule. Also note that the advantage of using 15% as our discount and compounding rates when modifying the cash flows is that the NPV of the modified cash flows at 15% is the same as the NPV of the true cash flows at 15%. Figure 8.6 also makes an important point: we are no longer evaluating the true cash flows of the project. Instead, we have modified them to force them to produce a single IRR. The NPV profile of the true cash flows of the project is given earlier in Figure 8.5 and is clearly different from the one produced by the modified cash flows in Figure 8.6.

FIGURE 8.6

NPV Profile of Modified Cash Flows for the Multiple-IRR Project from Figure 8.5

The graph presents the NPV profile for the modified project cash flows of −2164.46 in year 0 and 2875 in year 2. The modified cash flows have only one IRR: 15.25%. Given the 15% cost of capital, the IRR rule confirms that we should accept the project.



There is no set way to modify project cash flows to produce an MIRR. Two other approaches that each solve the multiple IRR problem are:

1. Discount all the negative cash flows to time 0 and leave the positive cash flows alone.
2. Leave the initial cash flow alone and compound all the remaining cash flows to the final period of the project. In this approach, you are implicitly reinvesting all the cash flows from the project at your compound rate until the project is complete.

Again, in either case if you use the project's cost of capital as your discount and compounding rate, you will not alter the NPV of the project at that discount rate. Further, a decision to accept or reject the project based on the modified IRR will be the same as the one based on the NPV decision rule.

MIRR: A Final Word. There is considerable debate about whether MIRR is truly better than IRR. Most of the argument centers on whether it is advisable to modify the cash flows of the project. The IRR is truly an internal rate of return based solely on the actual cash flows of the project. However, the IRR implicitly assumes that all cash flows generated by the project are reinvested at the project's IRR rather than at the firm's cost of capital until the project ends. For a project with a high IRR, this may be an unrealistic assumption. Further, there may be more than one IRR, which complicates its use. The MIRR avoids these problems, but is based on a set of cash flows modified through the use of a chosen discount and compounding rate. Thus, it is not really an internal rate of return and is no longer based solely on the actual cash flows of the project. Finally, MIRR still does not solve some of the other problems associated with using IRR when choosing among projects.

Concept Check

5. How do you apply the payback rule?
6. Under what conditions will the IRR rule lead to the same decision as the NPV rule?

8.4 Choosing Between Projects

Thus far, we have considered only decisions where the choice is either to accept or to reject a single, stand-alone project. Sometimes, however, a firm must choose just one project from among several possible projects. For example, a manager may be evaluating alternative package designs for a new product. The manager must choose only one of the designs. When choosing any one project excludes us from taking the other projects, we are facing **mutually exclusive projects**.

mutually exclusive projects Projects that compete with one another; by accepting one, you exclude the others.

When projects, such as the package designs, are mutually exclusive, it is not enough to determine which projects have positive NPVs. With mutually exclusive projects, the manager's goal is to rank the projects and choose only the best one. In this situation, the NPV rule provides a straightforward answer: *Pick the project with the highest NPV.*

EXAMPLE 8.3 NPV and Mutually Exclusive Projects

Problem

You own a small piece of commercial land near a university. You are considering what to do with it. You have been approached recently with an offer to buy it for $220,000. You are also considering three alternative uses yourself: a bar, a coffee shop, and an apparel store. You assume that you would operate your choice indefinitely, eventually leaving the business to your children. You have collected the following information about the uses. What should you do?

	Initial Investment	Cash Flow in the First Year (CF_1)	Growth Rate (g)	Cost of Capital (r)
Bar	$400,000	$60,000	3.5%	12%
Coffee shop	$200,000	$40,000	3%	10%
Apparel Store	$500,000	$75,000	3%	13%

Solution

Plan

Since you can only do one project (you only have one piece of land), these are mutually exclusive projects. In order to decide which project is most valuable, you need to rank them by NPV. Each of these projects (except for selling the land) has cash flows that can be valued as a growing perpetuity, so from Chapter 4, the present value of the inflows is $CF_1/(r - g)$. The NPV of each investment will be:

$$\frac{CF_1}{r - g} - \text{Initial Investment}$$

Execute

The NPVs are:

$$\text{Bar: } \frac{\$60{,}000}{0.12 - 0.035} - \$400{,}000 = \$305{,}882$$

$$\text{Coffee Shop: } \frac{\$40{,}000}{0.10 - 0.03} - \$200{,}000 = \$371{,}429$$

$$\text{Apparel Store: } \frac{\$75{,}000}{0.13 - 0.03} - \$500{,}000 = \$250{,}000$$

So, the ranking is

Alternative	NPV
Coffee Shop	$371,429
Bar	$305,882
Apparel Store	$250,000
Sell the Land	$220,000

and you should choose the coffee shop.

Evaluate

All the alternatives have positive NPVs, but you can take only one of them, so you should choose the one that creates the most value. Even though the coffee shop has the lowest cash flows, its lower start-up cost coupled with its lower cost of capital (it is less risky) make it the best choice.

Because the IRR is a measure of the expected return of investing in the project, you might be tempted to extend the IRR investment rule to the case of mutually exclusive projects by picking the project with the highest IRR. Unfortunately, picking one project over another simply because it has a larger IRR can lead to mistakes. Problems arise when the mutually exclusive investments have differences in scale (require different initial investments) and when they have different cash flow patterns. We discuss each of these situations in turn.

Differences in Scale

Would you prefer a 200% return on $1 or a 10% return on $1 million? The former return certainly sounds impressive and gives you great bragging rights, but at the end of the day you make only $2. The latter opportunity may sound much more mundane, but you make $100,000. This comparison illustrates an important shortcoming of IRR: Because it is a return, you cannot tell how much value has actually been created without knowing the basis for the return—a 10% IRR can have very different value implications for an initial investment of $1 million versus an initial investment of $100 million.

If a project has a positive NPV, then if we can double its size, its NPV will double: By the Valuation Principle, doubling the cash flows of an investment opportunity must make it worth twice as much. However, the IRR rule does not have this property—it is unaffected by the scale of the investment opportunity because the IRR measures the average return of the investment. Hence, the IRR rule cannot be used to compare projects of different scales. Let's illustrate this concept in the context of an example.

Identical Scale. We begin by considering two mutually exclusive projects with the same scale. Javier is evaluating two investment opportunities. If he went into business with his girlfriend, he would need to invest $10,000 and the business would generate incremental cash flows of $6000 per year for three years. Alternatively, he could start a two-computer Internet café. The computer setup will cost a total of $10,000 and will generate $5000 for three years. The opportunity cost of capital for both opportunities is 12% and both will require all his time, so Javier must choose between them. How valuable is each opportunity, and which one should Javier choose?

Let's consider both the NPV and IRR of each project. The timeline for the investment with Javier's girlfriend is:



The NPV of the investment opportunity when $r = 0.12$ is:

$$NPV = -10{,}000 + \frac{6000}{1.12} + \frac{6000}{1.12^2} + \frac{6000}{1.12^3} = \$4411$$

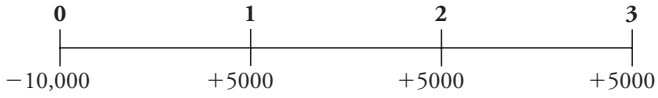
We can determine the IRR of this investment by using a financial calculator or spreadsheet:

	N	I/Y	PV	PMT	FV
Given:	3		−10,000	6000	0
Solve for:		36.3			

Excel Formula: =RATE(NPER,PMT,PV,FV)=RATE(3,6000,−10000,0)

Thus, the IRR for Javier's investment in his girlfriend's business is 36.3%.

The timeline for his investment in the Internet café is:



The NPV of the investment opportunity is:

$$NPV = -10{,}000 + \frac{5000}{1.12} + \frac{5000}{1.12^2} + \frac{5000}{1.12^3} = \$2009$$

The $2009 NPV of the Internet café is lower than the $4411 NPV for his girlfriend's business, so Javier should join his girlfriend in business. Luckily, it appears that Javier does not need to choose between his checkbook and his relationship!

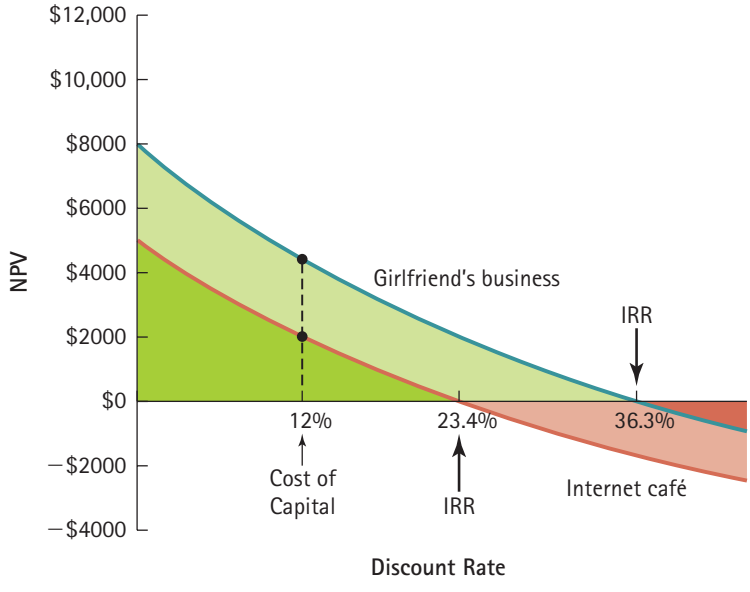
We could also compare IRRs. For the Internet café, we would find that the IRR is 23.4%. The Internet café has a lower IRR than the investment in his girlfriend's business. As Figure 8.7 shows, in this case the project with the higher IRR has the higher NPV.

Change in Scale. What happens if we change the scale of one of the projects? Javier's finance professor points out that, given the space available in the facility, he could just as

FIGURE 8.7

NPV of Javier's Investment Opportunities with the Two-Computer Internet Café

The NPV of his girlfriend's business is always larger than the NPV of the two-computer Internet café. The same is true for the IRR; the IRR of his girlfriend's business is 36.3%, while the IRR for the Internet café is 23.4%.



easily install five times as many computers in the Internet café. His setup cost would now be $50,000 and his annual cash flows would be $25,000. What should Javier do now?

Note that the IRR is unaffected by the scale. Because we are scaling all the cash flows

	N	I/Y	PV	PMT	FV
Given:	3		−50,000	25,000	0
Solve for:		23.4			

Excel Formula: =RATE(NPER,PMT,PV,FV)=RATE(3,25000,−50000,0)

up by a factor of 5, a ten-machine Internet café has exactly the same IRR as a two-machine Internet café, so his girlfriend's business still has a higher IRR than the Internet café:

However, the NPV of the Internet café does grow by the scale—it is five times larger:

$$NPV = -50{,}000 + \frac{25{,}000}{1.12} + \frac{25{,}000}{1.12^2} + \frac{25{,}000}{1.12^3} = \$10{,}046$$

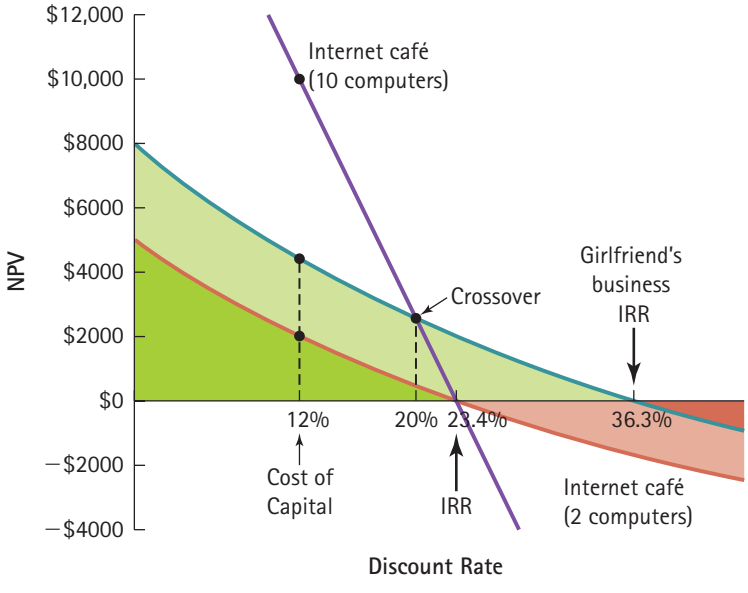
Now Javier should invest in the ten-computer Internet café. As Figure 8.8 shows, the NPV of the ten-computer Internet café exceeds the NPV of going into business with his girlfriend whenever the cost of capital is less than 20%. In this case, even though the IRR of going into business with his girlfriend exceeds the IRR of the Internet café, picking the investment opportunity with the higher IRR does not result in taking the opportunity with the higher NPV.

Percentage Return Versus Dollar Impact on Value. This result might seem counterintuitive and you can imagine Javier having a difficult time explaining to his girlfriend why he is choosing a lower return over going into business with her. Why would anyone turn

FIGURE 8.8

NPV of Javier's Investment Opportunities with the Ten-Computer Internet Café

As in Figure 8.7, the IRR of his girlfriend's business is 36.3%, while the IRR for the Internet café is 23.4%. But in this case, the NPV of his girlfriend's business is larger than the NPV of the ten-computer Internet café only for discount rates over 20%.



down an investment opportunity with a 36.3% return (IRR) in favor of one with only a 23.4% return? The answer is that the latter opportunity, the Internet café, makes more money. Recall the comparison at the beginning of this section: a 200% return on $1 versus a 10% return on $1 million. We agreed that ranking the returns was not the same as ranking the value created. The IRR is a measure of the average return, which can be valuable information. When you are comparing mutually exclusive projects of different scale, however, you need to know the dollar impact on value—the NPV.

INTERVIEW WITH DICK GRANNIS



Dick Grannis is Senior Vice President and Treasurer of QUALCOMM Incorporated, a world leader in digital wireless communications technology and semiconductors, headquartered in San Diego. He joined the company in 1991 and oversees the company's $10 billion cash investment portfolio. He works primarily on investment banking, capital structure, and international finance.

QUESTION: ***QUALCOMM has a wide variety of products in different business lines. How does your capital budgeting process for new products work?***

ANSWER: QUALCOMM evaluates new projects (such as new products, equipment, technologies, research and development, acquisitions, and strategic investments) by using traditional financial measurements including DCF models, IRR levels, peak funding requirements, the time needed to reach cumulative positive cash flows, and the short-term impact of the investment on our reported net earnings. For strategic investments, we consider the possible value of financial, competitive, technology and/or market value enhancements to our core businesses—even if those benefits cannot be quantified. Overall, we make capital budgeting decisions based on a combination of objective analyses and our own business judgment.

We do not engage in capital budgeting and analysis if the project represents an immediate and necessary requirement for our business operations. One example is new software or production equipment to start a project that has already received approval.

We are also mindful of the opportunity costs of allocating our internal engineering resources on one project vs. another project. We view this as a constantly challenging but worthwhile exercise, because we have many attractive opportunities but limited resources to pursue them.

QUESTION: ***How often does QUALCOMM evaluate its hurdle rates and what factors does it consider in setting them? How do you allocate capital across areas and regions and assess the risk of non-U.S. investments?***

ANSWER: QUALCOMM encourages its financial planners to utilize hurdle (or discount) rates that vary according to the risk of the particular project. We expect a rate of return commensurate with the project's risk. Our finance staff considers a wide range of discount rates and chooses one that fits the project's expected risk profile and time horizon. The range can be from 6.00% to 8.00% for relatively safe investments in the domestic market to 50% or more for equity investments in foreign markets that may be illiquid and difficult to predict. We re-evaluate our hurdle rates at least every year.

We analyze key factors including: (i) market adoption risk (whether or not customers will buy the new product or service at the price and volume we expect), (ii) technology development risk (whether or not we can develop and patent the new product or service as expected), (iii) execution risk (whether we can launch the new product or service cost effectively and on time), and (iv) dedicated asset risk (the amount of resources that must be consumed to complete the work).

QUESTION: ***How are projects categorized and how are the hurdle rates for new projects determined? What would happen if QUALCOMM simply evaluated all new projects against the same hurdle rate?***

ANSWER: We primarily categorize projects by risk level, but we also categorize projects by the expected time horizon. We consider short-term and long-term projects to balance our needs and achieve our objectives. For example, immediate projects and opportunities may demand a great amount of attention, but we also stay focused on long-term projects because they often create greater long-term value for stockholders.

If we were to evaluate all new projects against the same hurdle rate, then our business planners would, by default, consistently choose to invest in the highest risk projects because those projects would appear to have the greatest expected returns in DCF models or IRR analyses. That approach would probably not work well for very long.

EXAMPLE 8.4

Computing the Crossover Point

Problem

Solve for the crossover point for Javier from Figure 8.8.

Solution

Plan

The *crossover point* is the discount rate that makes the NPV of the two alternatives equal. We can find the discount rate by setting the equations for the NPV of each project equal to each other and solving for the discount rate. In general, we can always compute the effect of choosing the Internet café over his girlfriend's business as the difference of the NPVs. At the crossover point the difference is 0.

Execute

Setting the difference equal to 0:

$$NPV = -50{,}000 + \frac{25{,}000}{1+r} + \frac{25{,}000}{(1+r)^2} + \frac{25{,}000}{(1+r)^3} - \left(-10{,}000 + \frac{6{,}000}{(1+r)} + \frac{6{,}000}{(1+r)^2} + \frac{6{,}000}{(1+r)^3}\right) = 0$$

combining terms, we have

$$-40{,}000 + \frac{19{,}000}{(1+r)} + \frac{19{,}000}{(1+r)^2} + \frac{19{,}000}{(1+r)^3} = 0$$

As you can see, solving for the crossover point is just like solving for the IRR, so we will need to use a financial calculator or spreadsheet:

	N	I/Y	PV	PMT	FV
Given:	3		−40,000	19,000	0
Solve for:		20.04			

Excel Formula: =RATE(NPER,PMT,PV,FV)=RATE(3,19000,−40000,0)

And we find that the crossover occurs at a discount rate of 20.04%.

Evaluate

Just as the NPV of a project tells us the value impact of taking the project, so the difference of the NPVs of two alternatives tells us the *incremental* impact of choosing one project over another. The crossover point is the discount rate at which we would be indifferent between the two projects because the incremental value of choosing one over the other would be zero.

Timing of the Cash Flows

Even when projects have the same scale, the IRR may lead you to rank them incorrectly due to differences in the timing of the cash flows. The reason for this is that the IRR is expressed as a return, but the dollar value of earning a given return—and therefore the NPV—depends on how long the return is earned. Consider a high-IRR project with cash flows paid back quickly. It may have a lower NPV than a project with a lower IRR whose cash flows are paid back over a longer period. This sensitivity to timing is another reason why you cannot use the IRR to choose between mutually exclusive investments. To see this in the context of an example, let's return to Javier's Internet café.

Javier believes that after starting the Internet café, he may be able to sell his stake in the business at the end of the first year for $40,000 (he will continue to stay on and manage the business after he sells). Thus, counting his first-year profit of $25,000, he would earn a total of $65,000 after one year. In that case, the timeline is:

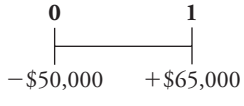


FIGURE 8.9
NPV With and Without Selling

The IRR from selling after one year (30%) is larger than the IRR without selling (23.4%). However, the NPV from selling after one year exceeds the NPV without selling only for discount rates that are in excess of 16.3% (see the yellow-shaded area vs. the blue-shaded area). Thus, given a cost of capital of 12%, it is better not to sell the Internet café after one year, despite the higher IRR.

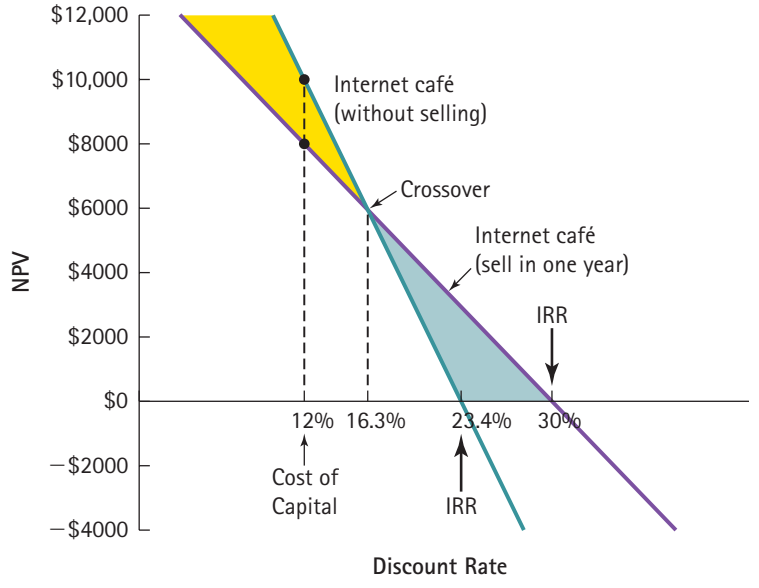


Figure 8.9 plots the NPV profile for the café with and without selling it after one year. If he sells, the NPV profile crosses the x-axis at 30%, which is its IRR. The IRR for the café if he does not sell is still 23.4%. Therefore, if Javier picks the alternative with the higher IRR, he will sell. However, since the height of each line indicates the NPV of that decision, we can see that his NPV given a 12% cost of capital is higher if he chooses not to sell. (In fact, the NPV is higher as long as the cost of capital is less than 16.3%.) The intuition is as follows: While the 30% IRR from selling is high, this return is only earned in the first year. While the 23.4% IRR from not selling is not as high, it is still attractive relative to the cost of capital, and it is earned over a longer period. Again, only by comparing the NPV can we determine which option is truly more valuable.

The Bottom Line on IRR. As these examples make clear, picking the investment opportunity with the largest IRR can lead to a mistake. In general, it is dangerous to use the IRR in cases where you are choosing between projects, or anytime when your decision to accept or reject one project would affect your decision on another project. In such a situation, always rely on NPV.

Concept Check

7. What is the most reliable way to choose between mutually exclusive projects?
8. For mutually exclusive projects, explain why picking one project over another because it has a larger IRR can lead to mistakes.

8.5 Evaluating Projects with Different Lives

Often, a company will need to choose between two solutions to the same problem. A complication arises when those solutions last for different periods of time. For example, a firm could be considering two vendors for its internal network servers. Each vendor offers the

5. Under what conditions will the IRR rule and the NPV rule give the same accept/reject decision?
6. When is it possible to have multiple IRRs?
7. How does the MIRR solve the problem of multiple IRRs?
8. Why is it generally a bad idea to use IRR to choose between mutually exclusive projects?
9. When should you use the equivalent annual annuity?
10. What is the intuition behind the profitability index?

Problems

*All problems in this chapter are available in MyFinanceLab. An asterisk * indicates problems with a higher level of difficulty. If a problem requires you to calculate IRR, you should use a spreadsheet or financial calculator.*

The NPV Decision Rule

1. You have an opportunity to invest $50,000 now in return for $60,000 in one year. If your cost of capital is 8%, what is the NPV of this investment?
2. You have an opportunity to invest $100,000 now in return for $80,000 in one year and $30,000 in two years. If your cost of capital is 9%, what is the NPV of this investment?
3. Your storage firm has been offered $100,000 in one year to store some goods for one year. Assume your costs are $95,000, payable immediately, and the cost of capital is 8%. Should you take the contract?
4. You run a construction firm. You have just won a contract to build a government office building. Building it will require an investment of $10 million today and $5 million in one year. The government will pay you $20 million in one year upon the building's completion. Assume the cost of capital is 10%.
   a. What is the NPV of this opportunity?
   b. How can your firm turn this NPV into cash today?
5. You have been offered a unique investment opportunity. If you invest $10,000 today, you will receive $500 one year from now, $1500 two years from now, and $10,000 ten years from now.
   a. What is the NPV of the opportunity if the cost of capital is 6% per year? Should you take the opportunity?
   b. What is the NPV of the opportunity if the cost of capital is 2% per year? Should you take it now?
6. Marian Plunket owns her own business and is considering an investment. If she undertakes the investment, it will pay $4000 at the end of each of the next three years. The opportunity requires an initial investment of $1000 plus an additional investment at the end of the second year of $5000. What is the NPV of this opportunity if the cost of capital is 2% per year? Should Marian take it?

Using the NPV Rule

7. Your factory has been offered a contract to produce a part for a new printer. The contract would last for three years and your cash flows from the contract would be $5 million per year. Your upfront setup costs to be ready to produce the part would

be $8 million. Your cost of capital for this contract is 8%.
a. What does the NPV rule say you should do?
b. If you take the contract, what will be the change in the value of your firm?

8. You are considering opening a new plant. The plant will cost $100 million upfront and will take one year to build. After that, it is expected to produce profits of $30 million at the end of every year of production. The cash flows are expected to last forever. Calculate the NPV of this investment opportunity if your cost of capital is 8%. Should you make the investment? Calculate the IRR and use it to determine the maximum deviation allowable in the cost of capital estimate to leave the decision unchanged.

9. Bill Clinton reportedly was paid $10 million to write his book *My Life*. The book took three years to write. In the time he spent writing, Clinton could have been paid to make speeches. Given his popularity, assume that he could earn $8 million per year (paid at the end of the year) speaking instead of writing. Assume his cost of capital is 10% per year.
a. What is the NPV of agreeing to write the book (ignoring any royalty payments)?
b. Assume that, once the book was finished, it was expected to generate royalties of $5 million in the first year (paid at the end of the year) and these royalties were expected to decrease at a rate of 30% per year in perpetuity. What is the NPV of the book with the royalty payments?



***10.** FastTrack Bikes, Inc., is thinking of developing a new composite road bike. Development will take six years and the cost is $200,000 per year. Once in production, the bike is expected to make $300,000 per year for ten years. The cash inflows begin at the end of year 7.

Assuming the cost of capital is 10%:
a. Calculate the NPV of this investment opportunity. Should the company make the investment?
b. Calculate the IRR and use it to determine the maximum deviation allowable in the cost of capital estimate to leave the decision unchanged.
c. How long must development last to change the decision?

Assuming the cost of capital is 14%:
a. Calculate the NPV of this investment opportunity. Should the company make the investment?
b. How much must this cost of capital estimate deviate to change the decision?
c. How long must development last to change the decision?



11. OpenSeas, Inc., is evaluating the purchase of a new cruise ship. The ship would cost $500 million, but would operate for 20 years. OpenSeas expects annual cash flows from operating the ship to be $70 million and its cost of capital is 12%.
a. Prepare an NPV profile of the purchase.
b. Identify the IRR on the graph.
c. Should OpenSeas go ahead with the purchase?
d. How far off could OpenSeas' cost of capital estimate be before your purchase decision would change?

Alternative Decision Rules

12. You are a real estate agent thinking of placing a sign advertising your services at a local bus stop. The sign will cost $5000 and will be posted for one year. You expect that it will generate additional revenue of $500 per month. What is the payback period?

13. Does the IRR rule agree with the NPV rule in Problem 7?

14. How many IRRs are there in part (a) of Problem 9? Does the IRR rule give the right answer in this case?

15. How many IRRs are there in part (b) of Problem 9? Does the IRR rule work in this case?

16. Professor Wendy Smith has been offered the following deal: A law firm would like to retain her for an upfront payment of $50,000. In return, for the next year the firm would have access to eight hours of her time every month. Smith's rate is $550 per hour and her opportunity cost of capital is 15% per year. What does the IRR rule advise regarding this opportunity? What about the NPV rule?

17. Innovation Company is thinking about marketing a new software product. Upfront costs to market and develop the product are $5 million. The product is expected to generate profits of $1 million per year for ten years. The company will have to provide product support expected to cost $100,000 per year in perpetuity. Assume all profits and expenses occur at the end of the year.

a. What is the NPV of this investment if the cost of capital is 6%? Should the firm undertake the project? Repeat the analysis for discount rates of 2% and 11%.
b. How many IRRs does this investment opportunity have?
c. What does the IRR rule indicate about this investment?

18. You own a coal mining company and are considering opening a new mine. The mine itself will cost $120 million to open. If this money is spent immediately, the mine will generate $20 million for the next ten years. After that, the coal will run out and the site must be cleaned and maintained at environmental standards. The cleaning and maintenance are expected to cost $2 million per year in perpetuity. What does the IRR rule say about whether you should accept this opportunity? If the cost of capital is 8%, what does the NPV rule say?

19. Your firm is considering a project that will cost $4.55 million upfront, generate cash flows of $3.5 million per year for three years, and then have a cleanup and shutdown cost of $6 million in the fourth year.

a. How many IRRs does this project have?
b. Calculate a modified IRR for this project discounting the outflows and leaving the inflows unchanged. Assume a discount and compounding rate of 10%.
c. Using the MIRR and a cost of capital of 10%, would you take the project?

20. You have just been offered a contract worth $1 million per year for five years. However, to take the contract, you will need to purchase some new equipment. Your discount rate for this project is 12%. You are still negotiating the purchase price of the equipment. What is the most you can pay for the equipment and still have a positive NPV?

***21.** You are getting ready to start a new project that will incur some cleanup and shutdown costs when it is completed. The project costs $5.4 million upfront and is expected to generate $1.1 million per year for ten years and then have some shutdown costs in year 11. Use the MIRR approach to find the maximum shutdown costs you could incur and still meet your cost of capital of 15% on this project.

***22.** You are considering investing in a new gold mine in South Africa. Gold in South Africa is buried very deep, so the mine will require an initial investment of $250 million. Once this investment is made, the mine is expected to produce revenues of $30 million per year for the next 20 years. It will cost $10 million per year to operate

the mine. After 20 years, the gold will be depleted. The mine must then be stabilized on an ongoing basis, which will cost $5 million per year in perpetuity. Calculate the IRR of this investment. (*Hint*: Plot the NPV as a function of the discount rate.)

23. You are considering making a movie. The movie is expected to cost $10 million upfront and take a year to make. After that, it is expected to make $5 million in the year it is released and $2 million for the following four years. What is the payback period of this investment? If you require a payback period of two years, will you make the movie? Does the movie have positive NPV if the cost of capital is 10%?

Choosing Between Projects



24. You are choosing between two projects, but can only take one. The cash flows for the projects are given in the following table:

	0	1	2	3	4
A	−$50	25	20	20	15
B	−$100	20	40	50	60

a. What are the IRRs of the two projects?
b. If your discount rate is 5%, what are the NPVs of the two projects?
c. Why do IRR and NPV rank the two projects differently?

25. You are deciding between two mutually exclusive investment opportunities. Both require the same initial investment of $10 million. Investment A will generate $2 million per year (starting at the end of the first year) in perpetuity. Investment B will generate $1.5 million at the end of the first year and its revenues will grow at 2% per year for every year after that.
a. Which investment has the higher IRR?
b. Which investment has the higher NPV when the cost of capital is 7%?
c. In this case, when does picking the higher IRR give the correct answer as to which investment is the better opportunity?



26. You are considering the following two projects and can only take one. Your cost of capital is 11%.

	0	1	2	3	4
A	−100	25	30	40	50
B	−100	50	40	30	20

a. What is the NPV of each project at your cost of capital?
b. What is the IRR of each project?
c. At what cost of capital are you indifferent between the two projects?
d. What should you do?

27. You need a particular piece of equipment for your production process. An equipment-leasing company has offered to lease you the equipment for $10,000 per year if you sign a guaranteed five-year lease. The company would also maintain the equipment for you as part of the lease. Alternatively, you could buy and maintain the equipment yourself. The cash flows (in thousands) from doing so are listed below (the equipment has an economic life of five years). If your discount rate is 7%, what should you do?

0	1	2	3	4	5
−40	−2	−2	−2	−2	−2

incremental earnings The amount by which a firm's earnings are expected to change as a result of an investment decision.

will *begin* our analysis in Section 9.2 by determining the **incremental earnings** of a project—that is, the amount by which the firm's earnings are expected to change as a result of the investment decision. The incremental earnings forecast tells us how the decision will affect the firm's reported profits from an accounting perspective. However, as we emphasized in Chapter 2, *earnings are not actual cash flows*. We need to estimate the project's cash flows to determine its NPV and decide whether it is a good project for the firm. Therefore, in Section 9.3, we demonstrate how to use the incremental earnings to forecast the actual cash flows of the project. Understanding how to compute the cash flow consequences of an investment based on its earning consequences is important for a number of reasons. First, as a practical matter, financial managers often begin by forecasting earnings. Second, if we are looking at historical data, accounting information is often the only information that is readily available.

Concept Check

1. What is capital budgeting, and what is its goal?
2. Why is computing a project's effect on the firm's earnings insufficient for capital budgeting?

9.2 Forecasting Incremental Earnings

Let's begin our discussion of incremental earnings with a simple example that we will examine throughout this section. Suppose you are considering whether to upgrade your manufacturing plant to increase its capacity by purchasing a new piece of equipment. The equipment costs $1 million, plus an additional $20,000 to transport it and install it. You will also spend $50,000 on engineering costs to redesign the plant to accommodate the increased capacity. What are the initial earnings consequences of this decision?

Operating Expenses Versus Capital Expenditures

Most projects require some form of upfront investment—we may need to conduct a marketing survey, develop a prototype, or launch an ad campaign. These types of costs are accounted for as operating expenses in the year that they are incurred. However, many projects also include investments in plant, property, and/or equipment, called *capital expenditures*. Recall from Chapter 2 that while investments in plant, property, and equipment are a cash expense, they are not directly listed as expenses when calculating *earnings*. Instead, the firm deducts a fraction of the cost of these items each year as depreciation. Financial managers use several different methods to compute depreciation. The simplest method is **straight-line depreciation**, in which the asset's cost is divided equally over its depreciable life (we discuss another common method in Section 9.4).

straight-line depreciation A method of depreciation in which an asset's cost is divided equally over its life.

In our example, the upfront costs associated with the decision to increase capacity have two distinct consequences for the firm's earnings. First, the $50,000 spent on redesigning the plant is an operating expense reported in year 0. For the $1,020,000 spent to buy, ship, and install the machine, accounting principles as well as tax rules require you to depreciate the $1,020,000 over the depreciable life of the equipment. Assuming that the equipment has a five-year depreciable life and that we use the straight-line method, we would expense $1,020,000/5 = $204,000 per year for five years. (The motivation for this accounting treatment is to match the cost of acquiring the machine to the timing of the revenues it will generate.)

1	Year	0	1	2	3	4	5
2	Operating Expenses (Plant Redesign)	−$50,000					
3	Depreciation (New Equipment)		−$204,000	−$204,000	−$204,000	−$204,000	−$204,000

As the timeline shows, the upfront cash outflow of $1,020,000 to purchase and set up the machine is not recognized as an expense in year 0. Instead, it appears as depreciation expenses in years 1 through 5. Remember that these *depreciation expenses do not correspond to actual cash outflows.* This accounting and tax treatment of capital expenditures is one of the key reasons why earnings are not an accurate representation of cash flows. We will return to this issue in Section 9.3.[1]

Incremental Revenue and Cost Estimates

Our next step is to estimate the ongoing revenues and costs for the project. Forecasting future revenues and costs is challenging. The most successful practitioners collect as much information as possible before tackling this task—they will talk to members of marketing and sales teams as well as company economists to develop an estimate of sales, and they will talk to engineering and production teams to refine their estimate of costs.

There are several factors to consider when estimating a project's revenues and costs, including the following:

1. A new product typically has lower sales initially, as customers gradually become aware of the product. Sales will then accelerate, plateau, and ultimately decline as the product nears obsolescence or faces increased competition.
2. The average selling price of a product and its cost of production will generally change over time. Prices and costs tend to rise with the general level of inflation in the economy. The prices of technology products, however, often fall over time as newer, superior technologies emerge and production costs decline.
3. For most industries, competition tends to reduce profit margins over time.

Our focus here is on how to get from these forecasts to incremental earnings and then to cash flows; Chapter 18 discusses forecasting methods in more detail.

All our revenue and cost estimates should be *incremental,* meaning that we only account for additional sales and costs generated by the project. For example, if we are evaluating the purchase of a faster manufacturing machine, we are only concerned with how many additional units of the product we will be able to sell (and at what price) and any additional costs created by the new machine. We do not forecast total sales and costs because those include our production using the old machine. *Remember, we are evaluating how the project will change the cash flows of the firm. That is why we focus on incremental revenues and costs.*

Let's return to our plant upgrade example. Assume that after we have bought and installed the machine and redesigned the plant, our additional capacity will allow us to generate incremental revenues of $500,000 per year for five years. Those incremental revenues will be associated with $150,000 per year in incremental costs. In that case our revenue, cost, and depreciation estimates for the project are as shown below (in thousands of dollars):

	Year	0	1	2	3	4	5
2	Incremental Revenues		500	500	500	500	500
3	Incremental Costs	−50	−150	−150	−150	−150	−150
4	Depreciation		−204	−204	−204	−204	−204

[1]Some students might note that for financial reporting, the asset should be depreciated to salvage value. For tax purposes, salvage value is always treated as zero, so this is the relevant method for capital budgeting.

Now that we have these estimates, we are ready to compute the consequences of our project for the firm's earnings. As we saw in Chapter 2, both depreciation expenses and the actual costs of producing (e.g., cost of goods sold) must be subtracted from revenues, so that:

$$\text{Incremental Earnings Before Interest and Taxes (EBIT)} = \text{Incremental Revenue} - \text{Incremental Costs} - \text{Depreciation} \quad (9.1)$$

Taxes

marginal corporate tax rate The tax rate a firm will pay on an incremental dollar of pre-tax income.

The final expense we must account for is corporate taxes. The correct tax rate to use is the firm's **marginal corporate tax rate**, which is the tax rate it will pay on an *incremental* dollar of pre-tax income. The incremental income tax expense is calculated as:

$$\text{Income Tax} = \text{EBIT} \times \text{The Firm's Marginal Corporate Tax Rate} \quad (9.2)$$

Incremental Earnings Forecast

We're now ready to put the pieces together for an incremental earnings forecast. Assume our firm faces a marginal tax rate of 40% and that the firm as a whole has at least $50,000 in profits in year 0 for the incremental costs in that year to offset. Then the incremental earnings (or net income) are as follows (in thousands of dollars):[2]

	Year	0	1	2	3	4	5
2	Incremental Revenues		500	500	500	500	500
3	Incremental Costs	−50	−150	−150	−150	−150	−150
4	Depreciation		−204	−204	−204	−204	−204
5	**EBIT**	−50	146	146	146	146	146
6	Income Tax at 40%	20	−58.4	−58.4	−58.4	−58.4	−58.4
7	**Incremental Earnings**	**−30**	**87.6**	**87.6**	**87.6**	**87.6**	**87.6**

We can also combine Eq. 9.1 and Eq. 9.2 to compute incremental earnings directly. For example, in years 1 through 5 we have:

$$\begin{aligned}\text{Incremental Earnings} &= (\text{Incremental Revenues} - \text{Incremental Cost} \\ &\quad - \text{Depreciation}) \times (1 - \text{Tax Rate}) \quad (9.3)\\ \text{Incremental Earnings} &= (500{,}000 - 150{,}000 - 204{,}000) \\ &\quad \times (1 - 0.4) = 87{,}600\end{aligned}$$

EXAMPLE 9.1

Incremental Earnings

Problem

Suppose that Linksys is considering the development of a wireless home networking appliance, called HomeNet, that will provide both the hardware and the software necessary to run an entire home from any Internet connection. In addition to connecting PCs and printers, HomeNet will control new Internet-capable stereos, digital video recorders, heating and air-conditioning units, major appliances, telephone and security systems, office equipment, and so on. The major competitor for HomeNet is a product being developed by Brandt-Quigley Corporation.

Based on extensive marketing surveys, the sales forecast for HomeNet is 50,000 units per year. Given the pace of technological change, Linksys expects the product will have a four-year life and an expected

[2]While revenues and costs occur throughout the year, the standard convention, which we adopt here, is to list revenues and costs in the year in which they occur. Thus, cash flows that occur at the end of one year will be listed in a different column than those that occur at the start of the next year, even though they may occur only weeks apart. When additional precision is required, cash flows are often estimated on a quarterly or monthly basis.

wholesale price of $260 (the price Linksys will receive from stores). Actual production will be outsourced at a cost (including packaging) of $110 per unit.

To verify the compatibility of new consumer Internet-ready appliances, as they become available, with the HomeNet system, Linksys must also establish a new lab for testing purposes. It will rent the lab space, but will need to purchase $7.5 million of new equipment. The equipment will be depreciated using the straight-line method over a five-year life. Linksys' marginal tax rate is 40%.

The lab will be operational at the end of one year. At that time, HomeNet will be ready to ship. Linksys expects to spend $2.8 million per year on rental costs for the lab space, as well as marketing and support for this product. Forecast the incremental earnings from the HomeNet project.

Solution

Plan

We need four items to calculate incremental earnings: (1) incremental revenues, (2) incremental costs, (3) depreciation, and (4) the marginal tax rate:

Incremental revenues are: Additional units sold $\times$ price $=$ 50,000 $\times$ $260 $=$ $13,000,000

Incremental costs are: Additional units sold $\times$ production costs
$=$ 50,000 $\times$ $110 $=$ $5,500,000
Selling, general, and administrative $=$ $2,800,000 for rent, marketing, and support

Depreciation is: Depreciable basis/Depreciable life $=$ $7,500,000/5 $=$ $1,500,000

Marginal tax rate: 40%

Note that even though the project lasts for four years, the equipment has a five-year life, so we must account for the final depreciation charge in the fifth year.

Execute (in $000s)

	Year	0	1	2	3	4	5
2	Revenues		13,000	13,000	13,000	13,000	–
3	Cost of Goods Sold		−5,500	−5,500	−5,500	−5,500	–
4	**Gross Profit**		7,500	7,500	7,500	7,500	–
5	Selling, General, and Administrative		−2,800	−2,800	−2,800	−2,800	–
6	Depreciation		−1,500	−1,500	−1,500	−1,500	−1,500
7	**EBIT**		3,200	3,200	3,200	3,200	−1,500
8	Income Tax at 40%		−1,280	−1,280	−1,280	−1,280	600
9	**Incremental Earnings**		**1,920**	**1,920**	**1,920**	**1,920**	**−900**

Evaluate

These incremental earnings are an intermediate step on the way to calculating the incremental cash flows that would form the basis of any analysis of the HomeNet project. The cost of the equipment does not affect earnings in the year it is purchased, but does so through the depreciation expense in the following five years. Note that the depreciable life, which is based on accounting rules, does not have to be the same as the economic life of the asset—the period over which it will have value. Here, the firm will use the equipment for four years, but will depreciate it over five years.

pro forma Describes a statement that is not based on actual data but rather depicts a firm's financials under a given set of hypothetical assumptions.

Pro Forma Statement. The table calculating incremental earnings that we produced for our plant upgrade, and again in Example 9.1, is often referred to as a **pro forma** statement, because it is not based on actual data but rather depicts the firm's financials under a given set of hypothetical assumptions. In the HomeNet example, the firm's forecasts of revenues and costs were assumptions that allowed Linksys to forecast incremental earnings in a pro forma statement.

Taxes and Negative EBIT. Notice that in year 0 of our plant upgrade project, and in year 5 of the HomeNet example, EBIT is negative. Why are taxes relevant in this case? Consider the HomeNet example. HomeNet will reduce Linksys's taxable income in year 5 by $1.5

million. As long as Linksys earns taxable income elsewhere in year 5 against which it can offset HomeNet's losses, Linksys will owe \$1.5 million × 40% = \$600,000 *less* in taxes in year 5 than if it were not undertaking the project. Because the tax savings come from the depreciation expense on equipment for the HomeNet project, the firm should credit this tax savings to the HomeNet project.

EXAMPLE 9.2
Taxing Losses for Projects in Profitable Companies

Problem

Kellogg Company plans to launch a new line of high-fiber, zero-trans-fat breakfast pastries. The heavy advertising expenses associated with the new product launch will generate operating losses of \$15 million next year for the product. Kellogg expects to earn pre-tax income of \$460 million from operations other than the new pastries next year. If Kellogg pays a 40% tax rate on its pre-tax income, what will it owe in taxes next year without the new pastry product? What will it owe with the new product?

Solution

Plan

We need Kellogg's pre-tax income with and without the new product losses and its tax rate of 40%. We can then compute the tax without the losses and compare it to the tax with the losses.

Execute

Without the new product, Kellogg will owe \$460 million × 40% = \$184 million in corporate taxes next year. With the new product, Kellogg's pre-tax income next year will be only \$460 million − \$15 million = \$445 million, and it will owe \$445 million × 40% = \$178 million in tax.

Evaluate

Thus, launching the new product reduces Kellogg's taxes next year by \$184 million − \$178 million = \$6 million. Because the losses on the new product reduce Kellogg's taxable income dollar for dollar, it is the same as if the new product had a tax bill of *negative* \$6 million.

What About Interest Expenses? In Chapter 2, we saw that to compute a firm's net income, we must first deduct interest expenses from EBIT. When evaluating a capital budgeting decision, however, we generally *do not include interest expenses*. Any incremental interest expenses will be related to the firm's decision regarding how to finance the project, which is a separate decision. Here, we wish to evaluate the earnings contributions from the project on its own, separate from the financing decision. Ultimately, managers may also look at the additional earnings consequences associated with different methods of financing the project.

Thus, we evaluate a project *as if* the company will not use any debt to finance it (whether or not that is actually the case), and we postpone the consideration of alternative financing choices until Parts 5 and 6 of this book. Because we calculate the net income assuming no debt (no leverage), we refer to the net income we compute using Eq. 9.3, as in the pro forma in Example 9.1, as the **unlevered net income** of the project, to indicate that it does not include any interest expenses associated with debt.

unlevered net income
Net income that does not include interest expenses associated with debt.

Concept Check

3. How are operating expenses and capital expenditures treated differently when calculating incremental earnings?
4. Why do we focus only on *incremental* revenues and costs, rather than all revenues and costs of the firm?

9.3 Determining Incremental Free Cash Flow

As discussed in Chapter 2, earnings are an accounting measure of the firm's performance. They do not represent real profits: The firm cannot use its earnings to buy goods, pay employees, fund new investments, or pay dividends to shareholders. To do those things,

free cash flow The incremental effect of a project on a firm's available cash.

the firm needs cash. Thus, to evaluate a capital budgeting decision, we must determine its consequences for the firm's available cash. The incremental effect of a project on the firm's available cash is the project's incremental **free cash flow**.

Converting from Earnings to Free Cash Flow

As discussed in Chapter 2, there are important differences between earnings and cash flow. Earnings include non-cash charges, such as depreciation, but do not include expenditures on capital investment. To determine a project's free cash flow from its incremental earnings, we must adjust for these differences.

Capital Expenditures and Depreciation. As we have noted, depreciation is not a cash expense that is paid by the firm. Rather, it is a method used for accounting and tax purposes to allocate the original purchase cost of the asset over its life. Because depreciation is not a cash flow, we do not include it in the cash flow forecast. However, that does not mean we can ignore depreciation. The depreciation expense reduces our taxable earnings and in doing so reduces our taxes. Taxes are cash flows, so because depreciation affects our cash flows, it still matters. Our approach for handling depreciation is to add it back to the incremental earnings to recognize the fact that we still have the cash flow associated with it.

For example, a project has incremental gross profit (revenues minus costs) of $1 million and a $200,000 depreciation expense. If the firm's tax rate is 40%, then the incremental earnings will be $(\$1{,}000{,}000 - \$200{,}000) \times (1 - 0.40) = \$480{,}000$. However, the firm will still have $680,000 because the $200,000 depreciation expense is not an actual cash outflow. Table 9.1 shows the calculation to get the incremental free cash flow in this case. Blue boxes indicate all of the actual cash flows in the column labeled "Correct." A good way to check to make sure the incremental free cash flow is correct is to sum the actual cash flows. In this case, the firm generated $1,000,000 in gross profit (a positive cash flow), paid $320,000 in taxes (a negative cash flow), and was left with $\$1{,}000{,}000 - \$320{,}000 = \$680{,}000$, which is the amount shown as the incremental free cash flow. In the last column, labeled "Incorrect," we show what would happen if you just ignored depreciation altogether. Because EBIT would be too high, the taxes would be too high as well and consequently, the incremental free cash flow would be too low. (Note that the difference of $80,000 between the two cases is entirely due to the difference in tax payments.)

TABLE 9.1
Deducting and Then Adding Back Depreciation

	Correct	Incorrect
Incremental Gross Profit	$1,000,000	$1,000,000
Depreciation	−200,000	
EBIT	$800,000	$1,000,000
Tax at 40%	−$320,000	−$400,000
Incremental Earnings	$480,000	$600,000
Add Back depreciation	$200,000	
Incremental Free Cash Flow	$680,000	$600,000

EXAMPLE 9.3
Incremental Free Cash Flows

Problem

Let's return to the HomeNet example. In Example 9.1, we computed the incremental earnings for HomeNet, but we need the incremental free cash flows to decide whether Linksys should proceed with the project.

Solution

Plan

The difference between the incremental earnings and incremental free cash flows in the HomeNet example will be driven by the equipment purchased for the lab. We need to recognize the $7.5 million cash outflow associated with the purchase in year 0 and add back the $1.5 million depreciation expenses from year 1 to 5 as they are not actually cash outflows.

Execute (in $000s)

We recognize the outflow for the equipment in row 11 and we add back the depreciation expenses in row 10.

1	Year	0	1	2	3	4	5
2	Revenues		13,000	13,000	13,000	13,000	–
3	Cost of Goods Sold		−5,500	−5,500	−5,500	−5,500	–
4	**Gross Profit**		7,500	7,500	7,500	7,500	–
5	Selling, General, and Administrative		−2,800	−2,800	−2,800	−2,800	–
6	Depreciation		−1,500	−1,500	−1,500	−1,500	−1,500
7	**EBIT**		3,200	3,200	3,200	3,200	−1,500
8	Income Tax at 40%		−1,280	−1,280	−1,280	−1,280	600
9	**Incremental Earnings**		1,920	1,920	1,920	1,920	−900
10	Add Back Depreciation		1,500	1,500	1,500	1,500	1,500
11	Purchase of Equipment	−7,500					
12	**Incremental Free Cash Flows**	**−7,500**	**3,420**	**3,420**	**3,420**	**3,420**	**600**

Evaluate

By recognizing the outflow from purchasing the equipment in year 0, we account for the fact that $7.5 million left the firm at that time. By adding back the $1.5 million depreciation expenses in years 1–5, we adjust the incremental earnings to reflect the fact that the depreciation expense is not a cash outflow.

Net Working Capital (NWC). Another way that incremental earnings and free cash flows can differ is if there are changes in net working capital. We defined net working capital in Chapter 2 as the difference between current assets and current liabilities. The main components of net working capital are cash, inventory, receivables, and payables:

$$\begin{aligned}\text{Net Working Capital} &= \text{Current Assets} - \text{Current Liabilities} \\ &= \text{Cash} + \text{Inventory} + \text{Receivables} - \text{Payables}\end{aligned} \tag{9.4}$$

Note that as discussed in Chapter 2, we do not include short-term financing such as notes payable or short-term debt because those represent financing decisions that we keep separate from our investment decisions. Most projects will require the firm to invest in net working capital. Firms may need to maintain a minimum cash balance[3] to meet unexpected expenditures, and inventories of raw materials and finished product to accommodate production uncertainties and demand fluctuations. Also, customers may not pay for the goods they purchase immediately. While sales are immediately counted as part of earnings, the firm does not receive any cash until the customers actually pay. In the interim, the firm includes the amount that customers owe in its receivables. Thus, the firm's receivables measure the total credit that the firm has extended to its customers. In the same way, payables measure the credit the firm has received from its suppliers. The difference between receivables and payables is the net amount of the firm's capital that is consumed as a result of these credit transactions, known as **trade credit**.

trade credit The difference between receivables and payables that is the net amount of a firm's capital consumed as a result of those credit transactions; the credit that a firm extends to its customers.

We care about net working capital because it reflects a short-term investment that ties up cash flow that could be used elsewhere. For example, when a firm holds a lot of

[3]The cash included in net working capital is cash that is *not* invested to earn a market rate of return. It includes cash held in the firm's checking account, in a company safe or cash box, in cash registers (for retail stores), and other sites.

unsold inventory or has a lot of outstanding receivables, cash flow is tied up in the form of inventory or in the form of credit extended to customers. It is costly for the firm to tie up that cash flow because it delays the time until the cash flow is available for reinvestment or distribution to shareholders. Since we know that money has time value, we cannot ignore this delay in our forecasts for the project. Thus, whenever net working capital increases, reflecting additional investment in working capital, it represents a reduction in cash flow that year.

It is important to note that only changes in net working capital impact cash flows. For example, consider a three-year project that causes the firm to build up initial inventory by $20,000 and maintain that level of inventory in years 1 and 2, before drawing it down as the project ends and the last product is sold. It is often necessary for the initial increase in inventory to occur prior to the first sale so that the higher level of inventory would be achieved by the end of year 0. The level of the incremental net working capital in each year, the associated change in net working capital and the cash flow implications, would be:

	Year	0	1	2	3
2	Level of Incremental NWC	20,000	20,000	20,000	0
3	*Change* in Incremental NWC	+20,000	0	0	−20,000
4	**Cash Flow from Change in NWC**	**−20,000**	**0**	**0**	**+20,000**

Note that the cash flow effect from a change in net working capital is always equal and opposite in sign to the change in net working capital. For example, an increase in inventory represents an investment or cash outflow, while a reduction in that inventory frees up that investment of capital and represents a cash inflow. Thus in capital budgeting we subtract changes in net working capital to arrive at the cash flows. Also notice that since the level of incremental net working capital did not change in years 1 and 2, there was no new cash flow effect. Intuitively, as the firm is using up inventory and replenishing it, the net new investment in inventory is zero, so no additional cash outflow is required. Finally, note that over the life of the project, the incremental net working capital returns to zero so that the changes (+20,000 in year 0 and −20,000 in year 3) sum to zero. Accounting principles ensure this by requiring the recapture of working capital over the life of the project.

More generally, we define the change in net working capital in year t as:

$$\text{Change in } NWC \text{ in Year } t = NWC_t - NWC_{t-1} \tag{9.5}$$

When a project causes a change in NWC, that change must be subtracted from incremental earnings to arrive at incremental free cash flows.

EXAMPLE 9.4

Incorporating Changes in Net Working Capital

Problem

Suppose that HomeNet will have no incremental cash or inventory requirements (products will be shipped directly from the contract manufacturer to customers). However, receivables related to HomeNet are expected to account for 15% of annual sales, and payables are expected to be 15% of the annual cost of goods sold (COGS). Fifteen percent of $13 million in sales is $1.95 million and 15% of $5.5 million in COGS is $825,000. HomeNet's net working capital requirements are shown in the following table:

	Year	0	1	2	3	4	5
2	**Net Working Capital Forecast ($000s)**						
3	Cash Requirements	0	0	0	0	0	0
4	Inventory	0	0	0	0	0	0
5	Receivables (15% of Sales)	0	1,950	1,950	1,950	1,950	0
6	Payables (15% of COGS)	0	−825	−825	−825	−825	0
7	**Net Working Capital**	**0**	**1,125**	**1,125**	**1,125**	**1,125**	**0**

How does this requirement affect the project's free cash flow?

Solution

Plan

Any increases in net working capital represent an investment that reduces the cash available to the firm and so reduces free cash flow. We can use our forecast of HomeNet's net working capital requirements to complete our estimate of HomeNet's free cash flow. In year 1, net working capital increases by $1.125 million. This increase represents a cost to the firm. This reduction of free cash flow corresponds to the fact that in year 1, $1.950 million of the firm's sales and $0.825 million of its costs have not yet been paid.

In years 2–4, net working capital does not change, so no further contributions are needed. In year 5, when the project is shut down, net working capital falls by $1.125 million as the payments of the last customers are received and the final bills are paid. We add this $1.125 million to free cash flow in year 5.

Execute (in $000s)

1	Year	0	1	2	3	4	5
2	Net Working Capital	0	1,125	1,125	1,125	1,125	0
3	Change in NWC		+1,125	0	0	0	−1,125
4	Cash Flow Effect		−1,125	0	0	0	+1,125

The incremental free cash flows would then be:

1	Year	0	1	2	3	4	5
2	Revenues		13,000	13,000	13,000	13,000	0
3	Costs of Goods Sold		−5,500	−5,500	−5,500	−5,500	0
4	**Gross Profit**		7,500	7,500	7,500	7,500	0
5	Selling, General, and Administrative		−2,800	−2,800	−2,800	−2,800	0
6	Depreciation		−1,500	−1,500	−1,500	−1,500	−1,500
7	**EBIT**		3,200	3,200	3,200	3,200	−1,500
8	Income Tax at 40%		−1,280	−1,280	−1,280	−1,280	600
9	**Incremental Earnings**		1,920	1,920	1,920	1,920	−900
10	Add Back Depreciation		1,500	1,500	1,500	1,500	1,500
11	Purchase of Equipment	−7,500					
12	Subtract Changes in NWC		−1,125	0	0	0	1,125
13	**Incremental Free Cash Flows**	**−7,500**	**2,295**	**3,420**	**3,420**	**3,420**	**1,725**

Evaluate

The free cash flows differ from unlevered net income by reflecting the cash flow effects of capital expenditures on equipment, depreciation, and changes in net working capital. Note that in the first year, free cash flow is lower than unlevered net income, reflecting the upfront investment in equipment. In later years, free cash flow exceeds unlevered net income because depreciation is not a cash expense. In the last year, the firm ultimately recovers the investment in net working capital, further boosting the free cash flow.

Calculating Free Cash Flow Directly

As we noted at the outset of this chapter, because practitioners usually begin the capital budgeting process by first forecasting earnings, we have chosen to do the same. However, we can calculate a project's free cash flow directly by using the following shorthand formula:

$$\text{Free Cash Flow} = \overbrace{(\text{Revenues} - \text{Costs} - \text{Depreciation}) \times (1 - \text{Tax Rate})}^{\textbf{Free Cash Flow}\ \text{Unlevered Net Income}} + \text{Depreciation} - \text{CapEx} - \text{Change in } NWC \tag{9.6}$$

Note that we first deduct depreciation when computing the project's incremental earnings and then add it back (because it is a non-cash expense) when computing free cash

flow. Thus, the only effect of depreciation is to reduce the firm's taxable income. Indeed, we can rewrite Eq. 9.6 as:

$$\text{Free Cash Flow} = (\text{Revenues} - \text{Costs}) \times (1 - \text{Tax Rate}) - \text{CapEx} - \text{Change in } NWC + \text{Tax Rate} \times \text{Depreciation} \quad (9.7)$$

depreciation tax shield The tax savings that result from the ability to deduct depreciation.

The last term in Eq. 9.7, Tax Rate × Depreciation, is called the **depreciation tax shield,** which is the tax savings that results from the ability to deduct depreciation. As a consequence, depreciation expenses have a *positive* impact on free cash flow. Returning to our example in Table 9.1, if the firm ignored depreciation, its taxes were $400,000 instead of $320,000, leaving it with incremental free cash flow of $600,000 instead of $680,000. Notice that the $80,000 difference is exactly equal to the tax rate (40%) multiplied by the depreciation expense ($200,000). Every dollar of depreciation expense saves the firm 40 cents in taxes, so the $200,000 depreciation expense translates into an $80,000 tax savings.

Firms often report a different depreciation expense for accounting and for tax purposes. Because only the tax consequences of depreciation are relevant for free cash flow, we should use the depreciation expense that the firm will use for tax purposes in our forecast. For tax purposes, many firms use a system called *Modified Accelerated Cost Recovery System,* which we discuss in the next section.

Calculating the NPV

The goal of forecasting the incremental free cash flows is to have the necessary inputs to calculate the project's NPV. To compute a project's NPV, we must discount its free cash flow at the appropriate cost of capital. As discussed in Chapter 5, the cost of capital for a project is the expected return that investors could earn on their best alternative investment with similar risk and maturity. We will develop the techniques needed to estimate the cost of capital in Part 4 of the text, when we discuss risk and return. For now, we take the cost of capital as given.

We compute the present value of each free cash flow in the future by discounting it at the project's cost of capital. As explained in Chapter 3, using r to represent the cost of capital, the present value of the free cash flow in year t (or FCF_t) is:

$$PV(FCF_t) = \frac{FCF_t}{(1+r)^t} = FCF_t \times \underbrace{\frac{1}{(1+r)^t}}_{t\text{-year discount factor}} \quad (9.8)$$

EXAMPLE 9.5

Calculating the Project's NPV

Problem

Assume that Linksys's managers believe that the HomeNet project has risks similar to its existing projects, for which it has a cost of capital of 12%. Compute the NPV of the HomeNet project.

Solution

Plan

From Example 9.4, the incremental free cash flows for the HomeNet project are (in $000s):

	Year	0	1	2	3	4	5
2	Incremental Free Cash Flows	−7,500	2,295	3,420	3,420	3,420	1,725

To compute the NPV, we sum the present values of all of the cash flows, noting that the year 0 cash outflow is already a present value.

Execute

Using Eq. 9.8,

$$NPV = -7500 + \frac{2295}{(1.12)^1} + \frac{3420}{(1.12)^2} + \frac{3420}{(1.12)^3} + \frac{3420}{(1.12)^4} + \frac{1725}{(1.12)^5} = 2862$$

Evaluate

Based on our estimates, HomeNet's NPV is $2.862 million. While HomeNet's upfront cost is $7.5 million, the present value of the additional free cash flow that Linksys will receive from the project is $10.362 million. Thus, taking the HomeNet project is equivalent to Linksys having an extra $2.862 million in the bank today.

Concept Check

5. If depreciation expense is not a cash flow, why do we have to subtract it and add it back? Why not just ignore it?
6. Why does an increase in net working capital represent a cash outflow?

9.4 Other Effects on Incremental Free Cash Flows

When computing the incremental free cash flows of an investment decision, we should include *all* changes between the firm's free cash flows with the project versus without the project. These include opportunities forgone due to the project and effects of the project on other parts of the firm. In this section, we discuss these other effects, some of the pitfalls and common mistakes to avoid, and the complications that can arise when forecasting incremental free cash flows.

Opportunity Costs

opportunity cost The value a resource could have provided in its best alternative use.

Many projects use a resource that the company already owns. Because the firm does not need to pay cash to acquire this resource for a new project, it is tempting to assume that the resource is available for free. However, in many cases the resource could provide value for the firm in another opportunity or project. The **opportunity cost** of using a resource is the value it could have provided in its best alternative use.[4] Because this value is lost when the resource is used by another project, we should include the opportunity cost as an incremental cost of the project. For example, your company may be considering building a retail store on some land that it owns. Even though it already owns the land, it is not free to the store project. If it did not put its store on the land, the company could sell the land, for example. This forgone market price for the land is an opportunity cost of the retail store project.

COMMON MISTAKE

The Opportunity Cost of an Idle Asset

A common mistake is to conclude that if an asset is currently idle, its opportunity cost is zero. For example, the firm might have a warehouse that is currently empty or a machine that is not being used. Often, the asset may have been idled in anticipation of taking on the new project, and would have otherwise been put to use by the firm. Even if the firm has no alternative use for the asset, the firm could choose to sell or rent the asset. The value obtained from the asset's alternative use, sale, or rental represents an opportunity cost that must be included as part of the incremental cash flows.

Project Externalities

project externalities Indirect effects of a project that may increase or decrease the profits of other business activities of a firm.

Project externalities are indirect effects of a project that may increase or decrease the profits of other business activities of the firm. For instance, some purchasers of Apple's iPhone would otherwise have bought Apple's iPod Nano. When sales of a new product

[4] In Chapter 5, we defined the opportunity cost of capital as the rate you could earn on an alternative investment with equivalent risk. We similarly define the opportunity cost of using an existing asset in a project as the cash flow generated by the next-best alternative use for the asset.

cannibalization When sales of a firm's new product displace sales of one of its existing products.

displace sales of an existing product, the situation is often referred to as **cannibalization**. The lost sales of the existing project are an incremental cost to the company of going forward with the new product.



Sunk Costs

sunk cost Any unrecoverable cost for which a firm is already liable.

A **sunk cost** is any unrecoverable cost for which the firm is already liable. Sunk costs have been or will be paid regardless of the decision whether or not to proceed with the project. Therefore, they are not incremental with respect to the current decision and should not be included in its analysis. You may hire a market research firm to do market analysis to determine whether there is demand for a new product you are considering and the analysis may show that there is not enough demand, so you decide not to go forward with the project. Does that mean you do not have to pay the research firm's bill? Of course you still have to pay the bill, emphasizing that the cost was sunk and incurred whether you went forward with the project or not.

A good rule to remember is that *if your decision does not affect a cash flow, then the cash flow should not affect your decision.* If the cash flow is the same regardless of the decision, then it is not relevant to your decision. The following are some common examples of sunk costs you may encounter.

overhead expenses Those expenses associated with activities that are not directly attributable to a single business activity but instead affect many different areas of a corporation.

Fixed Overhead Expenses. **Overhead expenses** are associated with activities that are not directly attributable to a single business activity but instead affect many different areas of the corporation. Examples include the cost of maintaining the company's headquarters and the salary of the CEO. These expenses are often allocated to the different business activities for accounting purposes. To the extent that these overhead costs are fixed and will be incurred in any case, they are not incremental to the project and should not be included. Only include as incremental expenses the *additional* overhead expenses that arise because of the decision to take on the project.

COMMON MISTAKE

The Sunk Cost Fallacy

Being influenced by sunk costs is such a widespread mistake that it has a special name: *sunk cost fallacy.* The most common problem is that people "throw good money after bad." That is, people sometimes continue to invest in a project that has a negative NPV because they have already invested a large amount in the project and feel that by not continuing it, the prior investment will be wasted. The sunk cost fallacy is also sometimes called the "Concorde effect," a term that refers to the British and French governments' decision to continue funding the joint development of the Concorde aircraft even after it was clear that sales of the plane would fall far short of what was necessary to justify its continued development. The project was viewed by the British government as a commercial and financial disaster. However, the political implications of halting the project—and thereby publicly admitting that all past expenses on the project would result in nothing—ultimately prevented either government from abandoning the project.

Past Research and Development Expenditures. A pharmaceutical company may spend tens of millions of dollars developing a new drug, but if it fails to produce an effect in trials (or worse, has only negative effects), should it proceed? The company cannot get its development costs back and the amount of those costs should have no bearing on whether to continue developing a failed drug.

When a firm has already devoted significant resources to develop a new product, there may be a tendency to continue investing in the product even if market conditions

have changed and the product is unlikely to be viable. The rationale that is sometimes given is that if the product is abandoned, the money that has already been invested will be "wasted." In other cases, a decision is made to abandon a project because it cannot possibly be successful enough to recoup the investment that has already been made. In fact, neither argument is correct: Any money that has already been spent is a sunk cost and therefore irrelevant. The decision to continue or abandon should be based only on the incremental costs and benefits of the product going forward.

Adjusting Free Cash Flow

Here, we describe a number of complications that can arise when estimating a project's free cash flow.

Timing of Cash Flows. For simplicity, we have treated the cash flows in our examples as if they occur at annual intervals. In reality, cash flows will be spread throughout the year. While it is common to forecast at the annual level, we can forecast free cash flow on a quarterly or monthly basis when greater accuracy is required. In practice, firms often choose shorter intervals for riskier projects so that they might forecast cash flows at the monthly level for projects that carry considerable risk. For example, cash flows for a new facility in Europe may be forecasted at the quarterly or annual level, but if that same facility were located in a politically unstable country, the forecasts would likely be at the monthly level.

Accelerated Depreciation. Because depreciation contributes positively to the firm's cash flow through the depreciation tax shield, it is in the firm's best interest to use the most accelerated method of depreciation that is allowable for tax purposes. By doing so, the firm will accelerate its tax savings and increase their present value. In the United States, the most accelerated depreciation method allowed by the IRS is *MACRS (Modified Accelerated Cost Recovery System) depreciation*. With **MACRS depreciation**, the firm first categorizes assets according to their recovery period. Based on the recovery period, MACRS depreciation tables assign a fraction of the purchase price that the firm can recover each year. We provide MACRS tables and recovery periods for common assets in the appendix at the end of this chapter. As we explain in the appendix, MACRS allows for partial depreciation in year 0, when the asset is purchased and put into service.

MACRS depreciation The most accelerated cost recovery system allowed by the IRS. Based on the recovery period, MACRS depreciation tables assign a fraction of the purchase price that the firm can depreciate each year.

EXAMPLE 9.6
Computing Accelerated Depreciation

Problem

What depreciation deduction would be allowed for HomeNet's $7.5 million lab equipment using the MACRS method, assuming the lab equipment is designated to have a five-year recovery period? (See this chapter's appendix for information on MACRS depreciation schedules.)

Solution

Plan

Table 9.4 (in the appendix) provides the percentage of the cost that can be depreciated each year. Under MACRS, we take the percentage in the table for each year and multiply it by the original purchase price of the equipment to calculate the depreciation for that year.

Execute

Based on the table, the allowable depreciation expense for the lab equipment is shown below (in thousands of dollars):

1	Year	0	1	2	3	4	5
2	**MACRS Depreciation**						
3	Lab Equipment Cost	−7,500					
4	MACRS Depreciation Rate	20.00%	32.00%	19.20%	11.52%	11.52%	5.76%
5	Depreciation Expense	−1,500	−2,400	−1,440	−864	−864	−432

Evaluate

Compared with straight-line depreciation, the MACRS method allows for larger depreciation deductions earlier in the asset's life, which increases the present value of the depreciation tax shield and thus will raise the project's NPV. In the case of HomeNet, computing the NPV using MACRS depreciation leads to an NPV of $3.179 million.

Liquidation or Salvage Value. Assets that are no longer needed often have a resale value, or some salvage value if the parts are sold for scrap. Some assets may have a negative liquidation value. For example, it may cost money to remove and dispose of the used equipment.

In the calculation of free cash flow, we include the liquidation value of any assets that are no longer needed and may be disposed of. When an asset is liquidated, any capital gain is taxed as income. We calculate the capital gain as the difference between the sale price and the book value of the asset:

$$\text{Capital Gain} = \text{Sale Price} - \text{Book Value} \tag{9.9}$$

The book value is equal to the asset's original cost less the amount it has already been depreciated for tax purposes:

$$\text{Book Value} = \text{Purchase Price} - \text{Accumulated Depreciation} \tag{9.10}$$

We must adjust the project's free cash flow to account for the after-tax cash flow that would result from an asset sale:

$$\text{After-Tax Cash Flow from Asset Sale} = \text{Sale Price} - (\text{Tax Rate} \times \text{Capital Gain}) \tag{9.11}$$

EXAMPLE 9.7

Computing After-Tax Cash Flows from an Asset Sale

Problem

As production manager, you are overseeing the shutdown of a production line for a discontinued product. Some of the equipment can be sold for a total price of $50,000. The equipment was originally purchased four years ago for $500,000 and is being depreciated according to the five-year MACRS schedule. If your marginal tax rate is 35%, what is the after-tax cash flow you can expect from selling the equipment?

Solution

Plan

In order to compute the after-tax cash flow, you will need to compute the capital gain, which, as Eq. 9.9 shows, requires you to know the book value of the equipment. The book value is given in Eq. 9.10 as the original purchase price of the equipment less accumulated depreciation. Thus, you need to follow these steps:

1. Use the MACRS schedule to determine the accumulated depreciation.
2. Determine the book value as purchase price minus accumulated depreciation.
3. Determine the capital gain as the sale price less the book value.
4. Compute the tax owed on the capital gain and subtract it from the sale price, following Eq. 9.11, and then subtract the tax owed from the sale price.

Execute

From the chapter's appendix, we see that the first five rates of the five-year MACRS schedule (including year 0) are:

	Year	0	1	2	3	4
2	Depreciation Rate	20.00%	32.00%	19.20%	11.52%	11.52%
3	Depreciation Amount	100,000	160,000	96,000	57,600	57,600

Thus, the accumulated depreciation is $100{,}000 + 160{,}000 + 96{,}000 + 57{,}600 + 57{,}600 = 471{,}200$, such that the remaining book value is $\$500{,}000 - \$471{,}200 = \$28{,}800$. (Note we could have also

calculated this by summing the rates for years remaining on the MACRS schedule: Year 5 is 5.76%, so .0576 × 500,000 = 28,800).

The capital gain is then \$50,000 − \$28,800 = \$21,200 and the tax owed is 0.35 × \$21,200 = \$7,420.

Your after-tax cash flow is then found as the sale price minus the tax owed: \$50,000 − \$7,420 = \$42,580.

Evaluate

Because you are only taxed on the capital gain portion of the sale price, figuring the after-tax cash flow is not as simple as subtracting the tax rate multiplied by the sale price. Instead, you have to determine the portion of the sale price that represents a gain and compute the tax from there. The same procedure holds for selling equipment at a loss relative to book value—the loss creates a deduction for taxable income elsewhere in the company.

Tax Carryforwards. A firm generally identifies its marginal tax rate by determining the tax bracket that it falls into based on its overall level of pre-tax income. Two additional features of the tax code, called **tax loss carryforwards** and **carrybacks**, allow corporations to take losses during a current year and offset them against gains in nearby years. Since 1997, companies can "carry back" losses for two years and "carry forward" losses for 20 years. This tax rule means that a firm can offset losses during one year against income for the last two years, or save the losses to be offset against income during the next 20 years. When a firm can carry back losses, it receives a refund for back taxes in the current year. Otherwise, the firm must carry forward the loss and use it to offset future taxable income. When a firm has tax loss carryforwards well in excess of its current pre-tax income, then additional income it earns today will simply increase the taxes it owes after it exhausts its carryforwards.

tax loss carryforwards and carrybacks Two features of the U.S. tax code that allow corporations to take losses during a current year and offset them against gains in nearby years. Since 1997, companies can "carry back" losses for two years and "carry forward" losses for 20 years.

Replacement Decisions

Often the financial manager must decide whether to replace an existing piece of equipment. The new equipment may allow increased production, resulting in incremental revenue, or it may simply be more efficient, lowering costs. The typical incremental effects associated with such a decision are salvage value from the old machine, purchase of the new machine, cost savings and revenue increases, and depreciation effects.

EXAMPLE 9.8
Replacing an Existing Machine

Problem

You are trying to decide whether to replace a machine on your production line. The new machine will cost \$1 million, but will be more efficient than the old machine, reducing costs by \$500,000 per year. Your old machine is fully depreciated, but you could sell it for \$50,000. You would depreciate the new machine over a five-year life using MACRS. The new machine will not change your working capital needs. Your tax rate is 35%, and your cost of capital is 9%. Should you replace the machine?

Solution

Plan

Incremental revenues: 0

Incremental costs: −500,000 (a reduction in costs will appear as a positive number in the costs line of our analysis)

Depreciation schedule (from the appendix):

1	Year	0	1	2	3	4	5
2	Depreciation Rate	20.00%	32.00%	19.20%	11.52%	11.52%	5.76%
3	Depreciation Amount	200,000	320,000	192,000	115,200	115,200	57,600

Capital gain on salvage = $50,000 − $0 = $50,000

Cash flow from salvage value: +50,000 − (50,000)(.35) = 32,500

Execute

(values in thousands)

	Year	0	1	2	3	4	5
2	Incremental Revenues						
3	Incremental Costs of Goods Sold		500	500	500	500	500
4	**Incremental Gross Profit**		500	500	500	500	500
5	Depreciation	−200	−320	−192	−115.2	−115.2	−57.6
6	**EBIT**	−200	180	308	384.8	384.8	442.4
7	Income Tax at 35%	70	−63	−107.8	−134.68	−134.68	−154.84
8	**Incremental Earnings**	−130	117	200.2	250.12	250.12	287.56
9	Add Back Depreciation	200	320	192	115.2	115.2	57.6
10	Purchase of Equipment	−1,000					
11	Salvage Cash Flow	32.5					
12	**Incremental Free Cash Flows**	**−897.5**	**437**	**392.2**	**365.32**	**365.32**	**345.16**

$$NPV = -897.50 + \frac{437.00}{1.09} + \frac{392.20}{1.09^2} + \frac{365.32}{1.09^3} + \frac{365.32}{1.09^4} + \frac{345.16}{1.09^5} = 598.75$$

Evaluate

Even though the decision has no impact on revenues, it still matters for cash flows because it reduces costs. Further, both selling the old machine and buying the new machine involve cash flows with tax implications. The NPV analysis shows that replacing the machine will increase the value of the firm by almost $599 thousand.

Concept Check

7. Should we include sunk costs in the cash flows of a project? Why or why not?
8. Explain why it is advantageous for a firm to use the most accelerated depreciation schedule possible for tax purposes.

9.5 Analyzing the Project

When evaluating a capital budgeting project, financial managers should make the decision that maximizes NPV. As we have discussed, to compute the NPV for a project you need to estimate the incremental free cash flows and choose a discount rate. Given these inputs, the NPV calculation is relatively straightforward. The most difficult part of capital budgeting is deciding how to estimate the cash flows and cost of capital. These estimates are often subject to significant uncertainty. In this section, we look at methods that assess the importance of this uncertainty and identify the drivers of value in the project.

Sensitivity Analysis

sensitivity analysis An important capital budgeting tool that determines how the NPV varies as a single underlying assumption is changed.

An important capital budgeting tool for assessing the effect of uncertainty in forecasts is *sensitivity analysis*. **Sensitivity analysis** breaks the NPV calculation into its component assumptions and shows how the NPV varies as the underlying assumptions change. In this way, sensitivity analysis allows us to explore the effects of errors in our NPV estimates for a project. By conducting a sensitivity analysis, we learn which assumptions are the most important; we can then invest further resources and effort to refine these assumptions. Such an analysis also reveals which aspects of a project are most critical when we are actually managing the project.

In fact, we have already performed a type of sensitivity analysis in Chapter 8 when we constructed an NPV profile. By graphing the NPV of a project as a function of the discount rate, we are assessing the sensitivity of our NPV calculation to uncertainty about the cor-

Other Effects on Incremental Free Cash Flows

18. You have just completed a $20,000 feasibility study for a new coffee shop in some retail space you own. You bought the space two years ago for $100,000, but if you sold it today, you would net $115,000 after taxes. Outfitting the space for a coffee shop would require a capital expenditure of $30,000 plus an initial investment of $5,000 in inventory. What is the correct initial cash flow for your analysis of the coffee shop opportunity?

19. You purchased a machine for $1 million three years ago and have been applying straight-line depreciation to zero for a seven-year life. Your tax rate is 35%. If you sell the machine right now (after three years of depreciation) for $700,000, what is your incremental cash flow from selling the machine?

20. The Jones Company has just completed the third year of a five-year MACRS recovery period for a piece of equipment it originally purchased for $300,000.
a. What is the book value of the equipment?
b. If Jones sells the equipment today for $180,000 and its tax rate is 35%, what is the after-tax cash flow from selling it?

21. Just before it is about to sell the equipment from Problem 20, Jones receives a new order. It can take the new order if it keeps the old equipment. Is there a cost to taking the order and if so, what is it? Explain.

22. Home Builder Supply, a retailer in the home improvement industry, currently operates seven retail outlets in Georgia and South Carolina. Management is contemplating building an eighth retail store across town from its most successful retail outlet. The company already owns the land for this store, which currently has an abandoned warehouse located on it. Last month, the marketing department spent $10,000 on market research to determine the extent of customer demand for the new store. Now Home Builder Supply must decide whether to build and open the new store.

Which of the following should be included as part of the incremental earnings for the proposed new retail store?
a. The original purchase price of the land where the store will be located.
b. The cost of demolishing the abandoned warehouse and clearing the lot.
c. The loss of sales in the existing retail outlet, if customers who previously drove across town to shop at the existing outlet become customers of the new store instead.
d. The $10,000 in market research spent to evaluate customer demand.
e. Construction costs for the new store.
f. The value of the land if sold.
g. Interest expense on the debt borrowed to pay the construction costs.

23. If Daily Enterprises uses MACRS instead of straight-line depreciation, which incremental free cash flows from Problem 10 would increase and which would decrease?



24. Markov Manufacturing recently spent $15 million to purchase some equipment used in the manufacture of disk drives. The firm expects that this equipment will have a useful life of five years, and its marginal corporate tax rate is 35%. The company plans to use straight-line depreciation.
a. What is the annual depreciation expense associated with this equipment?
b. What is the annual depreciation tax shield?
c. Rather than straight-line depreciation, suppose Markov will use the MACRS depreciation method for the five-year life of the property. Calculate the deprecia-

tion tax shield each year for this equipment under this accelerated depreciation schedule.

d. If Markov has a choice between straight-line and MACRS depreciation schedules, and its marginal corporate tax rate is expected to remain constant, which schedule should it choose? Why?

e. How might your answer to part (d) change if Markov anticipates that its marginal corporate tax rate will increase substantially over the next five years?



25. You are a manager at Percolated Fiber, which is considering expanding its operations in synthetic fiber manufacturing. Your boss comes into your office, drops a consultant's report on your desk, and complains, "We owe these consultants $1 million for this report, and I am not sure their analysis makes sense. Before we spend the $25 million on new equipment needed for this project, look it over and give me your opinion." You open the report and find the following estimates (in thousands of dollars) (*see MyFinanceLab for the data in Excel format*):

1	Year	1	2	...	9	10
2	Sales Revenue	30,000	30,000		30,000	30,000
3	Costs of Goods Sold	18,000	18,000		18,000	18,000
4	**Gross Profit**	12,000	12,000		12,000	12,000
5	General, Sales, and Administrative Expenses	2,000	2,000		2,000	2,000
6	Depreciation	2,500	2,500		2,500	2,500
7	**EBIT**	7,500	7,500		7,500	7,500
8	Income Tax	2,625	2,625		2,625	2,625
9	**Net Income**	**4,875**	**4,875**		**4,875**	**4,875**

All of the estimates in the report seem correct. You note that the consultants used straight-line depreciation for the new equipment that will be purchased today (year 0), which is what the accounting department recommended. They also calculated the depreciation assuming no salvage value for the equipment, which is the company's assumption in this case. The report concludes that because the project will increase earnings by $4.875 million per year for ten years, the project is worth $48.75 million. You think back to your glory days in finance class and realize there is more work to be done!

First, you note that the consultants have not factored in the fact that the project will require $10 million in working capital upfront (year 0), which will be fully recovered in year 10. Next, you see they have attributed $2 million of selling, general, and administrative expenses to the project, but you know that $1 million of this amount is overhead that will be incurred even if the project is not accepted. Finally, you know that accounting earnings are not the right thing to focus on!

a. Given the available information, what are the free cash flows in years 0 through 10 that should be used to evaluate the proposed project?

b. If the cost of capital for this project is 14%, what is your estimate of the value of the new project?

Analyzing the Project



26. Bauer Industries is an automobile manufacturer. Management is currently evaluating a proposal to build a plant that will manufacture lightweight trucks. Bauer plans to use a cost of capital of 12% to evaluate this project. Based on extensive research, it has prepared the incremental free cash flow projections shown on the next page (in millions of dollars) (*see MyFinanceLab for the data in Excel format*).

a. For this base-case scenario, what is the NPV of the plant to manufacture lightweight trucks?

Concept Check

5. What is the market portfolio?
6. What does beta (β) tell us?

12.4 Putting It All Together: The Capital Asset Pricing Model

One of our goals in this chapter is to compute the cost of equity capital for Apple, which is the best available expected return offered in the market on an investment of comparable risk and term. Thus, in order to compute the cost of capital, we need to know the relation between Apple's risk and its expected return. Over the last three sections, we have laid the foundation for a practical way of measuring this relation. In this section, we put all the pieces together to build a model for determining the expected return of any investment.

The CAPM Equation Relating Risk to Expected Return

As we have learned, only common, systematic risk determines expected returns—firm-specific risk is diversifiable and does not warrant extra return. In the introduction to this chapter, we stated that, intuitively, the expected return on any investment should come from two components:

1. A baseline risk-free rate of return that we would demand to compensate for inflation and the time value of money, even if there were no risk of losing our money.
2. A risk premium that varies with the amount of systematic risk in the investment.

$$\textit{Expected Return} = \textit{Risk-Free Rate} + \textit{Risk Premium for Systematic Risk}$$

We devoted the last section to measuring systematic risk. Beta is our measure of the amount of systematic risk in an investment:

$$\textit{Expected Return for Investment } i = \textit{Risk-Free Rate} + \beta_i \times \textit{Risk Premium per Unit of Systematic Risk } (\textit{per unit of } \beta).$$

market risk premium (equity risk premium) The historical average excess returns on the market portfolio.

But what is the risk premium per unit of systematic risk? Well, we know that the market portfolio, by definition, has exactly one unit of systematic risk (it has a beta of 1). So, a natural estimate of the risk premium per unit of systematic risk is the historical average excess return on the market portfolio, also known as the **market** or **equity risk premium**. Historically, the average excess return of the S&P 500 over the return on U.S. Treasury bonds (the risk-free rate) has been 5% to 7%, depending on the period of measurement (we'll discuss issues in determining the risk premium and risk-free rate in the next chapter). With this last piece of the puzzle, we can write down the equation for the expected return of an investment:

Capital Asset Pricing Model

$$E[R_i] = r_f + \underbrace{\beta_i (E[R_{Mkt}] - r_f)}_{\text{Risk premium for security } i} \tag{12.6}$$

Capital Asset Pricing Model (CAPM) An equilibrium model of the relationship between risk and return that characterizes a security's expected return based on its beta with the market portfolio.

This equation for the expected return of any investment is the **Capital Asset Pricing Model (CAPM)**. In words, the CAPM simply says that the return we should expect on any investment is equal to the risk-free rate of return plus a risk premium proportional to the amount of systematic risk in the investment. Specifically, the risk premium of an investment is equal to the market risk premium $(E[R_{Mkt}] - r_f)$ multiplied by the amount of systematic (market) risk present in the investment, measured by its beta with the market

Why Not Estimate Expected Returns Directly?

If we have to use historical data to estimate beta and determine a security's expected return (or an investment's cost of capital), why not just use the security's historical average return as an estimate for its expected return instead? This method would certainly be simpler and more direct.

The answer is that it is extremely difficult to infer the average return of individual stocks from historical data. Because stock returns are so volatile, with even 100 years of data we would have little confidence in our estimate of the true average. (Imagine drawing 100 numbers from a swimming pool full of widely ranging numbers and being asked to guess the average of all of the numbers in the pool.) Worse, few stocks have existed for 100 years, and those that have probably bear little resemblance today to what they were like 100 years ago. If we use less than ten years of data, we would have very little confidence in our estimate at all. In fact, if the volatility of the stock's return is 20%, it turns out that we would need 1600 years of data to be 95% confident that our estimate of its true average return was within +/−1% of being correct!

On the other hand, the linear regression technique allows us to infer beta from historical data reasonably accurately with just a few years of data. Thus, in theory at least, using beta and the CAPM can provide much more accurate estimates of expected returns for stocks than we could obtain from their historical average return.

required return The expected return of an investment that is necessary to compensate for the risk of undertaking the investment.

(β_i). Because investors will not invest in this security unless they can expect at least the return given in Eq. 12.6, we also call this return the investment's **required return**.

The CAPM is the main method used by most major corporations to determine the equity cost of capital. In a survey of CFOs, Graham and Harvey found that over 70% rely on the CAPM, and Bruner, Eades, Harris, and Higgins reported that 85% of a sample of large firms rely on it.[5] It has become the most important model of the relationship between risk and return, and for his contributions to the theory, William Sharpe was awarded the Nobel Prize in Economics in 1990.

EXAMPLE 12.6
Computing the Expected Return for a Stock

Problem

Suppose the risk-free return is 3% and you measure the market risk premium to be 6%. Apple has a beta of 1.6. According to the CAPM, what is its expected return?

Solution

Plan

We can use Eq. 12.6 to compute the expected return according to the CAPM. For that equation, we will need the market risk premium, the risk-free return, and the stock's beta. We have all of these inputs, so we are ready to go.

Execute

Using Eq. 12.6:

$$E[R_{AAPL}] = r_f + \beta_{AAPL}(E[R_{Mkt}] - r_f) = 3\% + 1.6(6\%)$$
$$= 12.6\%$$

Evaluate

Because of Apple's beta of 1.6, investors will require a risk premium of 9.6% over the risk-free rate for investments in its stock to compensate for the systematic risk of Apple stock. This leads to a total expected return of 12.6%.

[5] J. Graham and C. Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field." *Journal of Financial Economics* 60 (2001): 187–243; and R. Bruner, K. Eades, R. Harris, and R. Higgins, "Best Practices in Estimating the Cost of Capital: Survey and Synthesis." *Financial Practice and Education* 8 (1998): 13–28.

NOBEL PRIZE

William Sharpe

The CAPM was proposed as a model of risk and return by William Sharpe in a 1964 paper, and in related papers by Jack Treynor (1961), John Lintner (1965), and Jan Mossin (1966).*

Below is an excerpt from a 1998 interview with William Sharpe:

> *Portfolio Theory focused on the actions of a single investor with an optimal portfolio. I said what if everyone was optimizing? They've all got their copies of Markowitz and they're doing what he says. Then some people decide they want to hold more IBM, but there aren't enough shares to satisfy demand. So they put price pressure on IBM and up it goes, at which point they have to change their estimates of risk and return, because now they're paying more for the stock. That process of upward and downward pressure on prices continues until prices reach an equilibrium and everyone collectively wants to hold what's available. At that point, what can you say about the relationship between risk and return? The answer is that expected return is proportionate to beta relative to the market portfolio.*
>
> *The CAPM was and is a theory of equilibrium. Why should anyone expect to earn more by investing in one security as opposed to another? You need to be compensated for doing badly when times are bad. The security that is going to do badly just when you need money when times are bad is a security you have to hate, and there had better be some redeeming virtue or else who will hold it? That redeeming virtue has to be that in normal times you expect to do better. The key insight of the Capital Asset Pricing Model is that higher expected returns go with the greater risk of doing badly in bad times. Beta is a measure of that. Securities or asset classes with high betas tend to do worse in bad times than those with low betas.*

Source: Jonathan Burton, "Revisiting the Capital Asset Pricing Model." *Dow Jones Asset Manager* (May/June 1998): 20–28.

*W. F. Sharpe: "Capital Asset Prices: A Theory of Market Equilibrium Under Conditions of Risk." *Journal of Finance* 19 (September 1964): 425–442.

Jack Treynor, "Toward a Theory of the Market Value of Risky Assets." Unpublished manuscript (1961).

J. Lintner: "The Valuation of Risk Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets." *Review of Economics and Statistics* 47 (February 1965): 13–37.

J. Mossin "Equilibrium in a Capital Asset Market." *Econometrica* 34 (4) (1966): 768–783.

The Security Market Line

security market line (SML) The pricing implication of the CAPM, it specifies a linear relation between the risk premium of a security and its beta with the market portfolio.

Figure 12.9 graphs the relations between expected return and both total risk and systematic risk (beta) for the stocks in Table 12.4. Recall from Chapter 11 that there is no clear relation between a stock's standard deviation (total risk) and its expected return, as shown in panel (a). However, the CAPM equation (Eq. 12.6) implies that there is a linear relation between a stock's beta and its expected return. This line is graphed in panel (b) as the line through the risk-free investment (with a beta of zero) and the market (with a beta of one); it is called the **security market line (SML)**. We see that the relation between risk and return for individual securities is only evident when we measure market risk as in panel (b), rather than total risk as in panel (a).

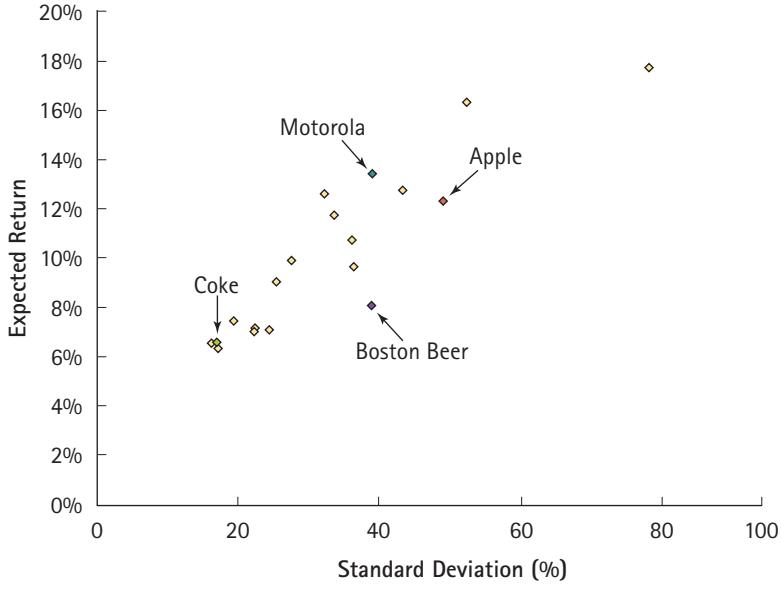
The security market line in Figure 12.9 raises the subject of negative beta stocks. While the vast majority of stocks have a positive beta, it is possible to have returns that co-vary negatively with the market. Firms that provide goods or services that are in greater demand in economic contractions than in booms fit this description.

FIGURE 12.9

Expected Return, Volatility, and Beta

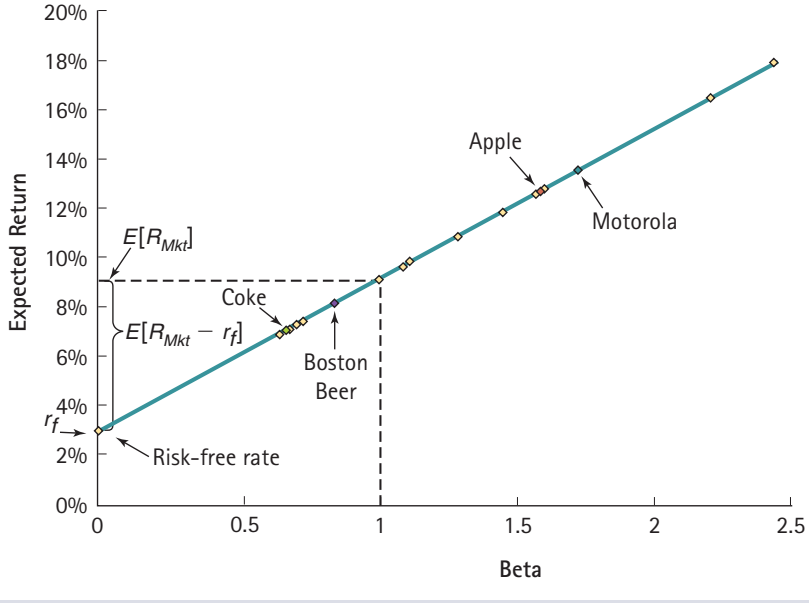
Panel (a) Expected Return and Total Risk (Standard Deviation)

The graph compares the standard deviation and expected returns of the stocks in Table 12.4. There is no relation between total risk and expected return. Some of the stocks are identified. It is clear that we could not predict Apple's expected return using its total risk (volatility).



Panel (b) Expected Return and Beta

The security market line shows the expected return for each security in Table 12.4 as a function of its beta with the market. According to the CAPM, all stocks and portfolios (including the market portfolio) should lie on the security market line. Thus, Apple's expected return can be determined by its beta, which measures its systematic risk.



EXAMPLE 12.7
A Negative Beta Stock

Problem

Suppose the stock of Bankruptcy Auction Services Inc. (BAS) has a negative beta of −0.30. How does its expected return compare to the risk-free rate, according to the CAPM? Does your result make sense?

Solution

Plan

We can use the CAPM equation, Eq. 12.6, to compute the expected return of this negative beta stock just like we would with a positive beta stock. We don't have the risk-free rate or the market risk premium, but the problem doesn't ask us for the exact expected return, just whether or not it will be more or less than the risk-free rate. Using Eq. 12.6, we can answer that question.

Execute

Because the expected return of the market is higher than the risk-free rate, Eq. 12.6 implies that the expected return of Bankruptcy Auction Services (BAS) will be *below* the risk-free rate. As long as the market risk premium is positive (as long as people demand a higher return for investing in the market than for a risk-free investment), then the second term in Eq. 12.6 will have to be negative if the beta is negative. For example, if the risk-free rate is 4% and the market risk premium is 6%:

$$E[R_{BAS}] = 4\% - 0.30(6\%) = 2.2\%.$$

(See Figure 12.9—the security market line drops below r_f for $\beta < 0$.)

Evaluate

This result seems odd—why would investors be willing to accept a 2.2% expected return on this stock when they can invest in a safe investment and earn 4%? The answer is that a savvy investor will not hold BAS alone; instead, the investor will hold it in combination with other securities as part of a well-diversified portfolio. These other securities will tend to rise and fall with the market. But because BAS has a negative beta, its correlation with the market is negative, which means that BAS tends to perform well when the rest of the market is doing poorly. Therefore, by holding BAS, an investor can reduce the overall market risk of the portfolio. In a sense, BAS is "recession insurance" for a portfolio, and investors will pay for this insurance by accepting a lower expected return.

The CAPM and Portfolios

Because the security market line applies to all securities, we can apply it to portfolios as well. For example, the market portfolio is on the SML, and according to the CAPM, other portfolios (such as mutual funds) are also on the SML. Therefore, the expected return of a portfolio should correspond to the portfolio's beta. We calculate the beta of a portfolio made up of securities each with weight w_i as follows:

$$\beta_P = w_1\beta_1 + w_2\beta_2 + \cdots + w_n\beta_n \tag{12.7}$$

That is, the *beta of a portfolio is the weighted average beta of the securities in the portfolio*.

EXAMPLE 12.8
The Expected Return of a Portfolio

Problem

Suppose drug-maker Pfizer (PFE) has a beta of 0.7, whereas the beta of Google (GOOG) is 1.1. If the risk-free interest rate is 3% and the market risk premium is 6%, what is the expected return of an equally weighted portfolio of Pfizer and Google, according to the CAPM?

Solution

Plan

We have the following information:

	$r_f = 3\%$,	$E[R_{Mkt}] - r_f = 6\%$
PFE:	$\beta_{PFE} = 0.7$,	$w_{PFE} = 0.50$
GOOG:	$\beta_{GOOG} = 1.1$,	$w_{GOOG} = 0.50$

We can compute the expected return of the portfolio two ways. First, we can use the CAPM (Eq. 12.6) to compute the expected return of each stock and then compute the expected return for the portfolio using Eq. 12.3.

Or, we could compute the beta of the portfolio using Eq. 12.7 and then use the CAPM (Eq. 12.6) to find the portfolio's expected return.

Execute

Using the first approach, we compute the expected return for PFE and GOOG:

$$E[R_{PFE}] = r_f + \beta_{PFE}(E[R_{Mkt}] - r_f) \qquad E[R_{GOOG}] = r_f + \beta_{GOOG}(E[R_{Mkt}] - r_f)$$

$$E[R_{PFE}] = 3\% + 0.7(6\%) = 7.2\% \qquad E[R_{GOOG}] = 3\% + 1.1(6\%) = 9.6\%$$

Then the expected return of the equally weighted portfolio P is:

$$E[R_P] = 0.5(7.2\%) + 0.5(9.6\%) = 8.4\%$$

Alternatively, we can compute the beta of the portfolio using Eq.12.7:

$$\beta_P = w_{PFE}\beta_{PFE} + w_{GOOG}\beta_{GOOG}$$

$$\beta_P = (0.5)(0.7) + (0.5)(1.1) = 0.9$$

We can then find the portfolio's expected return from the CAPM:

$$E[R_P] = r_f + \beta_P(E[R_{Mkt}] - r_f)$$

$$E[R_P] = 3\% + 0.9(6\%) = 8.4\%$$

Evaluate

The CAPM is an effective tool for analyzing securities and portfolios of those securities. You can compute the expected return of each security using its beta and then compute the weighted average of those expected returns to determine the portfolio's expected return. Or, you can compute the weighted average of the securities' betas to get the portfolio's beta and then compute the expected return of the portfolio using the CAPM. Either way, you will get the same answer.

Summary of the Capital Asset Pricing Model

The CAPM is a powerful tool that is widely used to estimate the expected return on stocks and investments within companies. To summarize the model and its use:

- Investors require a risk premium proportional to the amount of *systematic* risk they are bearing.
- We can measure the systematic risk of an investment by its β, which is the sensitivity of the investment's return to the market's return. For each 1% change in the market portfolio's return, the investment's return is expected to change by β percent due to risks that it has in common with the market.
- The most common way to estimate a stock's beta is to regress its historical returns on the market's historical returns. The stock's beta is the slope of the line that best explains the relation between the market's return and the stock's return.
- The CAPM says that we can compute the expected, or required, return for any investment using the following equation:

$$E[R_i] = r_f + \beta_i(E[R_{Mkt}] - r_f)$$

which when graphed is called the *security market line*.

The Big Picture

The CAPM marks the culmination of our examination of how investors in capital markets trade off risk and return. It provides a powerful and widely used tool to quantify the return that should accompany a particular amount of systematic risk. We have already reached

our goal (in Example 12.6) of estimating the cost of equity capital for Apple. While our finding that equity investors in Apple should reasonably expect (and therefore require) a return of 12.6% on their investments is an important piece of information to Apple's managers, it is not the whole picture. While some firms, like Apple and Microsoft, have only equity investors, most have debt investors as well. In the next chapter we will apply what we have learned here and in Chapters 6, 7 and 10 on bonds and stocks to develop the overall cost of capital for a company. The Valuation Principle tells us to use this cost of capital to discount the future expected cash flows of the firm to arrive at the value of the firm. Thus, the cost of capital is an essential input to the financial manager's job of analyzing investment opportunities, and so knowing this overall cost of capital is critical to the company's success at creating value for its investors.

Concept Check

7. What does the CAPM say about the required return of a security?
8. What is the security market line?

myfinancelab Here is what you should know after reading this chapter. MyFinanceLab will help you identify what you know, and where to go when you need to practice.

Key Points and Equations	Key Terms	Online Practice Opportunities
12.1 The Expected Return of a Portfolio • The portfolio weight is the initial fraction w_i of an investor's money invested in each asset. Portfolio weights add up to one: $$w_i = \frac{\text{Value of Investment } i}{\text{Total Value of Portfolio}} \quad (12.1)$$ • The expected return of a portfolio is the weighted average of the expected returns of the investments within it, using the portfolio weights: $$E[R_P] = w_1E[R_1] + w_2E[R_2] + \cdots + w_nE[R_n] \quad (12.3)$$	expected return of a portfolio, p. 349 portfolio weights, p. 347 return on a portfolio, p. 347	MyFinanceLab Study Plan 12.1
12.2 The Volatility of a Portfolio • To find the risk of a portfolio, we need to know the degree to which stock returns move together. Correlation measures the co-movement of returns. The correlation is always between -1 and $+1$. It represents the fraction of the volatility due to risk that is common to the securities. • The variance of a portfolio depends on the correlation of the stocks. For a portfolio with two stocks, the portfolio variance is: $$Var(R_p) = w_1^2SD(R_1)^2 + w_2^2SD(R_2)^2 + 2w_1w_2Corr(R_1, R_2)SD(R_1)SD(R_2) \quad (12.4)$$ • As we lower the correlation between the two stocks in a portfolio, we lower the portfolio variance. • Diversification eliminates independent, firm-specific risks, and the volatility of a large portfolio results from the common systematic risk between the stocks in the portfolio.	correlation, p. 352 equally weighted portfolio, p. 357 volatility of a portfolio, p. 350	MyFinanceLab Study Plan 12.2 Using Excel: Correlation Between Two Sets of Returns

Putting It All Together: The Capital Asset Pricing Model

21. Suppose the risk-free return is 4% and the market portfolio has an expected return of 10% and a standard deviation of 16%. Johnson & Johnson Corporation stock has a beta of 0.32. What is its expected return?

22. What is the sign of the risk premium of a negative-beta stock? Explain. (Assume the risk premium of the market portfolio is positive.)

23. EJH has a beta of 1.2, CSH has a beta of 0.6, and KMS has a beta of 1.0. If you put 25% of your money in EJH, 25% in CSH, and 50% in KMS, what is the beta of your portfolio?

24. Suppose Intel stock has a beta of 1.6, whereas Boeing stock has a beta of 1. If the risk-free interest rate is 4% and the expected return of the market portfolio is 10%, according to the CAPM,
   a. What is the expected return of Intel stock?
   b. What is the expected return of Boeing stock?
   c. What is the beta of a portfolio that consists of 60% Intel stock and 40% Boeing stock?
   d. What is the expected return of a portfolio that consists of 60% Intel stock and 40% Boeing stock (show both ways to solve this)?

***25.** You are thinking of buying a stock priced at $100 per share. Assume that the risk-free rate is about 4.5% and the market risk premium is 6%. If you think the stock will rise to $117 per share by the end of the year, at which time it will pay a $1 dividend, what beta would it need to have for this expectation to be consistent with the CAPM?

***26.** You are analyzing a stock that has a beta of 1.2. The risk-free rate is 5% and you estimate the market risk premium to be 6%. If you expect the stock to have a return of 11% over the next year, should you buy it? Why or why not?

27. You have risen through the ranks of a coffee company, from the lowly green-apron barista to the coveted black apron, and all the way to CFO. A quick Internet check shows you that your company's beta is 0.6. The risk free rate is 5% and you believe the market risk premium to be 5.5%. What is your best estimate of investors' expected return on your company's stock (its cost of equity capital)?

28. At the beginning of 2007, Apple's beta was 1.4 and the risk-free rate was about 4.5%. Apple's price was $84.84. Apple's price at the end of 2007 was 198.08. If you estimate the market risk premium to have been 6%, did Apple's managers exceed their investors' required return as given by the CAPM?

***29.** You want to invest $50,000 in a portfolio with a beta of no more than 1.4 and an expected return of 12.4%. Bay Corp. has a beta of 1.2 and an expected return of 11.2%, and City Inc. has a beta of 1.8 and an expected return of 14.8%. The risk-free rate is 4%. You can invest in Bay Corp. and City Inc. How much will you invest in each?

13.1 A First Look at the Weighted Average Cost of Capital

Most firms draw on some combination of equity, debt, and other securities to raise the funds they need for investment. In this section, we examine the role of financing sources in determining the firm's overall cost of capital. We begin by stepping back to assess these financing sources in the context of the firm's balance sheet.

The Firm's Capital Structure

capital A firm's sources of financing—debt, equity, and other securities that it has outstanding.

A firm's sources of financing, which usually consist of debt and equity, represent its **capital.** The typical firm raises funds to invest by selling shares to stockholders (its equity) and borrowing from lenders (its debt). Recall the most basic form of the balance sheet, as represented in Figure 13.1. The left side of the balance sheet lists the firm's assets, and the right side describes the firm's capital.

FIGURE 13.1
A Basic Balance Sheet

This figure provides a very basic balance sheet for reference. As discussed in Chapter 2, the two sides of the balance sheet must equal each other: Assets = Liabilities + Equity. The right side represents the way the assets are financed. In this chapter, we will focus on the required returns for the different forms of financing found on the right side of the balance sheet.

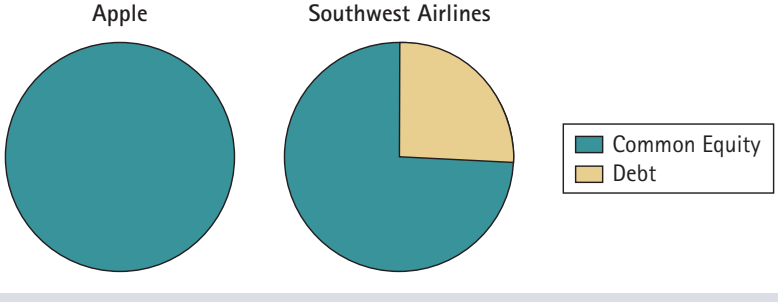
Assets	Liabilities and Equity
Current Assets	Debt
Long-Term Assets	Preferred Stock
	Equity

capital structure The relative proportions of debt, equity, and other securities that a firm has outstanding.

The relative proportions of debt, equity, and other securities that a firm has outstanding constitute its **capital structure**. When corporations raise funds from outside investors, they must choose which type of security to issue. The most common choices are financing through equity alone and financing through a combination of debt and equity. Figure 13.2 shows the capital structures of Apple and Southwest Airlines. Capital structures vary widely across firms. In Chapter 16, we will discuss how a firm sets its capital structure.

FIGURE 13.2
Two Capital Structures

This figure shows the capital structures of two real firms. Apple is financed 100% with common equity, shown in green, while Southwest Airlines is financed 74% with common equity and 26% with debt, shaded in yellow.



Source: Authors' calculations based on publicly available data in 2010.

Opportunity Cost and the Overall Cost of Capital

Financial managers take into account each component of the firm's capital structure when determining the firm's overall cost of capital. Throughout the discussion that follows, keep in mind the intuition behind the term "cost of capital." When investors buy the stock or bonds of a company, they forgo the opportunity to invest that money elsewhere. The expected return from those alternative investments constitutes an opportunity cost to them. Thus, to attract their investments as capital to the firm, the firm must offer potential investors an expected return equal to what they could expect to earn elsewhere for assuming the same level of risk. Providing this return is the cost a company bears in exchange for obtaining capital from investors.

Weighted Averages and the Overall Cost of Capital

Intuitively, the firm's overall cost of capital should be a blend of the costs of the different sources of capital. In fact, we calculate the firm's overall cost of capital as a weighted average of its equity and debt costs of capital, known as the firm's **weighted average cost of capital (WACC)**.

weighted average cost of capital (WACC) The average of a firm's equity and debt costs of capital, weighted by the fractions of the firm's value that correspond to equity and debt, respectively.

But what should the weights be? Imagine you owned all of the stock and all of the debt of the firm. If that was all you had in your portfolio, the return on your portfolio would be the total return of the firm. As we showed in Chapter 12, a portfolio return is the weighted average of the returns of the securities in the portfolio. In this case, the return on your portfolio—the total return of the firm—is a weighted average of the return you earn holding all the stock of the firm and the return you earn holding all of the debt. Since you hold all of each, your portfolio weights are just the relative amount of debt and equity issued by the firm. Thus the weights we use in the WACC are the proportions of debt and equity used in the firm's capital structure. For example, if the firm is financed 30% by debt and 70% by equity, then the weights used in its WACC would be 30% on the debt cost of capital and 70% on the equity cost of capital.

This example suggests that you can determine the weights by looking at the right side of the firm's balance sheet. That assumption is correct, with one important modification: You must use the *market values* of the debt and equity to determine the proportions, not the accounting-based *book values* listed on the balance sheet. Recall from Chapter 2 that book values reflect historical costs, but market values are forward-looking, based on what the assets are expected to produce in the future. Holders of the firm's financial claims—equity and, if the firm has it, debt—assess the firm based on the market value of its assets, not the book value.

market-value balance sheet Similar to an accounting balance sheet, but all values are current market values rather than historical costs.

In fact, it is useful to think about the **market-value balance sheet,** where the assets, debt, and equity are all listed in terms of their market values, instead of their book values. Of course, the market-value balance sheet must still balance:

$$\text{Market Value of Equity} + \text{Market Value of Debt} = \text{Market Value of Assets} \quad (13.1)$$

Equation 13.1 states that the total market value of all the claims (equity and debt) issued by the firm must be equal to the total market value of all its assets. This equality drives home the point that the equity and debt issued by the firm derive their value from the underlying assets they claim. The risk, and hence the required return, of the debt and equity of the firm are determined by the risk of the firm's assets. This point will be useful as we derive the firm's WACC.

Weighted Average Cost of Capital Calculations

In this section, we will develop the intuition behind the use of market-value weights as well as the link between the risk of the assets and the risk of the debt and equity claims on those assets.

unlevered A firm that does not have debt outstanding.

levered A firm that has debt outstanding.

leverage The relative amount of debt on a firm's balance sheet.

We begin with the straightforward case of the firm that does not issue debt—the **unlevered** firm that pays out all of the free cash flows generated by its assets to its equity holders. When some of a firm's financing comes from debt, we say the firm is **levered**. Just as a lever allows you to lift a heavy object by exerting relatively little force, so borrowing money through debt allows equity holders to control highly valued assets with relatively little investment of their own money. We refer to the relative amount of debt on the balance sheet as the firm's **leverage**.

The Weighted Average Cost of Capital: Unlevered Firm. If a firm is unlevered, so that it has no debt, all of the free cash flows generated by its assets are ultimately paid out to its equity holders. Because the free cash flows to the equity holders are the same as the free cash flows from the assets, the Valuation Principle tells us that the market value, risk, and cost of capital for the firm's equity are equal to the corresponding amounts for its assets. Given this relationship, we can estimate the firm's equity cost of capital using the Capital Asset Pricing Model (CAPM). The resulting estimate is the cost of capital for the firm as a whole. For example, both Cisco and Apple do not issue debt, so the cost of capital for Cisco's or Apple's assets is the same as the firms' costs of equity.

The Weighted Average Cost of Capital: Levered Firm. But what if the firm has debt? How should we incorporate the cost of this debt to determine the cost of capital for the firm's assets as a whole? The market-value balance sheet provides the answer. We can interpret the equality in Eq. 13.1 in terms of a portfolio: By holding a portfolio of the firm's equity and debt, we can get the same cash flows as if we held the assets directly. Because the return of a portfolio is equal to the weighted average of the returns of the securities in it, this equality implies the following relationship between the required returns (costs) of equity, debt, and assets:

Weighted Average Cost of Capital (Pre-tax)

$$\begin{aligned} r_{wacc} &= \begin{pmatrix}\text{Fraction of Firm Value}\\ \text{Financed by Equity}\end{pmatrix}\begin{pmatrix}\text{Equity}\\ \text{Cost of Capital}\end{pmatrix} \\ &\quad + \begin{pmatrix}\text{Fraction of Firm Value}\\ \text{Financed by Debt}\end{pmatrix}\begin{pmatrix}\text{Debt}\\ \text{Cost of Capital}\end{pmatrix} \\ &= \begin{pmatrix}\text{Asset}\\ \text{Cost of Capital}\end{pmatrix} \end{aligned} \tag{13.2}$$

We now have the justification for our intuition that the overall cost of capital for a firm should be a weighted average of its equity and debt costs of capital. Eq. 13.2 shows that we can calculate the cost of capital of the firm's assets by computing the weighted average of the firm's equity and debt cost of capital. In the next section, we explore how to estimate the firm's costs of equity and debt capital.

EXAMPLE 13.1
Calculating the Weights in the WACC

Problem

Suppose Kenai Corp. has debt with a book (face) value of $10 million, trading at 95% of face value. It also has book equity of $10 million, and 1 million shares of common stock trading at $30 per share. What weights should Kenai use in calculating its WACC?

Solution

Plan

Equation 13.2 tells us that the weights are the fractions of Kenai financed with debt and financed with equity. Furthermore, these weights should be based on market values because the cost of capital is based on investors' current assessment of the value of the firm, not on accounting-based book values. As a consequence, we can ignore the book values of debt and equity.

Execute

$10 million in debt trading at 95% of face value is $9.5 million in market value. One million shares of stock at $30 per share is $30 million in market value. So, the total market value of the firm is $39.5 million.

The weights are

$$9.5 \div 39.5 = 24.1\% \text{ for debt and } 30 \div 39.5 = 75.9\% \text{ for equity.}$$

Evaluate

When calculating its overall cost of capital, Kenai will use a weighted average of the cost of its debt capital and the cost of its equity capital, giving a weight of 24.1% to its cost of debt and a weight of 75.9% to its cost of equity.

Concept Check

1. Why does a firm's capital have a cost?
2. Why do we use market-value weights in the weighted average cost of capital?

13.2 The Firm's Costs of Debt and Equity Capital

Section 13.1 made it clear that to measure the firm's overall cost of capital, we need to start by determining the cost of each type of capital a firm might use. We now turn to how a company measures the costs of its debt, preferred stock, and common stock. We will use DuPont, a global chemical company, as an example.

Cost of Debt Capital

We will start at the top of the right side of the balance sheet with the cost of the firm's debt. A firm's cost of debt is the interest rate it would have to pay to refinance its existing debt, such as through new bond issues. This rate differs from the coupon rate on the firm's existing debt, which reflects the interest rate the firm had to offer at the time the debt was issued.

Yield to Maturity and the Cost of Debt. Existing debt trades in the marketplace, so its price fluctuates to reflect both changes in the overall credit environment and changes in the risk specifically associated with the firm. As we learned in Chapter 6, the market price of the firm's existing debt implies a yield to maturity, which is the return that current purchasers of the debt would earn if they held the debt to maturity and received all of the payments as promised. So, we can use the yield to maturity to estimate the firm's current cost of debt: It is the yield that investors demand to hold the firm's debt (new or existing).[1]

Suppose DuPont has debt due in 2020 with a coupon rate of 4.63% priced at $1076.60 per $1000 face value. Because the market price of the debt is above its face value, investors in debt earn a yield that is lower than the 4.63% coupon rate. In fact, based on the timing of the remaining payments, this price implies a yield to maturity of 3.66%, which is an estimate of DuPont's current cost of debt. You would not need to actually compute the yield to maturity yourself because prices and their implied yields to maturity are always quoted together in the bond market.[2]

[1] In fact, the yield to maturity is the *most* the firm will pay because there is some risk the firm may not repay its debt. The expected return on the bond can be significantly less than the *promised* yield to maturity during times when the firm might be close to financial distress. However, outside of these times, the yield on an A-rated bond such as DuPont's provides a reasonable estimate of its expected return.

[2] Chapter 6 demonstrated how to find current prices and yields to maturity for corporate bonds online using the Web site http://www.finra.org/marketdata.

TABLE 13.3
Expected Free Cash Flow from DuPont's Facility Project

	Year	0	1	2	3	4
2	**Incremental Earnings Forecast ($million)**					
3	Sales	—	**60.00**	**60.00**	**60.00**	**60.00**
4	Cost of Goods Sold	—	**−25.00**	**−25.00**	**−25.00**	**−25.00**
5	**Gross Profit**	—	35.00	35.00	35.00	35.00
6	Operating Expenses	**−6.67**	**−9.00**	**−9.00**	**−9.00**	**−9.00**
7	Depreciation	—	**−6.00**	**−6.00**	**−6.00**	**−6.00**
8	**EBIT**	−6.67	20.00	20.00	20.00	20.00
9	Income Tax at 35%	2.33	−7.00	−7.00	−7.00	−7.00
10	**Unlevered Net Income**	−4.34	13.00	13.00	13.00	13.00
11	**Incremental Free Cash Flow ($ million)**					
12	Plus: Depreciation	—	6.00	6.00	6.00	6.00
13	Less: Capital Expenditures	**−24.00**	—	—	—	—
14	Less: Increases in NWC	—	—	—	—	—
15	**Incremental Free Cash Flow**	**−28.34**	**19.00**	**19.00**	**19.00**	**19.00**

flows, V_0^L, using the WACC method and DuPont's WACC of 9.11%, which we computed in Section 13.3:

$$V_0^L = \frac{19}{1.0911} + \frac{19}{1.0911^2} + \frac{19}{1.0911^3} + \frac{19}{1.0911^4} = \$61.41 \text{ million}$$

Because the upfront after-tax cost of extending the facility's life is only $28.34 million, this project is a good idea. Taking the project results in an NPV of $61.41 million − $28.34 million = $33.07 million for the firm.

Summary of the WACC Method

To summarize, the key steps in the WACC valuation method are as follows:

1. Determine the incremental free cash flow of the investment.
2. Compute the weighted average cost of capital using Eq. 13.6.
3. Compute the value of the investment, including the tax benefit of leverage, by discounting the incremental free cash flow of the investment using the WACC.

In many firms, the corporate treasurer performs the second step, calculating the firm's WACC. This rate can then be used throughout the firm as the companywide cost of capital for new investments *that are of comparable risk to the rest of the firm and that will not alter the firm's debt-equity ratio*. Employing the WACC method in this way is very simple and straightforward. As a result, this method is the most commonly used in practice for capital budgeting purposes.

Concept Check

7. What are the main assumptions you make when you use the WACC method?
8. What inputs do you need to be ready to apply the WACC method?

13.5 Project-Based Costs of Capital

Up to this point we have assumed that both the risk and the leverage of the project under consideration matched those characteristics for the firm as a whole. This assumption allowed us, in turn, to assume that the cost of capital for a project matched the firm's cost of capital.

In reality, specific projects often differ from the average investment made by the firm. Consider General Electric Company (GE), a large firm with many divisions that operate in completely different lines of business. Projects in GE's health-care division are likely to have different market risk than projects in its air transportation equipment division or in its appliance division. Projects may also vary in terms of the amount of leverage they will support—for example, acquisitions of real estate or capital equipment are often highly levered, while investments in intellectual property are not. We will study the effect of leverage on the cost of capital when we cover the leverage decision in Chapter 16. In this section, we show how to calculate the cost of capital for the project's cash flows when a project's risk differs from the firm's overall risk.

Cost of Capital for a New Acquisition

We begin by explaining how to calculate the cost of capital of a project with market risk that is different from the risk for the rest of the firm. Suppose DuPont wants to enter the forest products business. To do so, it is considering acquiring Weyerhaeuser, a company that is focused on timber, paper, and other forest products. Weyerhaeuser faces different market risks than DuPont does in its chemicals business. What cost of capital should DuPont use to value a possible acquisition of Weyerhaeuser?

Because the risks are different, DuPont's WACC would be inappropriate for valuing Weyerhaeuser. Instead, DuPont should calculate and use Weyerhaeuser's WACC when assessing the acquisition. In Figure 13.3, we find the following information for Weyerhaeuser, where the WACC is based on a tax rate of 35%:

	Beta	Cost of Equity	Cost of Debt	% Equity	% Debt	WACC
Weyerhaeuser	1.45	11.7%	6.1%	63%	37%	8.8%

Assuming that DuPont will find it appropriate to continue to finance Weyerhaeuser with the same mix of debt and equity after it buys Weyerhaeuser,[7] we can use Weyerhaeuser's WACC as the cost of capital for acquiring it. Thus, DuPont would use a cost of capital of 8.8% to value Weyerhaeuser for purchase.

Divisional Costs of Capital

Now assume DuPont makes a different decision: It decides to create a forest products division internally, rather than buying Weyerhaeuser. What should the cost of capital for the new division be? If DuPont plans to finance the division with the same proportion of debt as is used by Weyerhaeuser, then DuPont would use Weyerhaeuser's WACC as the WACC for its new division. Because Weyerhaeuser's WACC is the right cost of capital given the risks of forest products and 37% debt financing, it has to be the right cost of capital for an internally created forest products division that is financed 37% with debt.

In most cases firms with more than one division should not use a single company-wide WACC to evaluate projects. More typically, they perform analyses similar to DuPont's analysis of Weyerhaeuser. Multidivisional firms benchmark their own divisions off of companies that compete with their division and are focused in that single line of business. By performing the same analysis as we did in Figure 13.3, the multidivisional firm can estimate the WACCs of its divisions' competitors—adjusting for different financing if necessary—to estimate the cost of capital for each division.

[7]We consider what to do if DuPont wants to change Weyerhaeuser's financing mix in Chapter 16.

EXAMPLE 13.6
A Project in a New Line of Business

Problem

You are working for Microsoft evaluating the possibility of selling digital video recorders (DVRs). Microsoft's WACC is 8.8% (it is merely a coincidence that Microsoft's WACC is the same as Weyerhauser's). DVRs would be a new line of business for Microsoft, however, so the systematic risk of this business would likely differ from the systematic risk of Microsoft's current business. As a result, the assets of this new business should have a different cost of capital. You need to find the cost of capital for the DVR business. Assuming that the risk-free rate is 3% and the market risk premium is 6%, how would you estimate the cost of capital for this type of investment?

Solution

Plan

The first step is to identify a company operating in Microsoft's targeted line of business. TiVo, Inc., is a well-known marketer of DVRs. In fact, that is all TiVo does. Thus, the cost of capital for TiVo would be a good estimate of the cost of capital for Microsoft's proposed DVR business. Many Web sites are available that provide betas for traded stocks, including http://finance.yahoo.com. Suppose you visit that site and find that the beta of TiVo stock is 1.3. With this beta, the risk-free rate, and the market risk premium, you can use the CAPM to estimate the cost of equity for TiVo. Fortunately for us, TiVo has no debt, so its cost of equity is the same as its cost of capital for its assets.

Execute

Using the CAPM, we have

$$\begin{aligned}\text{TiVo's Cost of Equity} &= \text{Risk-Free Rate} + \text{TiVo's Equity Beta} \times \text{Market Risk Premium}\\ &= 3\% + 1.3 \times 6\% = 10.8\%\end{aligned}$$

Because TiVo has no debt, its WACC is equivalent to its cost of equity.

Evaluate

The correct cost of capital for evaluating a DVR investment opportunity is 10.8%. If we had used the 8.8% cost of capital that is associated with Microsoft's *existing* business, we would have mistakenly used too low of a cost of capital. That could lead us to go ahead with the investment, even if it truly had a negative NPV.

INTERVIEW WITH Shelagh Glaser



Shelagh Glaser is the Finance Director for Intel's Mobility Group, which provides solutions for the mobile computing market. Prior to that she was Group Controller for Sales & Marketing and co-Group Controller for Digital Enterprise Group.

QUESTION: ***Does Intel set the discount rate at the corporate or project level?***

ANSWER: We typically set the discount rate at the corporate level. As a company, Intel makes a broad set of products that sell into similar markets, so one hurdle rate makes sense for our core business. To justify an investment, every project has to earn or exceed that level of return for our shareholders.

We may use a different discount rate for mergers and acquisitions. For example, recently we've done more software acquisitions. That industry is very different from semiconductors and has different risk factors, so we take those considerations into account to set the hurdle rate.

QUESTION: ***How does Intel compute the cost of capital for new investment opportunities?***

ANSWER: We reexamine our weighted average cost of capital (WACC) each year to see that we have the right inputs and if any have changed: What is the current market risk premium? Are we using the right risk-free rate? How should we weight historical

data? We use the CAPM to determine beta but consider whether to continue using the five-year weekly beta or to change to a daily or monthly beta in the calculation. We also look at the latest studies from academia on what the inputs should be.

Once we have estimated the WACC, we think about the appropriate hurdle rate for our circumstances. We have not changed the hurdle rate in recent years—even though the WACC may have changed—and continue to use a hurdle rate that is above the WACC. This higher hurdle rate reflects our ability to time our investments and helps us choose projects that maximize our expected returns. Intel has more projects with returns above our hurdle than we can invest in, so assessing opportunity cost is a significant aspect of decision making. We may invest in some projects with NPVs below the hurdle rate if they are of strategic importance.

QUESTION: ***How do project-specific considerations affect Intel's cost of capital calculation?***

ANSWER: When deciding whether to invest billions in wafer fabrication plants, Intel considers both the physical plant and the product line. We calculate the margin we would need from the product and develop a comprehensive set of metrics to justify the large capital investment. Typically we use our standard hurdle rate and also look at risk factors and the timing of the product launch. Intel's business is driven by the economics of Moore's Law, allowing us to double transistors every two years. Each generation of new technology creates cost reductions that enable new fabrication plants to clear the hurdle rate. These plants produce our leading-edge product, which earn our highest margins. To get a premium price, our customers require a product that takes advantage of the performance and power efficiency of the latest technology.

QUESTION: ***Did the 2008 financial crisis affect how you evaluate investment opportunities?***

ANSWER: In 2008, the market was very depressed. We continued making R&D investments and reduced spending in other areas, such as marketing and short-term promotion. Cutting R&D would have left us with a gap in our product line in several years, because product development cycles typically run four years. In this industry, not keeping the R&D machine flowing, to boost short-term earnings, will harm the company's long-term viability. R&D is critical to our fundamental business model, which may require long-term decisions that negatively impact short-term results.

Intel has a long history of investing to maintain performance leadership during downturns. Intel carries almost no debt. Our capital policy is to maintain sufficient cash for fabrication plants and multiple years of R&D expenses, to avoid depending on capital markets to finance ongoing operations. A painful experience in the 1980s, when Intel's access to capital markets was curtailed, highlighted how this conservative capital policy pays off. Going into this crisis with no debt and an extremely strong cash position has served us well.

Concept Check

9. When evaluating a project in a new line of business, which assumptions for the WACC method are most likely to be violated?
10. How can you estimate the WACC to be used in a new line of business?

13.6 When Raising External Capital Is Costly

So far, we have assumed that there are no important factors to consider in seeking capital other than taxes. Among other things, this implies that we can raise external capital without any extra costs associated with the capital-raising transaction. As a consequence, we have no reason to treat a project financed with new external funds any differently than a project financed with internal funds (retained earnings).

In reality, issuing new equity or bonds carries a number of costs. These costs include the costs of filing and registering with the Securities and Exchange Commission and the fees charged by investment bankers to place the securities. We will discuss the process for issuing equity and bonds in detail in the next two chapters. Here, we mention it briefly in the context of the cost of capital.

Because of these issuing costs, a project that can be financed from internal funds will be less costly overall than the same project if it were financed with external funds. One approach would be to adjust the costs of equity and debt capital in the WACC to incorporate the issuing costs. A better and far more direct route is to simply treat the issuing costs as what they are—cash outflows that are necessary to the project. We can then incorporate this additional cost as a negative cash flow in the NPV analysis.

9. How should you value a project in a line of business with risk that is different than the average risk of your firm's projects?
10. What is the right way to adjust for the costs of raising external financing?

Problems

All problems in this chapter are available in MyFinanceLab.

A First Look at the Weighted Average Cost of Capital

1. MV Corporation has debt with market value of $100 million, common equity with a book value of $100 million, and preferred stock worth $20 million outstanding. Its common equity trades at $50 per share, and the firm has 6 million shares outstanding. What weights should MV Corporation use in its WACC?
2. Andyco, Inc., has the following balance sheet and an equity market-to-book ratio of 1.5. Assuming the market value of debt equals its book value, what weights should it use for its WACC calculation?

Assets	Liabilities and Equity	
1000	Debt	400
	Equity	600

3. Book Co. has 1 million shares of common equity with a par (book) value of $1, retained earnings of $30 million, and its shares have a market value of $50 per share. It also has debt with a par value of $20 million that is trading at 101% of par.
   a. What is the market value of its equity?
   b. What is the market value of its debt?
   c. What weights should it use in computing its WACC?
4. Consider a simple firm that has the following market-value balance sheet:

Assets	Liabilities and Equity	
1000	Debt	400
	Equity	600

   Next year, there are two possible values for its assets, each equally likely: $1200 and $960. Its debt will be due with 5% interest. Because all of the cash flows from the assets must go to either the debt or the equity, if you hold a portfolio of the debt and equity in the same proportions as the firm's capital structure, your portfolio should earn exactly the expected return on the firm's assets. Show that a portfolio invested 40% in the firm's debt and 60% in its equity will have the same expected return as the assets of the firm. That is, show that the firm's pre-tax WACC is the same as the expected return on its assets.

The Firm's Costs of Debt and Equity Capital

5. Avicorp has a $10 million debt issue outstanding, with a 6% coupon rate. The debt has semi-annual coupons, the next coupon is due in six months, and the debt matures in five years. It is currently priced at 95% of par value.
   a. What is Avicorp's pre-tax cost of debt?
   b. If Avicorp faces a 40% tax rate, what is its after-tax cost of debt?

6. Laurel, Inc., has debt outstanding with a coupon rate of 6% and a yield to maturity of 7%. Its tax rate is 35%. What is Laurel's effective (after-tax) cost of debt?

7. Dewyco has preferred stock trading at $50 per share. The next preferred dividend of $4 is due in one year. What is Dewyco's cost of capital for preferred stock?

8. Steady Company's stock has a beta of 0.20. If the risk-free rate is 6% and the market risk premium is 7%, what is an estimate of Steady Company's cost of equity?

9. Wild Swings Inc.'s stock has a beta of 2.5. Given the information in Problem 8, what is an estimate of Wild Swings' cost of equity?

10. HighGrowth Company has a stock price of $20. The firm will pay a dividend next year of $1, and its dividend is expected to grow at a rate of 4% per year thereafter. What is your estimate of HighGrowth's cost of equity capital?

11. Slow 'n Steady, Inc., has a stock price of $30, will pay a dividend next year of $3, and has expected dividend growth of 1% per year. What is your estimate of Slow 'n Steady's cost of equity capital?

12. Mackenzie Company has a price of $36 and will issue a dividend of $2 next year. It has a beta of 1.2, the risk-free rate is 5.5%, and it estimates the market risk premium to be 5%.
    a. Estimate the equity cost of capital for Mackenzie.
    b. Under the CGDM, at what rate do you need to expect Mackenzie's dividends to grow to get the same equity cost of capital as in part (a)?

A Second Look at the Weighted Average Cost of Capital

13. CoffeeCarts has a cost of equity of 15%, has an effective cost of debt of 4%, and is financed 70% with equity and 30% with debt. What is this firm's WACC?

14. Pfd Company has debt with a yield to maturity of 7%, a cost of equity of 13%, and a cost of preferred stock of 9%. The market values of its debt, preferred stock, and equity are $10 million, $3 million, and $15 million, respectively, and its tax rate is 40%. What is this firm's WACC?

15. Growth Company's current share price is $20 and it is expected to pay a $1 dividend per share next year. After that, the firm's dividends are expected to grow at a rate of 4% per year.
    a. What is an estimate of Growth Company's cost of equity?
    b. Growth Company also has preferred stock outstanding that pays a $2 per share fixed dividend. If this stock is currently priced at $28, what is Growth Company's cost of preferred stock?
    c. Growth Company has existing debt issued three years ago with a coupon rate of 6%. The firm just issued new debt at par with a coupon rate of 6.5%. What is Growth Company's pre-tax cost of debt?
    d. Growth Company has 5 million common shares outstanding and 1 million preferred shares outstanding, and its equity has a total book value of $50 million. Its liabilities have a market value of $20 million. If Growth Company's common and preferred shares are priced as in parts (a) and (b), what is the market value of Growth Company's assets?
    e. Growth Company faces a 35% tax rate. Given the information in parts (a)–(d), and your answers to those problems, what is Growth Company's WACC?

Using the WACC to Value a Project

16. A retail coffee company is planning to open 100 new coffee outlets that are expected to generate, in total, $15 million in free cash flows per year, with a growth rate of 3% in perpetuity. If the coffee company's WACC is 10%, what is the NPV of this expansion?

17. RiverRocks, Inc., is considering a project with the following projected free cash flows:

0	1	2	3	4
–50	10	20	20	15

The firm believes that, given the risk of this project, the WACC method is the appropriate approach to valuing the project. RiverRocks' WACC is 12%. Should it take on this project? Why or why not?

Project-Based Costs of Capital

18. RiverRocks (whose WACC is 12%) is considering an acquisition of Raft Adventures (whose WACC is 15%). What is the appropriate discount rate for RiverRocks to use to evaluate the acquisition? Why?

19. RiverRocks' purchase of Raft Adventures (from Problem 18) will cost $100 million, but will generate cash flows that start at $15 million in one year and then grow at 4% per year forever. What is the NPV of the acquisition?

20. CoffeeStop primarily sells coffee. It recently introduced a premium coffee-flavored liquor. Suppose the firm faces a tax rate of 35% and collects the following information. If it plans to finance 11% of the new liquor-focused division with debt and the rest with equity, what WACC should it use for its liquor division? Assume a cost of debt of 4.8%, a risk-free rate of 3%, and a risk premium of 6%.

	Beta	% Equity	% Debt
CoffeeStop	0.61	96%	4%
BF Liquors	0.26	89%	11%

21. Your company has two divisions: One division sells software and the other division sells computers through a direct sales channel, primarily taking orders over the Internet. You have decided that Dell Computer is very similar to your computer division, in terms of both risk and financing. You go online and find the following information: Dell's beta is 1.21, the risk-free rate is 4.5%, its market value of equity is $67 billion, and it has $700 million worth of debt with a yield to maturity of 6%. Your tax rate is 35% and you use a market risk premium of 5% in your WACC estimates.

a. What is an estimate of the WACC for your computer sales division?
b. If your overall company WACC is 12% and the computer sales division represents 40% of the value of your firm, what is an estimate of the WACC for your software division?

When Raising External Capital Is Costly

22. RiverRocks realizes that it will have to raise the financing for the acquisition of Raft Adventures (described in Problem 19) by issuing new debt and equity. The firm

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Financial Markets And Corporate Strategy{Mark Grinblatt, She

- Effect of leverage on comparison-firm betas; implementing the RADR formula: Ch. 11, §11.3–11.4 (p. 379-

The Tracking Portfolio Method Is Implicit in the Risk-Adjusted Discount Rate Method

To understand the relation between the risk-adjusted discount rate method and tracking, assume that the CAPM is applied to value the Hilton casino cash flow considered in the last section. To do this, we discount the expected cash flow, assumed to be $11.3 million, at the discount rate implied by the CAPM.

The average of the betas of the traded equity of a group of comparison casinos is estimated to be .5. Thus, the Hilton casino cash flow is tracked by a portfolio that is invested

50 percent in the market portfolio and
50 percent in the risk-free asset.

If the expected return on the market is 20 percent and the risk-free rate is 6 percent, the appropriate discount rate is

$$.06 + .5(.20 - .06) = .13.$$

Hence, the present value of the expected future cash flow is

$$\$11.3 \text{ million}/1.13 = \$10 \text{ million}.$$

The 13 percent used to discount the $11.3 million expected future cash flow from the Hilton casino is the expected return of the tracking portfolio. Hence, if one buys enough of the tracking portfolio to have an expected future value of $11.3 million, the tracking portfolio will cost $10 million today. This tracking portfolio consists of $5 million in the market portfolio (50 percent) and $5 million in the risk-free asset (50 percent). Thus, the discounted (or present) value of the expected future cash flow is nothing more than the cost of acquiring the tracking portfolio's cash flows ($10 million) while the CAPM beta (.5) represents the proportion of the tracking portfolio allocated to the market portfolio, which is assumed to be the tangency portfolio.

11.3 The Effect of Leverage on Comparisons

Since the risk-adjusted discount rate method uses the traded stocks of comparison firms to estimate the betas of projects, it is important that the beta risk of the equity of the comparison firms be truly comparable. However, it is not enough to merely use firms in the same line of business as the project. Since the amount of debt financing a firm takes on can dramatically affect equity betas, it is necessary to adjust for differing amounts of debt financing in order to make appropriate beta risk comparisons.

In this section, we explore the effect of debt financing, also known as leverage, on the beta and standard deviation of a firm's equity. We begin by performing a financing experiment in which a firm issues risk-free debt, but does not alter the operations of the firm. Hence, the proceeds of the debt issue are used to retire outstanding equity while sales, EBIT, marketing, production, and so forth remain the same.

The Balance Sheet for an All-Equity-Financed Firm

Begin with an all-equity-financed firm. Exhibit 11.2 illustrates the familiar accounting balance sheet in T-account form. The left-hand side reflects the assets of the firm and the right-hand side reflects the debt (liabilities) and equity. In contrast to an

EXHIBIT 11.2 Balance Sheet for an All-Equity-Financed Firm

Assets	Liabilities and Equity	
A	Debt	0 (i.e., $D = 0$)
	Equity	E

EXHIBIT 11.3 Balance Sheet for a Firm with Leverage

Assets	Liabilities and Equity	
A	Debt	D ($D > 0$)
	Equity	E

accounting balance sheet, which reflects book values, the T-account in Exhibit 11.2 describes market values. Hence, A represents the market value of the firm's assets and E the market value of the firm's equity. In the absence of leverage, A equals E because the two sides of the balance sheet add up to the same number; that is, they *balance*. This identity implies that the *risk* of A and E in the all equity-financed firm must be the same, whether risk is measured as standard deviation (in which case $\sigma_A = \sigma_E$), or as beta risk (in which case $\beta_A = \beta_E$).

The Balance Sheet for a Firm Partially Financed with Debt

Now, let us introduce debt. In this case, the balance sheet looks like the T-account depicted in Exhibit 11.3. Because the two sides of the balance sheet must balance,

$$A = D + E.$$

In other words, the market value of the assets of the firm must equal the sum of the market values of the debt and equity. This makes sense. All of the future cash flows from the assets of the firm must at some point flow to the cash flow claimants. An investor who buys up all the debt and equity of the firm has a right to all the cash flows produced by the assets. Hence, the sum of the market values of the debt and equity must be the same as the market value of the firm's assets.

The Right-Hand Side of the Balance Sheet as a Portfolio

Now, view the right-hand side of the balance sheet in Exhibit 11.3 as a portfolio of investments, with the weight on debt as $D/(D + E)$, and the weight on equity as $E/(D + E)$. To understand the relation between asset risk and equity risk, we now use the portfolio mathematics developed in Chapters 4 and 5 to analyze risk measures of each of the components, A, D, and E. With risk-free debt, all of the risk is born by the equity holders. In this case, σ_D and β_D are both zero for risk-free debt, implying by the portfolio formulas developed earlier

$$\sigma_A = \left(\frac{D}{D + E}\right)0 + \left(\frac{E}{D + E}\right)\sigma_E$$

and

$$\beta_A = \left(\frac{D}{D+E}\right)0 + \left(\frac{E}{D+E}\right)\beta_E \tag{11.2a}$$

Inverting these equations to place the equity risk measures on the left-hand side results in

$$\sigma_E = \left(1 + \frac{D}{E}\right)\sigma_A$$

and

$$\beta_E = \left(1 + \frac{D}{E}\right)\beta_A \tag{11.2b}$$

Recall that the experiment performed is one in which altering the mix of financing between debt and equity does not change A, the operating assets of the firm. Hence, equation (11.2b) and the equation above it imply:

Result 11.3 Increasing the firm's debt (raising D and reducing E) increases the (beta and standard deviation) risk per dollar of equity investment. It will increase linearly in the D/E ratio if the debt is risk free.

Result 11.3 stems from the equivalence of the assets of the firm to a portfolio of the firm's debt and equity, with respective positive portfolio weights of $D/(D + E)$ and $E/(D + E)$. Thus

$$\tilde{r}_A = \left(\frac{D}{D+E}\right)\tilde{r}_D + \left(\frac{E}{D+E}\right)\tilde{r}_E \tag{11.3a}$$

This also means that each dollar of equity can be thought of as a portfolio-weighted average of the firm's assets and debt, with a long position in the assets and a short position in debt. Here the respective portfolio weights are $(1 + D/E)$ on the assets and $-D/E$ on the debt; that is, upon rearranging equation (11.3a) to place $\tilde{r}_E$ on the left-hand side, we obtain

$$\tilde{r}_E = \left(1 + \frac{D}{E}\right)\tilde{r}_A - \left(\frac{D}{E}\right)\tilde{r}_D \tag{11.3b}$$

Distinguishing Risk-Free Debt from Default-Free Debt

In the previous subsection, we assumed that the debt of the firm was risk-free. Risk-free debt is necessarily default-free debt but the reverse is not necessarily true. There are two sources of risk associated with debt. One is interest rate risk, which is associated with general changes in long-term interest rates, and the other is credit risk, which is associated with the possibility of default. All long-term debt, even if it is default-free, will see its value move inversely with long-term interest rates (bond yields), as Chapter 2 observed. Asset values, however, also tend to move inversely with long-term interest rates, which means that long-term debt, even if it is default-free, tends to have a positive beta (typically, around 0.2 when computed with respect to broad-based stock portfolios like the S&P 500). Risk-free debt is necessarily short-term, because it cannot have its value altered by changes in long-term interest rates. Short-term default-free debt tends to have a beta near zero and, for our purposes, can also be regarded as risk-free debt.

The main goal of this section is to analyze the effect of debt on beta and expected return. In particular, and as the next section illustrates, it is useful to isolate the impact of credit risk on bond and equity betas for high-leverage ratios. To avoid confounding the effect of interest rate risk and credit risk, we have deliberately chosen to analyze risk-free debt, which has a beta of zero.

It is straightforward to modify the analysis of the effect of leverage when debt has a positive beta. Again, viewing the firm as a portfolio, equation (11.2a) would generalize to

$$\beta_A = \left(\frac{D}{D+E}\right)\beta_D + \left(\frac{E}{D+E}\right)\beta_E$$

implying that equation (11.2b) generalizes to

$$\beta_E = \left(1 + \frac{D}{E}\right)\beta_A - \left(\frac{D}{E}\right)\beta_D$$

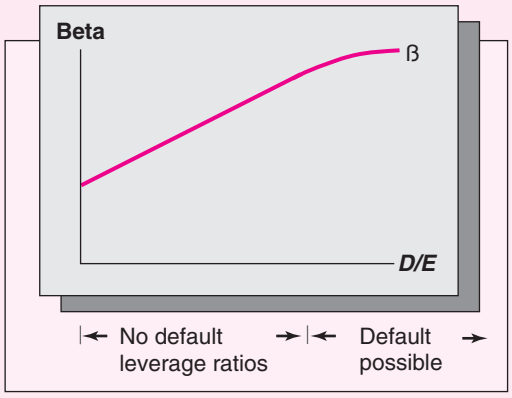
Equation (11.3b) remains unchanged.

Graphs and Numerical Illustrations of the Effect of Debt on Risk

Exhibit 11.4 illustrates the beta of equity as a function of the **leverage ratio**, D/E. When the debt is default free, the firm's equity holders bear all of the risk from swings in asset values. By equation (11.2b), the beta of equity as a function of D/E should graph as a straight line in this case. When debt default is possible, debt holders bear part of the risk, implying that equity holders bear less risk. As a result, the beta risk of equity becomes a curved line.

To explore Result 11.3 further, we illustrate numerically why equity holder (beta or standard deviation) risk per dollar invested increases with leverage. First, note that if a firm's total cash flows are independent of its **capital structure**—that is, its mix of debt and equity financing—the total risk borne by the aggregation of the investors of a firm, debt holders plus equity holders, does not change when the firm changes its

EXHIBIT 11.4 Equity Beta as a Function of the Firm's Leverage Ratio



capital structure. Thus, an all-equity financed firm with assets that change from a value of $100 million in 2001, to $110 million in 2002, and then to $88 million in 2003, has equity that experiences the same value changes. Per dollar invested, such equity holders experience value changes amounting to a 10 percent increase from 2001 to 2002 and a 20 percent decrease from 2002 to 2003.

However, if the same assets had been financed with $75 million in risk-free debt, implying an initial debt-to-equity ratio of 3, the equity jumps from $25 million (= $100 million − $75 million) in 2001 to $35 million (= $110 million − $75 million) in 2002. The equity increase of 40 percent ($25 million to $35 million) is four times larger per dollar invested than the 10 percent equity increase of the all-equity firm ($100 million to $110 million). Thus, leverage benefits the firm's stockholders when asset values appreciate but, from 2001 to 2002, when the assets drop from $110 million to $88 million, a 20 percent decrease, the equity of the levered firm goes from $35 million in 2001 to $13 million in 2002, a 63 percent decrease.[8]

The Cost of Equity, Cost of Debt, and Cost of Capital as a Function of the Leverage Ratio. The **cost of equity** for a firm is the expected return required by investors to induce them to hold the equity. Since the firm's equity beta increases linearly as the amount of default-free debt financing increases, it should not be surprising that the firm's cost of equity capital also increases linearly as a function of the leverage ratio, D/E. Specifically, taking expectations of equation (11.3b) and grouping terms yields:

Result 11.4 The cost of equity

$$\bar{r}_E = \bar{r}_A + (D/E)\,(\bar{r}_A - \bar{r}_D)$$

increases as the firm's leverage ratio D/E increases. It will increase linearly in the ratio D/E if the debt is default free and if $\bar{r}_A$, the expected return of the firm's assets, does not change as the leverage ratio increases.[9]

In Result 11.4, the expected (and actual) return of a firm's debt $\bar{r}_D$, its **cost of debt**, equals r_f for moderate leverage ratios. As long as the firm's debt is default free, the cost of equity capital increases linearly in the firm's leverage ratio. However, when firms take on extreme amounts of debt, $\bar{r}_D$, the expected return on risky debt, rises as D/E increases. Since debt holders share part of the risk in this case, the cost of equity increases more slowly as D/E rises than it does with default-free debt. Again, the risks of the assets are shared by both debt holders and equity holders when default is possible, but they are borne only by equity holders when default is not possible.

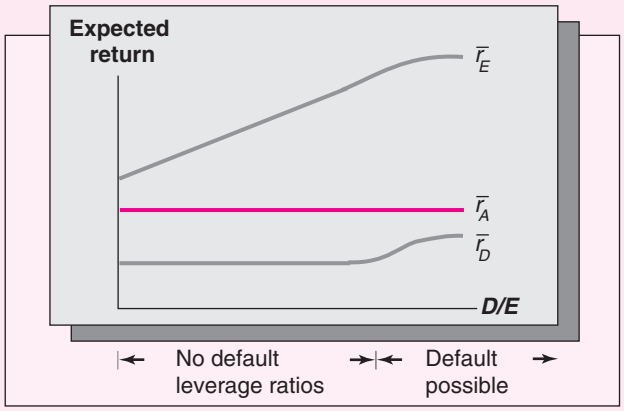
Exhibit 11.5 summarizes the findings of this section. It plots the cost of equity $\bar{r}_E$, the cost of debt $\bar{r}_D$, and the cost of capital $\bar{r}_A$, as functions of the leverage ratio D/E.

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[8]The 63 percent decrease for the leveraged firm is not quite four times the 20 percent decrease experienced by the assets because the debt-equity ratio is smaller than three to one in 2002.

[9]Result 11.4 also applies when expected returns are determined by factor betas in a multifactor setting. In this vein, note that interest rate risk, discussed earlier in this chapter, may make the beta of default-free fixed-rate debt non-zero, but only default risk can generate the curvature seen in Exhibits 11.4 and 11.5.

EXHIBIT 11.5 Cost of Debt, Equity, and Capital as a Function of D/E



11.4 Implementing the Risk-Adjusted Discount Rate Formula with Comparison Firms

Suppose that the CAPM is correct and that HSCC (from Example 11.1) has operations similar to those of Dell Computer, a stock traded on Nasdaq. HSCC is a wholly owned subsidiary of Novel, Inc., so it is not traded, but Dell is. In this case, it might be possible to use Dell's beta—computed by regressing Dell's past stock returns on the returns of a market proxy—to determine beta and, thus, the expected return for HSCC. However, one needs to be cautious about drawing this connection because the way in which a firm is financed can affect its equity beta, as indicated in Section 11.3. It is important not only for Dell Computer and HSCC to have similar lines of business but also for both to have similar leverage ratios.

The CAPM, the Comparison Method, and Adjusting for Leverage

If an acquisition (or project) and its comparison firm(s) are financed differently, it may be possible to adjust the comparison firm's beta for the difference in leverage ratios. However, making this type of adjustment can be tricky, especially when one takes corporate taxes into account. We will discuss beta adjustments in the absence of taxes below, and will examine how these adjustments are affected by taxes in Chapter 13. (In the absence of taxes, the *weighted average cost of capital* and the *unlevered cost of capital,* terms commonly used in the financial services industry, are identical to the expected return on assets, $\bar{r}_A$.)

An Illustration of the Necessary Leverage Adjustment without Taxes. Example 11.2 illustrates the simpler task of how to adjust for leverage differences in the absence of taxes.

Example 11.2: Using the Comparison Approach to Obtain Beta and $\bar{r}$

This example is based on a Harvard case about the Marriott Corporation,[10] although the data here are fictitious. In this case, Marriott has identified three comparison firms for its

[10]Ruback (1992).

restaurant division. For these comparison firms, the table below provides CAPM-based equity beta estimates (β_E), book values of debt (D), and market values of equity (E).

	β_E	D ($billions)	E ($billions)
Church's Chicken	.75	.004	.096
McDonald's	1.00	2.300	7.700
Wendy's	1.08	.210	.790

Assume that the risk-free rate is 4 percent per year, the risk premium on the market portfolio is 8.4 percent per year, the CAPM holds, the debt of the comparison firms is risk free, and all three firms provide equally good comparisons for Marriott's restaurant division. Estimate the cost of capital for Marriott's restaurant division.

Answer: Using equation (11.2a), $\beta_A = \dfrac{E}{D+E}\beta_E$, we first find the three firms' asset betas:

	β_A
Church's Chicken	$.72 = \dfrac{.096}{.100} \times .75$
McDonald's	$.77 = \dfrac{7.7}{10} \times 1.00$
Wendy's	$.85 = \dfrac{.79}{1.0} \times 1.08$
Marriott (average of above)	$.78 = \dfrac{.72 + .77 + .85}{3}$

Applying the CAPM risk-expected return equation, using the .78 estimate of Marriott's restaurant asset beta gives the restaurant cost of capital, 10.55 percent per year:

$$.1055 = .04 + .78(.084)$$

In Example 11.2, β_E, the equity beta, determines the cost of equity capital. Recall from the last section that if β_E is positive, this beta and the associated expected return on the equity increase as the leverage of the firm increases. The cash flows that are discounted by the risk-adjusted discount rate method are the cash flows from the project's assets, which do not have debt interest payments subtracted from them. Hence, it is inappropriate to discount these cash flows at a rate used for discounting the cash flows that belong to the leveraged equity of comparison firms.

By multiplying the betas by $E/(D + E)$, Example 11.2 identifies the beta of the portfolio of debt and equity of the comparison firms (with $\beta_D = 0$). Since assets equal debt plus equity, this portfolio beta is indeed the asset beta that generates an appropriate discount rate for the cash flows of the assets.

As an alternative to the process of unlevering the betas described in Example 11.2, one can compute the CAPM-based expected returns of the equity of the comparison firms. Use the equation

$$\bar{r}_A = \left(\frac{D}{D+E}\right)\bar{r}_D + \left(\frac{E}{D+E}\right)\bar{r}_E$$

(obtained from the expected values of both sides of (equation 11.3a)) to unlever the equity expected returns and obtain the expected asset return of the comparable firms.

Weighting the Betas of Comparison Firms. Note that Example 11.2 averaged the betas of the three comparison firms to estimate the beta of Marriott's restaurant assets. This averaging is appropriate if each comparison firm provides an equally valid estimate of the Marriott restaurant asset beta. However, if some firms provide better comparisons than others, their betas should be weighted more than those of the less closely matched firms. For example, if Church's Chicken was a less appropriate match than McDonald's or Wendy's, we might give the .72 beta of Church's Chicken a lower weight than the betas of McDonald's or Wendy's.

Obtaining a Cost of Capital from the Arbitrage Pricing Theory (APT)

Earlier, we learned that when the tangency portfolio is the market portfolio, the cost of capital—that is, the discount rate in the denominator of the present value formula, equation (11.1)—is obtained from the Capital Asset Pricing Model. An alternative to the CAPM is the Arbitrage Pricing Theory (APT), developed in Chapter 6, which is the correct theory to use when a combination of factor portfolios, instead of the market portfolio, is the tangency portfolio.

The Multifactor APT Version of the Risk-Adjusted Discount Rate Formula. When computing costs of capital using the expected returns generated by the APT, the present value of the project's future cash flow is

$$PV = \frac{E(\tilde{C})}{1 + r_f + \lambda_1\beta_1 + \lambda_2\beta_2 + \cdots + \lambda_K\beta_K} \tag{11.4}$$

The project's net present value is computed by subtracting the project's initial cost from this present value.

The discount rates provided by the APT generally differ from those of the CAPM. Thus, they can generate different capital allocation decisions. The Marriott Corporation, for example, has an APT-based cost of capital of 9.3 percent, resulting in a denominator of 1.093 for equation (11.4) and hence equation (11.1) as well.[11] However, Marriott's CAPM-based cost of capital is 10 percent, resulting in a denominator of 1.1 for equation (11.1). If the APT is correct and the CAPM is incorrect, Marriott would be missing out on some good projects by using the higher discount rate from the CAPM. As the following case study illustrates, such differences in the cost of capital between the two models are not uncommon.

Arbitrage Pricing Theory versus Capital Asset Pricing Model

Alcar's APT!, a consulting firm, provides its clients with the costs of equity capital and costs of capital for a variety of firms, using both the CAPM and the APT. The Alcar version of the APT is based on a five-factor model, where the five prespecified factors are changes in:

[11] Based on third-quarter 1995 data for the Marriott Corporation.

EXHIBIT 11.6 Cost of Equity Capital

	CAPM		Arbitrage Pricing Theory (APT)					
				Premiums from Sensitivity to Five Factors				
Firm	*Equity Beta*	*Equity Expected Returns (%)*	*Equity Expected Returns (%)*	*SINF*	*LINF*	*INT*	*PREM*	*GDP*
Coca Cola	1.03	11.42	12.61	0.83	1.25	1.39	0.95	1.22
Con Edison NY	0.60	9.57	10.41	0.52	0.76	0.86	0.56	0.74
CSX Corp.	1.24	12.34	11.89	0.57	1.13	1.24	0.82	1.15
Fed Nat Mtg Assn	1.52	13.56	11.79	0.39	1.16	1.31	0.69	1.26
Microsoft Corp.	1.05	11.53	8.95	−0.03	0.54	0.61	0.22	0.64
Northrop	0.98	11.19	8.54	−0.07	0.45	0.49	0.17	0.52

EXHIBIT 11.7 CAPM and APT Costs of Capital with Leverage Ratios (D/E) for Six Firms

Firm	*Debt to Equity Ratio (%)*	*CAPM Cost of Capital (%)*	*APT Cost of Capital (%)*	*Difference between APT and CAPM Cost of Capital (%)*
Coca-Cola	6.36	11.02	12.14	1.12
Consolidated Edison	46.90	7.93	8.51	0.58
CSX Corp.	51.31	9.85	9.55	−0.30
Federal National Mortgage Association	778.74	5.65	5.45	−0.20
Microsoft Corp.	0.00	11.53	8.95	−2.58
Northrop	27.50	10.12	8.04	−2.08

- Short-term inflation (SINF).
- Long-term inflation (LINF).
- The level of short-term interest rates (INT).
- The premium for default risk (PREM).
- The monthly Gross Domestic Product (GDP).

Exhibit 11.6 presents Alcar's equity expected returns from both the CAPM and APT, as well as the CAPM beta, for six well-known firms, as of the third quarter of 1995.

For each row, the sum of the numbers in the five right-hand columns in Exhibit 11.6 represent the APT risk premiums. APT equity expected returns (the costs of equity capital) are computed as the sum of these risk premiums plus the September 7, 1995 risk-free rate of 6.98 percent.

With adjustments for risky debt and taxes, these numbers translate into Exhibit 11.7's comparative costs of capital for the typical existing project of these firms.[12] The difference in the cost of capital computed with the CAPM and APT in Exhibit 11.7 is as large as 2.58 percent per year, as you'll note in the case of Microsoft (2.58% = 11.53% − 8.95%). Many projects that are similar to the existing projects of Microsoft typically have large investments

[12]Adjustments for taxes are discussed in Chapter 13.

in the early years—primarily wages for programmers and expenses for advertising and promotion—and substantial revenues from sales of software which are not likely to occur until much later, perhaps 5 to 15 years after the initial investment. Therefore, adoption decisions about any prospective project that resembles Microsoft's existing collection of projects will be greatly affected by whether one selects the CAPM or the APT to compute Microsoft's discount rate.

Costs of Capital Computed with Alternatives to CAPM and APT: Dividend Discount Models

The Capital Asset Pricing Model and the Arbitrage Pricing Theory are the two most popular models for determining risk-adjusted discount rates. Although both models are applied in practice, their applications have been criticized because of the difficulties associated with estimating their essential inputs.

Impediments to Using the CAPM and APT. Specifically, the CAPM requires knowledge of not only the covariance (or beta) of the return of an investment with the return of the market portfolio, but also an estimate of the expected return of the market portfolio. The APT requires multiple factor sensitivities and the corresponding expected returns on multiple factor portfolios.

The Dividend Discount Model. A number of financial analysts estimate required rates of return using analysts' forecasts of future earnings with a special case of what Chapter 9 referred to as the *dividend discount model*. This special case, where dividends grow at a constant rate, is sometimes known as the **Gordon Growth Model** because it was first developed by Gordon (1962). According to this model, the equity of a firm with a dividend stream growing at a constant rate can be valued as follows:

$$S_0 = \frac{div_1}{(\bar{r}_E - g)} \tag{11.5a}$$

where

S_0 = the firm's current stock price per share
div_1 = the expected dividend per share one year from now
$\bar{r}_E$ = the market required rate of return of the firm's stock (its cost of equity capital)
g = the expected growth rate of dividends

Equation (11.5a) is an application of the growing perpetuity formula developed in Chapter 9. By rearranging this equation, one sees that the expected rate of return of a stock can be expressed as the sum of the growth rate and the dividend yield

$$\bar{r}_E = g + \frac{div_1}{S_0} \tag{11.5b}$$

Using Analyst Forecasts to Estimate the Expected Dividend Growth Rate. To compute the risk-adjusted discount rate for equity from this equation, only g, the expected rate of growth of the firm's dividends, and div_1/S_0, the firm's dividend yield, need to be estimated.[13] Analysts' forecasts of the growth rate of a firm's earnings

[13] A historical average of the ratio of dividend per share to prior year stock price per share, sometimes over a period of five years, can be used if the coming year's dividend payout is expected to be unusual.

provide one estimate for g. Under the assumption that a firm pays a fixed percentage of its earnings as dividends, the expected growth rate in dividends equals the forecasted growth rate in earnings. This growth rate can then be added to the existing dividend yield to derive the expected return on the firm's stock.[14]

For example, suppose that Value Line, an investment advisory service, forecasts a 12.9 percent rate of growth for IBM's earnings. Adding this to the firm's 0.5 percent dividend yield (as of early 2001) implies an expected rate of return on IBM stock of 13.4 percent. To obtain a cost of capital for an IBM-like project, it is necessary to adjust this 13.4 percent rate of return for debt in IBM's capital structure. For example, in the risk-free debt no taxes case of Section 11.2, one can obtain $\bar{r}_A$ by multiplying $\bar{r}_E$ by $E/(D + E)$.

Using the Plowback Ratio Formula to Estimate the Expected Dividend Growth Rate. An alternative method for estimating g, the growth rate in dividends, employs accounting data. This method estimates the growth rate as

$$g = b \times ROE \tag{11.6}$$

where

b = the **plowback ratio**, the fraction of earnings retained in the firm
ROE = **book return on equity**, that is, earnings divided by last year's (midyear) book equity[15]

The intuition for the **plowback ratio formula**, given in equation (11.6), is that the book return on equity (ROE) represents the rate of growth of capital invested in the firm. When a firm has an ROE of 10 percent, every \$1 invested in the firm returns \$1.00 of equity capital and \$0.10 of earnings next year, or \$1.10. If this \$1.10 is entirely reinvested, it will grow another 10 percent to \$1.21 one year later. However, if 75 percent of the earnings are paid out in the form of dividends, implying a plowback ratio of .25, the capital will only grow at a rate of $(1 - .75)(\$.10)$—that is, at 25 percent of the 10 percent growth rate, or 2.5 percent. In this case, at the end of the first year, 75 percent of \$0.10 would be paid out in dividends, implying that only \$1.025 $[= \$1.10 - .75(\$.10)]$ is left in the firm for reinvestment. This would grow to $\$1.025(1.1)$, but if 75 percent of the amount over \$1.025 (the earnings) is paid out as a dividend, the amount to be reinvested is just $\$1.025(1.1) - .75(\$1.025)(.1) = \$1.025^2$. Thus, paying out a fixed proportion of a company's earnings as dividends slows the growth rate of the funds available for reinvestment. Moreover, since earnings and dividends are a constant proportion of the reinvestment amount, their growth rates will be the same as the growth rate of the funds available for investment in the firm.

Assumptions and Pitfalls of the Dividend Discount Model. The plowback ratio formula, equation (11.6), uses the book return on equity in lieu of the return on new investment, the return that theoretically should be used but which is more difficult to measure accurately. If old assets and new assets have different returns, ROE in the plowback ratio formula should be the *book return of equity for new asset investment*. If the project is a positive *NPV* project, the appropriate book return of equity will exceed the project's cost of capital.

[14]Note that using the current dividend yield in equation (11.5b) gives the wrong answer. The formula requires next year's expected dividend in the numerator. Multiplying the current dividend per share by $1 + g$ and dividing by the current stock price gives the appropriate dividend yield estimate.

[15]Alternatively, it is possible to use forecasts of next year's earnings divided by this year's (midyear) book equity.

The implicit assumptions of the dividend discount model's estimate of the cost of capital are that:

- The earnings growth forecasts, whether from analysts or equation (11.6), are unbiased; that is, they do not tend to systematically underestimate or overestimate the earnings growth rate.
- The earnings growth forecasts are based on the same information that investors use to value the firm's stock.
- The firm's earnings and dividends grow at the same constant rate, forever.

To the extent that these assumptions are valid, the dividend discount model may provide a better estimate of the expected rate of return on a firm's stock or project than either the CAPM or the APT because it does not require estimates of beta or estimates of the expected return of the market portfolio or of factor portfolios. However, these assumptions, particularly that of a constant growth rate, are stringent and may not apply to many of the firms or projects that an analyst wants to value.

What If No Pure Comparison Firm Exists?

Many firms are large diversified entities that have many lines of business. In this instance, the equity returns of potential comparison firms are distorted by other lines of business and cannot easily be used as comparison firms for projects that represent only a single line of business. Unfortunately, in many situations, there is no appropriate comparison firm with a single line of business. A financial manager in this situation still may be able to obtain an appropriate comparison by forming portfolios of firms that generate a "pure" line of business. The mathematics behind the approach taken in Example 11.3, which illustrates how to create comparison investments in a pure line of business when none initially exists, is similar in spirit to the formation of pure factor portfolios in Chapter 6.

Example 11.3: Finding a Comparison Firm from a Portfolio of Firms

Assume that AOL-Time Warner is interested in acquiring the ABC television network from Disney. It has estimated the expected incremental future cash flows from acquiring ABC and desires an appropriate beta in order to compute a discount rate to value those cash flows. However, the two major networks that are most comparable, NBC and CBS, are owned by General Electric and Viacom—respectively—which have substantial cash flows from other sources. For these comparison firms, the table below presents hypothetical equity betas, debt to asset ratios, and the ratios of the market values of the network assets to all assets:

	β_E	$\frac{D}{D+E}$	$\frac{\text{Network Assets}}{\text{All Assets}} = \frac{N}{A}$
General Electric	1.1	.1	.25
Viacom	1.3	.4	.50

Estimate the appropriate beta for the ABC acquisition. Assume that the debt of each of the two comparison firms is risk free. Also assume that the non-network assets of General Electric and Viacom are substantially similar and thus have the same beta.

Exercises

11.1. A project has an expected cash flow of $1 million one year from now. The standard deviation of this cash flow is $250,000. If the expected return of the market portfolio is 10 percent, the risk-free rate is 5 percent, the standard deviation of the market return is 5 percent, and the correlation between this future cash flow and the return on the market is .5, what is the present value of the cash flow? Assume the CAPM holds. *Hint:* Use the certainty equivalent method.

Exercises 11.2–11.6 make use of the following information.

Assume that Marriott's restaurant division has the following joint distribution with the market return:

Market Scenario	*Probability*	*Market Return (%)*	*Year 1 Restaurant Cash Flow Forecast*
Bad	.25	−15	$40 million
Good	.50	5	$50 million
Great	.25	25	$60 million

Assume also that the CAPM holds.

11.2. Compute the expected year 1 restaurant cash flow for Marriott.

11.3. Find the covariance of the cash flow with the market return and its cash flow beta.

11.4. Assuming that historical data suggests that the market risk premium is 8.4 percent per year and the market standard deviation is 40 percent per year, find the certainty equivalent of the year 1 cash flow. What are the advantages and disadvantages of using such historical data for market inputs as opposed to inputs from a set of scenarios, like those given in the table above exercise 11.2?

11.5. Discount your answer in exercise 11.4 at a risk-free rate of 4 percent per year to obtain the present value.

11.6. Explain why the answer to exercise 11.5 differs from the answer in Example 11.2.

11.7. Start with the risk-adjusted discount rate formula. Derive the certainty equivalent formula by rearranging terms and noting that $b = \beta \times PV$.

11.8. In Section 11.3's illustration, asset values increased 10 percent from 2001 to 2002, from $100 million to $110 million.

a. Compute the percentage increase in the value of equity if the firm is financed with $50 million in debt.

b. Compute the leverage ratio of this firm in 2002.

11.9. Explain intuitively why the certainty equivalent of a cash flow with a negative beta exceeds the cash flow's expected value.

Exercises 11.10–11.14 make use of the following data.

In 1989, General Motors (GM) was evaluating the acquisition of Hughes Aircraft Corporation. Recognizing that the appropriate discount rate for the projected cash flows of Hughes was different than its own cost of capital, GM assumed that Hughes had approximately the same risk as Lockheed or Northrop, which had low-risk defense contracts and products that were similar to Hughes. Specifically, assume the following inputs:

Comparison	β_E	*D/E*
GM	1.20	.40
Lockheed	0.90	.90
Northrop	0.85	.70

Target $\frac{D}{E}$ for Hughes's acquisition = 1

Hughes's expected cash flow next year = $300 million

Growth rate of Hughes's cash flows = 5 percent per year

Appropriate discount rate on debt (riskless rate) = 8 percent

Expected return of the tangency portfolio = 14 percent

11.10. Analyze the Hughes acquisition (which took place) by first computing the betas of the comparison firms, Lockheed and Northrop, as if they were all equity financed. Assume no taxes.

11.11. Compute the beta of the assets of the Hughes acquisition, assuming no taxes, by taking the average of the asset betas of Lockheed and Northrop.

11.12. Compute the cost of capital for the Hughes acquisition, assuming no taxes.

11.13. Compute the value of Hughes with the cost of capital estimated in exercise 11.12.

Higgins

Jennifer L. Koski_ Robert C. Higgins_ Todd Mitton - Analysis

- Cost of capital: WACC, CAPM cost of equity, market weights: Ch. 8, §8.4 (p. 294–304)
- Enterprise vs equity perspective: Ch. 8, §8.6.1 (p. 307–310)
- DCF valuation: free cash flow and terminal value: Ch. 9, §9.2 (p. 341–348)

Note how the risk-adjusted discount rate reduces the investment's appeal. If the investment were riskless, its NPV at a 5 percent discount rate would be $5.4 million, but because a higher risk-adjusted rate is deemed appropriate, NPV falls by over $4 million. In essence, management requires an inducement of at least this amount before it is willing to make the investment.

A virtue of risk-adjusted discount rates is that most executives have at least a rough idea of how an investment's required rate of return should vary with risk. Stated differently, they have a basic idea of the position of the market line in Figure 8.1. For instance, they know from the historical data in Table 5.1 of Chapter 5 that over many years, common stocks have yielded an average annual return about 6.4 percentage points higher than the return on government bonds. If the present return on government bonds is 2 percent, it is plausible to expect an investment that is about as risky as common stocks to yield a return of about 8.4 percent. Similarly, executives know that an investment promising a return of 40 percent is attractive unless its risk is extraordinarily high. Granted, such reasoning is imprecise; nonetheless, it does lend some objectivity to risk assessment.

The Cost of Capital

Now that we have introduced risk-adjusted discount rates and illustrated their use, the remaining challenge is to identify the appropriate rate for a specific investment. Do we just add 7 percentage points to the risk-free rate, or is there a more objective process?

There is a more objective process, and it rests on the notion of the *cost of capital.* When creditors and owners invest in a business, they incur opportunity costs equal to the returns they could have earned on alternative, similar-risk investments. Together these opportunity costs define the minimum rate of return the company must earn on existing assets to meet the expectations of its capital providers. This is the firm's cost of capital. If we can estimate this minimum required rate of return, we have an objectively determined risk-adjusted discount rate suitable for evaluating typical, or average risk, investments undertaken by a firm. Rather than relying on managers' "gut feelings" about investment risk, the cost of capital methodology enables us to look to financial markets for valuable information about the appropriate risk-adjusted discount rate.

Moreover, once we know how to estimate one company's cost of capital, we can use the technique to estimate the risk-adjusted discount rate applicable to a wide variety of project risks. The trick is to reason by analogy as follows. If Project A appears to be about as risky as investments undertaken by Company 1, use Company 1's cost of capital as the required return for Project A, or better yet, use an average of the cost of capital to Company 1 and all its

industry peers. Thus, if a traditional pharmaceutical company is contemplating an investment in the biotech industry, a suitable required rate of return for the decision is the average cost of capital to existing biotech firms. In the following paragraphs, we define the cost of capital more precisely, estimate Polaris's cost of capital, and discuss its use as a risk-adjustment vehicle.

The Cost of Capital Defined

Suppose we want to estimate the cost of capital to XYZ Corporation and we have the following information:

	XYZ Liabilities and Owners' Equity	Opportunity Cost of Capital
Debt	$100	10%
Equity	200	19

We will discuss the origins of the opportunity costs of capital a little later. For now, just assume we know that given alternative investment opportunities, creditors expect to earn at least 10 percent on their loans and shareholders expect to earn at least 19 percent on their ownership of XYZ shares. With this information, we need answer only two simple questions to calculate XYZ's cost of capital:

1. *How much money must XYZ earn annually on existing assets to meet the expectations of creditors and owners?*

The creditors expect a 10 percent return on their $100 loan, or $10. However, because interest payments are tax deductible, the effective after-tax cost to a profitable company with, say, a 30 percent tax rate is only $7. The owners expect 19 percent on their $200 investment, or $38. So in total, XYZ must earn $45 [$\$45 = (1 - 0.3)(10\%)\$100 + (19\%)\$200$].

2. *What rate of return must the company earn on existing assets to meet the expectations of creditors and owners?*

A total of $300 is invested in XYZ on which the company must earn $45, so the required rate of return is 15 percent ($45/$300). This is XYZ's cost of capital.

Let's repeat the preceding reasoning using symbols. The money XYZ must earn annually on existing capital is

$$(1 - t)K_D D + K_E E$$

where t is the tax rate, K_D is the expected return on debt or the cost of debt, D is the amount of interest-bearing debt in XYZ's capital structure, K_E is the expected return on equity or the cost of equity, and E is the amount of equity in XYZ's capital structure. Similarly, the annual return XYZ must earn on existing capital is

$$K_W = \frac{(1 - t)K_D D + K_E E}{D + E} \tag{8.1}$$

where K_W is the cost of capital.

From the preceding example,

$$15\% = \frac{(1 - 30\%)\,10\% \times \$100 + 19\% \times \$200}{\$100 + \$200}$$

In words, a company's cost of capital is the cost of the individual sources of capital, weighted by their importance in the firm's capital structure. The subscript W appears in the expression to denote that the cost of capital is a weighted-average cost. This is also why the cost of capital is often denoted by the acronym WACC for weighted-average cost of capital. To demonstrate that K_W is a weighted-average cost, note that one-third of XYZ's capital is debt and two-thirds is equity, so its WACC is one-third the cost of debt plus two-thirds the cost of equity:

$$15\% = (1/3 \times 7\%) + (2/3 \times 19\%)$$

The Cost of Capital and Stock Price

An important tie exists between a company's cost of capital and its stock price. To see the linkage, ask yourself what happens when XYZ Corporation earns a return on existing assets greater than its cost of capital. Because the return to creditors is fixed by contract, the excess return accrues entirely to shareholders. Because the company can earn more than shareholders' opportunity cost of capital, XYZ's stock price will rise as new investors are attracted by the excess return. Conversely, if XYZ earns a return below its cost of capital on existing assets, shareholders will not receive their expected return, and its stock price will fall. The price will continue falling until the prospective return to new buyers again equals equity investors' opportunity cost of capital. Another definition of the cost of capital, therefore, is *the return a firm must earn on existing assets to keep its stock price constant.* Finally, from a shareholder value perspective, we can say that management creates value when it earns returns above the firm's cost of capital and destroys value when it earns returns below this target.

The Cost of Capital for Polaris

To use the cost of capital as a risk-adjusted discount rate, we must be able to measure it. This involves assigning values to all of the quantities on the right side of Equation 8.1. To illustrate the process, let's estimate Polaris's cost of capital at year-end 2020.

The Weights

We begin by measuring the weights, D and E. There are two common ways to do this, only one of which is correct: Use the book values of debt and equity appearing on the company's balance sheet, or use the market values. By *market value,* I mean the price of the company's bonds and common shares in

TABLE 8.2 **Book and Market Values of Debt and Equity for Polaris Inc. (December 31, 2020)**

Source: Polaris annual reports and Compustat.

	Book Value		Market Value	
Source	Amount ($ millions)	Percentage of Total	Amount ($ millions)	Percentage of Total
Debt	$1,436	50.0%	$1,436	19.6%
Equity	1,434	50.0%	5,898	80.4%
Total	$2,870	100.0%	$7,334	100.0%

securities markets multiplied by the number of each security type outstanding. As Table 8.2 shows, the book values of Polaris's debt and equity at the end of 2020 were $1,436 million and $1,434 million, respectively. The figure for debt includes only interest-bearing debt because other liabilities are either the result of tax accruals that are subsumed in the estimation of after-tax cash flow (ATCF) or spontaneous sources of cash that are part of working capital in the investment's cash flows. The table also indicates that the market value of Polaris's debt and equity on the same date were $1,436 million and $5,898 million, respectively.

Consistent with common practice, I have assumed here that the market value of Polaris's debt equals its book value. This assumption is almost certainly incorrect, but just as certainly the difference between the book and market values of debt is quite small compared to that for equity. The market value of Polaris's equity is its price per share at year-end of $95.28 times 61.9 million common shares outstanding. The market value of equity exceeds the book value by a ratio of over 4 to 1 because investors are upbeat about the company's future prospects.

To decide whether book weights or market weights are appropriate for measuring the cost of capital, consider the following. Suppose that 10 years ago you invested $20,000 in a portfolio of common stocks that, through no doing of your own, is now worth $50,000. After talking to stockbrokers and investment consultants, you believe a reasonable return on the portfolio, given present market conditions, is 10 percent a year. Would you be satisfied with a 10 percent return on the original $20,000 cost of the portfolio, or would you expect to earn 10 percent on the current $50,000 market value? Obviously, the current market value is relevant for decision making; the original cost is sunk and therefore irrelevant. Similarly, Polaris's owners and creditors have investments worth $5,898 million and $1,436 million, respectively, on which they expect to earn competitive returns. Thus, the market values of debt and equity are appropriate for measuring the cost of capital.

The Cost of Debt

This is an easy one. Bonds with risk and maturity similar to Polaris's were yielding a return of approximately 2.7 percent in December 2020, and the company's marginal tax rate is about 21 percent. To get the after-tax cost of debt to Polaris, we simply multiply the pre-tax cost of debt (2.7 percent) by (1 − 21%) to arrive at the after-tax cost of debt of 2.1 percent. Some financial neophytes are tempted to use the coupon rate on the debt rather than the prevailing market rate in this calculation. But the coupon rate is old news; it reflects a market rate from the time when the debt was first issued. Because we want to use the cost of capital to evaluate new investments, we want the cost of new debt.

The Cost of Equity

Estimating the cost of equity is as hard as estimating debt was easy. With debt, or preferred stock, the company promises the holder a specified stream of future payments. Knowing these promised payments and the current price of the security, it is a simple matter to calculate the expected return. This is what we did in the last chapter when we calculated the yield to maturity on a bond. With common stock, the situation is more complex. Because the company makes no promises about future payments to shareholders, there is no simple way to calculate the return expected.

Assume a Perpetuity

One way out of this dilemma recalls the story of the physicist, the chemist, and the economist trapped at the bottom of a 40-foot pit. After failing with a number of schemes based on their knowledge of physics and chemistry to extract themselves from the pit, the two finally turn to the economist in desperation and ask if there isn't anything in his professional training that might help them devise a means of escape. "Why, yes," he replies. "The problem is really quite elementary. Simply assume a ladder." Here our "ladder" is an assumption about the future payments shareholders expect. From this heroic beginning, the problem really does become quite elementary. To illustrate, suppose equity investors expect to receive an annual dividend of $\$d$ per share forever. Because we know the current price, P, and have assumed a future payment stream, all that remains is to find the discount rate that makes the present value of the payment stream equal the current price. From the last chapter, we know that the present value of such a perpetuity at a discount rate of K_E is

$$P = \frac{d}{K_E}$$

and, solving for the discount rate,

$$K_E = \frac{d}{P}$$

In words, if you are willing to assume investors expect a company's stock to behave like a perpetuity, the cost of equity capital is simply the dividend yield.

Perpetual Growth

A somewhat more plausible assumption is that shareholders expect a per share dividend next year of $\$d$ and expect this dividend to grow at the rate of g percent per annum *forever.* Fortunately, it turns out that this cash flow stream also has an unusually simple solution. Without boring you with the arithmetic details, the present value of the assumed payment stream at a discount rate of K_E is

$$P = \frac{d}{K_E - g}$$

and, solving for the discount rate,

$$K_E = \frac{d}{P} + g$$

This equation says that if the perpetual growth assumption is correct, the cost of equity capital equals the company's dividend yield (d/P), plus the growth rate in dividends. This is known as the *perpetual growth equation* for K_E.

The problem with the perpetual growth estimate of K_E is that it is only as good as the assumption on which it is based. For mature companies such as railroads, electric utilities, and steel mills, it may be reasonable to assume that observed growth rates will continue indefinitely. And in these cases, the perpetual growth equation yields a plausible estimate of the cost of equity capital. In all other instances, when it is implausible to think the company can maintain its current rate of growth indefinitely, the equation overestimates the cost of equity.

Let History Be Your Guide

A second and generally more fruitful approach to estimating the cost of equity capital looks at the determinants of expected returns on risky investments. In general, the expected return on any risky asset is composed of three factors:

$$\begin{matrix}\text{Expected return} \\ \text{on risky asset}\end{matrix} = \begin{matrix}\text{Risk-free} \\ \text{interest rate}\end{matrix} + \begin{matrix}\text{Inflation} \\ \text{premium}\end{matrix} + \begin{matrix}\text{Risk} \\ \text{premium}\end{matrix}$$

The equation says that the owner of a risky asset should expect to earn a return from three sources. The first is compensation for the opportunity cost incurred in holding the asset. This is the risk-free interest rate. The second is compensation for the declining purchasing power of the currency over time. This is the inflation premium. The third is compensation for bearing the asset's systematic risk. This is the risk premium. Fortunately, we do not need to treat the first two terms as separate factors because together they equal the expected return on a default-free bond such as a government bond. Because we can readily determine the government bond interest rate, the only challenge is to estimate the risk premium.

When the risky asset is a common stock, it is useful to let history be our guide and recall from Table 5.1 that, on average, over the last century, the annual return on U.S. common stocks has exceeded that on government bonds by 6.4 percentage points. As a reward for bearing the added systematic risk, common stockholders earned a 6.4 percentage point higher annual return than government bondholders. Treating this as a risk premium and adding it to a 2020 long-term government bond rate of 1.7 percent yields an estimate of 8.1 percent as the cost of equity capital for a typical company.

What is the logic of treating the 6.4-percentage-point historical excess return as a risk premium? Essentially, it is that over a long enough time, the return investors receive and what they expect to receive should approximate each other. For example, suppose investors expect a 20-percentage-point excess return on common stocks but the actual return keeps turning out to be 3 percentage points. Then, two things should happen: Investors should lower their expectations, and selling by disappointed investors should increase subsequent realized returns. Eventually expectations and reality should come into rough parity.

We now have an estimate of the cost of capital to an "average-risk" company, but of course few companies are precisely average risk. How, then, can we customize our average-cost expression to reflect the risk of a specific firm? The answer is to insert a "customization factor," known as the company's *equity beta,* into the expression so that it becomes

$$\begin{pmatrix}\text{Cost of equity}\\ \text{capital}\end{pmatrix} = \begin{pmatrix}\text{Interest rate on}\\ \text{government bond}\end{pmatrix} + \beta_e\left(\begin{matrix}\text{Historical excess return}\\ \text{on common stocks}\end{matrix}\right)$$

or in symbols,

$$K_E = i_g + \beta_e \times Rp \tag{8.2}$$

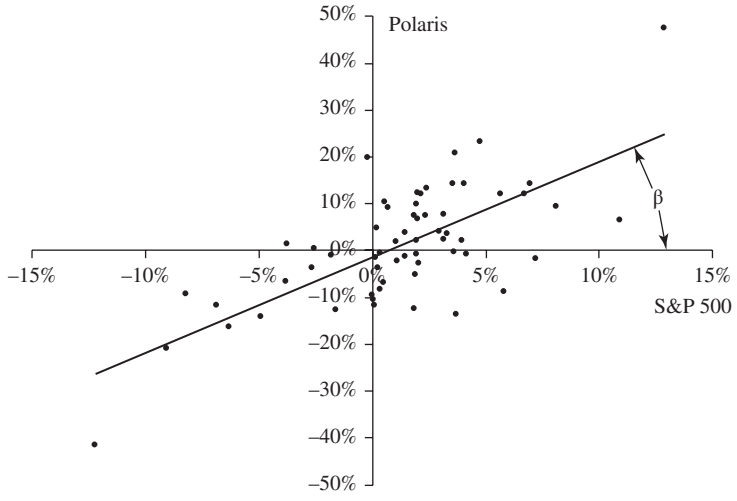
where i_g is a government bond rate, β_e is the equity beta of the target company, and Rp is the excess return on common stocks. You can think of β_e as a scale factor reflecting the systematic risk of a specific company's shares relative to that of an average share. When the stock's systematic risk equals that of an average share, β_e equals 1.0, and the historical risk premium applies directly. But for above-average-risk shares, β_e exceeds 1.0, and the risk premium grows accordingly. Conversely, for below-average-risk shares, β_e is below 1.0, and only a fraction of the historical risk premium applies.

Estimating Beta

But, you might well ask, how do we estimate a company's beta? Actually, it's pretty simple. Figure 8.4 provides everything required to estimate Polaris's beta. It shows the monthly realized returns on Polaris's common stock relative to returns on the S&P 500 stock index over the past 60 months. For example, in July 2020, the S&P index increased 6 percent, while Polaris's stock rose

FIGURE 8.4 Polaris Inc.'s Beta Is the Slope of the Best-Fit Line Below

Source: Monthly Returns of Polaris Inc. v. S&P 500, 60 Months through December 2020. Data from Center for Research in Security Prices (CRSP).



A Virtue of Statistics

Many of the concepts in this chapter can be described quite simply with the aid of a little statistics. As already noted, an investment's *total risk* refers to its dispersion in possible returns, commonly represented by the standard deviation of returns, while its *systematic risk* depends on the extent to which the investment's returns correlate with those on a broadly diversified portfolio. We can thus represent the systematic risk of investment *j* as

$$\text{Systematic risk} = \rho_{jm}\sigma_j$$

where ρ_{jm} is the correlation coefficient between investment *j* and well-diversified portfolio *m,* and σ_j is the standard deviation of returns on investment *j.* The correlation coefficient is, of course, a dimensionless number ranging between + 1 and −1, with + 1 characterizing perfectly positively correlated returns and −1 perfectly inversely correlated returns. For most business investments, ρ_{jm} is in the range of 0.5 to 0.8, meaning that 20 to 50 percent of the investment's total risk can be diversified away.

A common stock's *equity beta* equals its systematic risk relative to that of a well-diversified portfolio, or in symbols, stock *j*'s equity beta is

$$\beta_j = \frac{\rho_{jm}\sigma_j}{\rho_{mm}\sigma_m}$$

But because any variable must be perfectly positively correlated with itself, this expression reduces to

$$\beta_j = \frac{\rho_{jm}\sigma_j}{\sigma_m}$$

In addition to representing stock *j*'s equity beta, this expression also equals the slope coefficient of the regression of r_j on r_m, where r_j and r_m are realized returns on stock *j* and the diversified portfolio, respectively.

12 percent. This return pair constitutes one point on the graph. The S&P 500 index is a broadly diversified portfolio containing many common shares, so its systematic risk is a reasonable surrogate for the systematic risk of an average share, and of the market as a whole. Also appearing in the figure is a best-fit, straight line indicating the average relationship among the paired returns. (If you are familiar with regression analysis, this is a simple regression line.)

The slope of this line is the beta estimate we seek. It measures the sensitivity of Polaris's equity returns to movements in the S&P index. The indicated slope of 2.05 means that, on average, Polaris's stock price changes by 2.05 percent for every 1 percent change in the index, indicating that Polaris's equity is higher risk than average. Clearly, if this line were less steeply sloped, Polaris's stock would be less sensitive to market movements, or alternatively to economywide events, and thus less risky. A more steeply sloped line would imply just the reverse. The fact that all of the return pairs plotted in the figure do not lie precisely on the straight line reflects the importance of unsystematic risk in determining Polaris's monthly returns. Remember that because unsystematic risk can be eliminated through diversification, it should play no role in determining required returns or prices.

Fortunately, you do not need to worry about calculating betas yourself. Beta risk is so important a factor in security analysis that many financial websites regularly publish the betas of publicly traded common stocks. Table 8.3

TABLE 8.3 Representative Company Betas

Source: Yahoo! Finance, 2021.

Company	Beta	Company	Beta
Advanced Micro Devices	2.06	Duke Energy	0.24
Alphabet	1.00	eBay	1.21
Amazon	1.12	Facebook	1.29
American Electric Power	0.25	Ford Motor	1.20
American International Group	1.38	H&R Block	0.90
Apple	1.22	Hormel Foods	−0.05
Bank of America	1.58	IBM	1.23
Baxter International	0.68	Intel	0.66
Berkshire Hathaway	0.90	Johnson & Johnson	0.72
Boeing	1.69	Lowe's	1.36
Camping World	3.36	Microsoft	0.79
Cemex	1.07	Overstock	4.47
Coca-Cola	0.62	PetroChina	0.87
Costco Wholesale	0.65	Southwest Airlines	1.26
Dean Foods	−0.83	Walmart	0.48

presents recent betas for a representative sample of firms. Observe that betas range from a high of 4.47 for online retailer Overstock, to a low of −0.83 for food and beverage company Dean Foods. (Negative betas are uncommon; they indicate that a stock moves in the opposite direction of the market, on average.) Note, too, that most of the numbers are intuitively plausible, with high-risk businesses such as some technology companies having high betas, while low-risk companies such as utilities have lower betas.

Inserting Polaris's estimated equity beta of 2.05 into Equation 8.2 yields the following cost of equity capital:

$$K_E = 1.7\% + 2.05 \times 6.4\% = 14.8\%$$

Before leaving our discussion of beta, I should mention that while the motivation offered for Equation 8.2 has been largely intuitive, the equation actually rests on a solid conceptual foundation known as the Capital Asset Pricing Model, or the CAPM. If you have not been dazzled while reading about beta, you should know that the CAPM was considered revolutionary enough that in 1990 the most prominent developer of the model, William Sharpe, was awarded the Nobel Prize for his efforts. According to the CAPM, Equation 8.2 is nothing less than the equation of the market line shown earlier in Figure 8.1. As such, it describes the equilibrium relationship between the expected return on any risky asset and its systematic risk. Said differently, Equation 8.2 defines the minimum acceptable rate of return an investor should demand on any risky asset.

Polaris's Weighted-Average Cost of Capital

All that remains now is the figure work. Table 8.4 presents my estimate of Polaris's cost of capital in tabular form. Polaris's weighted-average cost of capital is 12.3 percent. This means that at year-end 2020, Polaris needed to earn at least this percentage return on the market value of existing assets to meet the expectations of creditors and shareholders and, by inference, to maintain its stock price. In equation form,

$$Kw = \frac{(1 - t)K_D D + K_E E}{D + E}$$

$$Kw = \frac{(1 - 0.21)(2.7\%)(\$1{,}436) + (14.8\%)(\$5{,}898)}{\$1{,}436 + \$5{,}898}$$

$$Kw = 12.3\%$$

TABLE 8.4 **Calculation of Polaris Inc.'s Weighted-Average Cost of Capital**

Source	Amount ($ millions)	Percentage of Total	Cost after Tax	Weighted Cost
Debt	$1,436	19.6%	2.1%	0.4%
Equity	$ 5,898	80.4%	14.8%	11.9%
			Weighted-Average Cost of Capital =	**12.3%**

The Cost of Capital in Investment Appraisal

The fact that the cost of capital is the return a company must earn on *existing assets* to meet creditor and shareholder expectations is an interesting detail, but we are after bigger game here: We want to use the cost of capital as an acceptance criterion for *new investments.*

Are there any problems in applying a concept derived for existing assets to new investments? Not if one critical assumption holds: The new investment must have the same risk existing assets do. If it does, the new investment is essentially a "carbon copy" of existing assets, and the cost of capital is the appropriate risk-adjusted discount rate. If it does not, we must proceed more carefully.

The market line in Figure 8.5 clearly illustrates the importance of the equal-risk assumption. It emphasizes that the rate of return risk-averse individuals anticipate rises with risk. This means, for example, that management should demand a higher expected return when introducing a new product than when replacing aged equipment, because the new product is presumably riskier and therefore warrants a higher return. The figure also shows that a company's cost of capital is but one of many possible risk-adjusted discount rates, the one corresponding to the risk of the firm's existing assets. We conclude that the cost of capital is an appropriate acceptance criterion only when the risk of a new investment equals that of existing assets. For all other investments, the cost of capital is inappropriate. But do not despair, for even when inappropriate itself, the cost of capital concept is central to identifying a correct risk-adjusted rate.

Multiple Hurdle Rates

Companies adjust their hurdle rates for differing investment risks in at least three possible ways. The first two are straightforward extensions of the cost of capital. For large projects, the approach is to identify an industry in which the contemplated investment would be considered average risk, estimate the weighted-average cost of capital for several companies in the industry, and

Type of Investment	Discount Rate (%)
Replacement or repair	8.0
Cost reduction	9.0
Expansion	12.3
New product	15.0

Investments to expand capacity in existing products are essentially carbon-copy investments, so their hurdle rates equal Polaris's cost of capital. Other types of investments have a higher or lower hurdle rate, depending on their risk relative to expansion investments. Replacement or repair investments are the safest because virtually all of the cash flows are well known from past experience. Cost reduction investments are somewhat riskier, because the magnitude of potential savings is uncertain. New-product investments are the riskiest type of all, because both revenues and costs are uncertain.

Multiple hurdle rates are consistent with risk aversion and with the market line, but the amount by which the hurdle rate should be adjusted for each level of risk is largely arbitrary. Whether the hurdle rate for new product investments should be 3 or 6 percentage points above Polaris's cost of capital cannot be determined objectively.

Four Pitfalls in the Use of Discounted Cash Flow Techniques

You now know the basics of investment appraisal: Estimate the opportunity's annual, expected ATCFs and discount them to the present at a risk-adjusted discount rate appropriate to the risk of the cash flows. When the opportunity is a "carbon-copy" investment, the firm's weighted-average cost of capital is the appropriate discount rate. In other instances, an upward or downward adjustment to the firm's cost of capital is necessary.

In the interest of full disclosure, I will now gingerly mention four pitfalls in the practical application of discounted cash flow techniques. The first two are easily avoided once you are aware of them; the last two highlight important limitations of discounted cash flow techniques as conventionally applied. Collectively, these pitfalls mean you need to master several more topics before attempting to pass as an expert.

The Enterprise Perspective vs. the Equity Perspective

Any corporate investment partially financed with debt can be analyzed from either of two perspectives: that of the company, commonly known as the *enterprise* perspective, or that of its owners, often referred to as the *equity* perspective. As the following example demonstrates, these two perspectives

The Fallacy of the Marginal Cost of Capital

Some readers, especially engineers, look at Equation 8.1 and naively conclude that it is possible to reduce a company's weighted-average cost of capital by using more of the cheap source of financing, debt, and less of the expensive source, equity. In other words, they conclude that increasing leverage will reduce the cost of capital. This reasoning, however, evidences an incomplete understanding of leverage. As we observed in Chapter 6, increasing leverage increases the risk borne by shareholders. Because they are risk averse, shareholders react by demanding a higher return on their investment. Thus, K_E and, to a lesser extent, K_D rise as leverage increases. This means that increasing leverage affects a company's cost of capital in two opposing ways: Increasing use of cheap debt reduces K_W, but the rise in K_E and K_D that accompanies added leverage increases it.

To review this reasoning, ask yourself how you would respond to a subordinate who made the following argument in favor of an investment: "I know the company's cost of capital is 12 percent and the IRR of this carbon-copy investment is only 10 percent. But at the last directors' meeting, we decided to finance this year's investments with new debt. Because new debt has a cost of only about 4 percent after tax, it is clearly in our shareholders' interest to invest 4 percent money to earn a 10 percent return."

The subordinate's reasoning is incorrect. Financing with debt means increasing leverage and increasing K_E. Adding the change in K_E to the 4 percent interest cost means the true *marginal* cost of the debt is well above the interest cost. In fact, it is probably quite close to K_W.

are functionally equivalent in the sense that when properly applied they yield the same investment decision—but woe be to him who confuses the two.

Suppose ABC Industries has a capital structure composed of 40 percent debt, costing 5 percent after tax, and 60 percent equity, costing 20 percent. Its WACC is therefore

$$K_W = 5\% \times 0.40 + 20\% \times 0.60 = 14\%$$

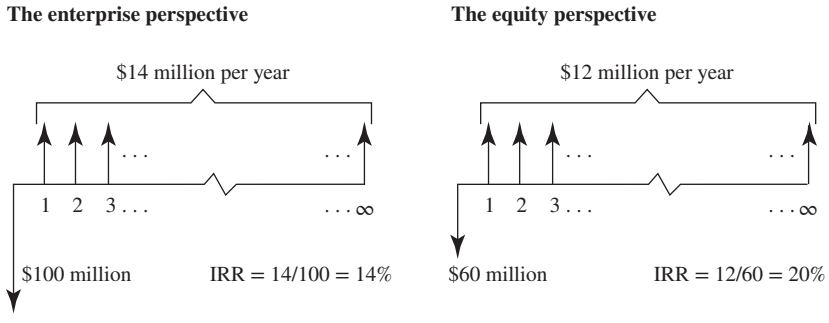
The company is considering an average-risk investment costing $100 million and promising an ATCF of $14 million a year in perpetuity. If undertaken, ABC plans to finance the investment with $40 million in new borrowings and $60 million in equity. Should ABC make the investment?

The Enterprise Perspective

The left side of the following diagram shows the investment's cash flows from the enterprise perspective. Applying our now standard approach, the investment is a perpetuity with a 14 percent internal rate of return. Comparing this return to ABC's weighted-average cost of capital, also 14 percent, we conclude that the investment is marginal. Undertaking it will neither create nor destroy shareholder value.

The Equity Perspective

The right side of the diagram shows the same investment from the owners' viewpoint, or the equity perspective. Because $40 million of the initial cost will be financed by debt, the equity outlay is only $60 million. Similarly, because $2 million after tax must be paid to creditors each year as interest, the residual cash flow to equity will be only $12 million. The investment's internal rate of return from the equity perspective is therefore 20 percent.



Does the fact that the return is now 20 percent mean the investment is suddenly an attractive one? Clearly, no. Because the equity cash flows are levered, they are riskier than the original cash flows and hence require a higher risk-adjusted discount rate. Indeed, the appropriate acceptance criterion for these equity cash flows is ABC's cost of equity capital, or 20 percent. (Remember, the discount rate should reflect the risk of the cash flows to be discounted.) Comparing the project's 20 percent IRR to equity with ABC's cost of equity, we again conclude that the investment is only marginal.

It is not an accident that the enterprise and equity perspectives yield the same result. Because the weighted-average cost of capital is defined to ensure that each supplier of capital receives a return equal to her opportunity cost, we know that an investment by ABC earning 14 percent, from the enterprise perspective, will earn just enough to service the debt and generate a 20 percent IRR on invested equity. Problems arise only when you mix the two perspectives, using K_E to discount enterprise cash flows or, more commonly, using K_W to discount equity cash flows.

Which perspective is better? Some of my best friends use the equity perspective, but I believe the enterprise perspective is easier to apply in practice. The problem with the equity perspective is that both the IRR to equity and the appropriate risk-adjusted discount rate vary with the amount of leverage used. The IRR to equity on ABC's investment is 20 percent with $40 million of debt financing but jumps to 95 percent with $90 million of debt and rises to infinity with all-debt financing.

The interdependency between the means of financing and the risk-adjusted discount rate is easily handled in a classroom, but when real money is on the line, we often become so enthralled by the return-enhancing aspect of debt that we forget the required rate of return rises as well. Moreover, even when we remember that leverage increases risk as well as return, it is devilishly hard to estimate exactly how much the cost of equity should change with leverage.

Life is short. I recommend you avoid unnecessary complications by using the enterprise perspective whenever possible. Assess the economic merit of the investment without regard to how it will be financed or how you will divvy up the spoils. If the investment meets this fundamental test, you can then turn to the nuances of how best to finance it.

Inflation

The second pitfall involves the improper handling of inflation. Too often, managers ignore inflation when estimating an investment's cash flows but inadvertently include it in their discount rate. The effect of this mismatch is to make companies overly conservative in their investment appraisal, especially with regard to long-lived assets. Table 8.5 illustrates the point. A company with a 15 percent cost of capital is considering a $10 million, carbon-copy investment. The investment has a four-year life and is expected to increase production capacity by 10,000 units annually. Because the product sells for $900, the company estimates that annual revenues will rise $9 million ($900 × 10,000 units), which, after subtracting production costs, yields an increase in annual after-tax cash flows of $3.3 million. The IRR of the investment is calculated to be 12 percent, which is below the firm's cost of capital.

TABLE 8.5 **When Evaluating Investments under Inflation, Always Compare Nominal Cash Flows to a Nominal Discount Rate or Real Cash Flows to a Real Discount Rate ($ millions)**

(a) Incorrect Investment Evaluation Comparing Real Cash Flows to a Nominal Discount Rate

	2020	2021	2022	2023	2024
After-tax cash flow	$(10.0) IRR = 12% K_W = 15% Decision: **Reject**	$3.3	$3.3	$3.3	$3.3

(b) Correct Investment Evaluation Comparing Nominal Cash Flows to a Nominal Discount Rate

	2020	2021	2022	2023	2024
After-tax cash flow	$(10.0) IRR =18% K_W =15% Decision: **Accept**	$3.5	$3.6	$3.8	$4.0

you believe this project is not a wise investment. What logical arguments would you use to convince your boss to forego the project despite its high rate of return? Is it possible that making investments with returns higher than the firm's cost of capital will destroy value? If so, how?

5. ABC Corporation and XYZ Corporation are both bidding for an existing food processing plant located in Monterrey, Mexico. Both firms are highly profitable and have similar debt ratios and costs of debt, but XYZ operates in a more volatile industry than ABC and thus has a higher beta. As ABC and XYZ separately prepare their valuations of the plant, what difference would you expect to see in the appropriate weighted-average cost of capital each firm will use to discount the projected cash flows from the plant?
6. Looking at Figure 8.1, explain why a company should reject investment opportunities lying below the market line and accept those lying above the line.
7. What is the weighted-average cost of capital for SKYE Corporation given the following information?

Equity shares outstanding	1 million
Stock price per share	$30.00
Yield to maturity on debt	7.68%
Book value of interest-bearing debt	$10 million
Coupon interest rate on debt	9%
Interest rate on government bonds	6%
SKYE's equity beta	0.75
Historical excess return on stocks	6.3%
Tax rate	40%

8. You have the following information about Burgundy Basins, a sink manufacturer.

Equity shares outstanding	20 million
Stock price per share	$40.00
Yield to maturity on debt	7.5%
Book value of interest-bearing debt	$320 million
Coupon interest rate on debt	4.8%
Market value of debt	$290 million
Book value of equity	$500 million
Cost of equity capital	14%
Tax rate	35%

Burgundy is contemplating what for the company is an average-risk investment costing $40 million and promising an annual ATCF of $6.4 million in perpetuity.

a. What is the internal rate of return on the investment?
b. What is Burgundy's weighted-average cost of capital?
c. If undertaken, would you expect this investment to benefit shareholders? Why or why not?

9. How will an increase in financial leverage affect a company's cost of equity capital, if at all? How will it affect a company's equity beta?
10. What is the present value of a cash flow stream of $1,000 per year annually for 15 years that then grows at 4 percent per year forever when the discount rate is 13 percent?
11. You are a commercial real estate broker eager to sell an office building. An investor is interested but demands a 20 percent return on her equity investment. The building's selling price is $25 million, and it promises free cash flows of $3 million annually in perpetuity. Interest-only financing is available at 8 percent interest; that is, the debt is outstanding forever and requires no principal payments. The tax rate is 50 percent.
    a. Propose an investment-financing package that meets the investor's return target.
    b. Propose an investment-financing package that meets the investor's target when she demands a 90 percent return on equity.
    c. Why would an investor settle for a 20 percent return on this investment when she can get as high as 90 percent?
12. A security analyst has regressed the monthly returns on Exxon Mobil equity shares over the past five years against those on the S&P 500 stock index over the same period. The resulting regression equation is $r_{EM} = 0.05 + 1.41\ r_{SP}$. Use this equation to estimate Exxon Mobil's equity beta.
13. An all-equity financed company has a cost of capital of 10 percent. It owns one asset: a mine capable of generating $100 million in free cash flow every year for five years, at which time it will be abandoned. A buyout firm proposes to purchase the company for $400 million financed with $350 million in debt to be repaid in five, equal, end-of-year payments and carrying an interest rate of 6 percent.
    a. Calculate the annual debt-service payments required on the debt.
    b. Ignoring taxes, estimate the rate of return to the buyout firm on the acquisition after debt service.
    c. Does the buyout look attractive? Why, or why not?
14. The following information is available about an investment opportunity. Investment will occur at year 0 and sales will occur from year 1 to year 8. Use a nominal discount rate, calculated as $i_n = (1 + i_r)(1 + p) - 1$, where i_r is the real discount rate, and p is expected inflation.

Initial cost	$28 million
Unit sales	400,000
Selling price per unit, this year	$60.00
Variable cost per unit, this year	$42.00
Life expectancy	8 years
Salvage value	$0
Depreciation	Straight-line
Tax rate	37%
Real discount rate	10.0%
Inflation rate	0.0%

a. Prepare a spreadsheet to estimate the project's annual after-tax cash flows.
b. Calculate the investment's internal rate of return and its net present value assuming zero inflation.
c. How do the internal rate of return and net present value change when you assume an inflation rate of 8 percent per year in price and variable cost per unit?
d. How do you explain the fact that inflation causes the internal rate of return to increase and the NPV to decrease?
e. Does inflation make this investment more attractive or less attractive? Why?

15. The chapter discusses General Design's option to expand its diamond film project.
    a. Is the option a call or a put?
    b. In qualitative terms, what is the option's strike price?
16. Karamazov Semiconductors is considering an investment to expand its existing line of business. The investment will cost $10 million and is expected to produce after-tax cash flows of $1 million per year in perpetuity. Karamazov's capital structure is composed of 50 percent debt and 50 percent equity, and it plans to finance the investment using this same ratio of financing. Karamazov's pre-tax cost of debt is 8 percent, its tax rate is 25 percent, and its cost of equity is 14 percent.
    a. What is Karamazov's weighted-average cost of capital?
    b. Calculate the IRR of the project from the enterprise perspective.
    c. Calculate the IRR of the project from the equity perspective.
    d. Explain the difference between your two answers in parts (b) and (c).
17. Reconsider General Design's diamond film project from Table 8.6(b). Suppose that General Design now believes it also has an option to grow the project if things go well. Initial success in diamond film semiconductors will open the door to a stage 2, follow-on investment in two years that will be five times as large as the initial stage 1. Stage 2 will be undertaken only if stage 1 is successful, and management believes the chance stage 2

In all other instances, market value may provide a useful frame of reference, but cannot by itself answer most interesting valuation questions. For this, we need to think more carefully about the determinants of business value.

Discounted Cash Flow Valuation

Having examined business valuation in the large, we turn now to the specific task of estimating a company's going-concern value. For simplicity, we will begin by considering the value of a minority interest in a privately held firm.

Absent market prices, the most direct way to estimate going-concern value, if not always the most practical, is to think of the target company as if it were nothing more than a giant capital expenditure opportunity. Just as with any piece of capital equipment, investing in a company requires the expenditure of money today in anticipation of future benefits, and the central issue is whether tomorrow's benefits justify today's costs. As in capital expenditure analysis, we can answer this question by calculating the present value of expected future cash flows accruing to owners and creditors. When this number exceeds the acquisition price, the purchase has a positive net present value and is therefore attractive. Conversely, when the present value of future cash flows is less than the acquisition price, the purchase is unattractive.

In equation form,

$$\text{FMV of firm} = \text{PV (Expected cash flows to owners and creditors)}$$

This formula states that the maximum price one should pay for a business equals the present value of expected future cash flows to capital suppliers discounted at an appropriate risk-adjusted discount rate. Moreover, as in any other application of risk-adjusted discount rates, we know the rate should reflect the risk of the cash flows being discounted. Because the cash flows here are to owners and creditors of the target firm, it follows that the discount rate should be the target company's weighted-average cost of capital.

A legitimate question at this point is: Why waste energy estimating firm value when the ultimate goal of the exercise is usually to value equity? Why not just value the equity directly? The answer is simple once you recall that the value of equity is closely tied to the value of the firm. In equation form, we have our old friend

$$\text{Value of equity} = \text{Value of firm} - \text{Value of debt}$$

To determine the value of a company's equity, therefore, we need only estimate firm value and subtract interest-bearing debt. Moreover, because the market value and the book value of debt are usually about equal to each other, estimating the value of debt amounts to nothing more than grabbing a few

numbers off the company's balance sheet.[3] If the fair market value of a business is $4 million and the firm has $1.5 million in debt outstanding, its equity is worth $2.5 million. It's that simple.[4] (We ignore non-interest-bearing debt such as accounts payable and deferred taxes here because they are treated as part of free cash flow, to be described momentarily.)

Free Cash Flow

As in all capital expenditure decisions, the biggest practical challenge in business valuation is estimating the relevant cash flows to be discounted. In Chapter 7, we said the relevant cash flows are the project's annual free cash flows (FCFs), defined as EBIT after tax plus depreciation, less investment. When valuing a company, this translates into the following:

$$\begin{matrix}\text{Free}\\ \text{cash flow}\end{matrix} = \text{EBIT}(1 - \text{Tax rate}) + \text{Depreciation} - \begin{matrix}\text{Capital}\\ \text{expenditures}\end{matrix} - \begin{matrix}\text{Working capital}\\ \text{investments}\end{matrix}$$

where EBIT is earnings before interest and taxes.

The rationale for using free cash flow goes like this. EBIT is the income the company earns without regard to how the business is financed; so EBIT(1 − Tax rate) is income after tax excluding the effects of debt financing. Adding depreciation and any other noncash items yields after-tax cash flow. If management were prepared to run the company into the ground, it could distribute this cash flow to owners and creditors, and that would be the end of it. But in most companies, management retains some or all of this cash flow in the business to pay for new capital expenditures and additions to short-term assets. The annual cash flow available for distribution to owners and creditors is thus operating cash flow after tax less capital expenditures and working capital investments.

The working capital term in this expression can be tricky. Working capital investment equals the increase in current assets necessary to support operations, less any accompanying increases in non-interest-bearing current liabilities, or what I referred to in Chapter 7 as "spontaneous sources." This difference equals the net investment in current assets that must be financed by creditors and owners. A second challenge is how to treat any excess cash a

[3] There are two instances in which the market value and the book value of debt will differ significantly: Default risk has changed significantly since issue, and the debt is fixed rate and market interest rates have changed significantly since issue. In these instances, it pays to estimate the market value of the debt independently.

[4] An alternative approach to equity valuation is to estimate the present value of expected cash flows to equity discounted at the target's cost of equity capital. Executed correctly, this equity approach yields the same answer as the enterprise approach described above; however, I find it more difficult to apply in practice. See the section The Enterprise Perspective vs. the Equity Perspective in Chapter 8 for details.

company accumulates over and above the amount necessary to support operations. My advice is to omit excess cash from the discounted cash flow valuation and treat it as a separate add-on term.

The Terminal Value

We now come to a serious practical problem. Our equation says that the FMV of a business equals the present value of all future free cash flows. Yet because companies typically have an indefinitely long life expectancy, the literal application of this equation would have us estimating free cash flows for perhaps hundreds of years into the far distant future—a clearly unreasonable task.

The standard way around this impasse is to think of the target company's future as composed of two discrete periods. During the first period, of some 5 to 15 years, we presume the company has a unique cash flow pattern and growth trajectory that we seek to capture by estimating individual, annual free cash flows just as the equation suggests. However, by the end of this forecast period, we assume the company has lost its individuality—has grown up, if you will—and become a stable, slow-growth business. From this date forward, we cease worrying about annual cash flows and instead estimate a single *terminal value* representing the worth of all subsequent free cash flows. If the initial forecast period is, say, 10 years, our valuation equation becomes

$$\text{FMV of firm} = \text{PV}(\text{FCF years } 1 - 10 + \text{Terminal value at year } 10)$$

Introduction of a terminal value, of course, only trades one problem for another, for now we need to know how to estimate a company's terminal value. I wish I could assure you that financial economists have solved this problem and present a simple, accurate expression for a company's terminal value, but I can't. Instead, the best I can offer are several plausible alternative estimates and some general advice on how to proceed.

Following are five alternative ways to estimate a company's terminal value with accompanying explanatory comments and observations. To use these estimates effectively, note that no single estimate is always best; rather, each is more or less appropriate depending on circumstances. Thus, liquidation value may be highly relevant when valuing a mining operation with 10 years of reserves but quite irrelevant when valuing a rapidly growing software company. Resist the natural temptation to pick what appears to be the best technique for the situation at hand, ignoring all others. Avoid too the simple averaging of several estimates. Instead, calculate a number of terminal value estimates and begin by asking why they differ. In some instances, the differences will be readily explainable; in others, you may find it necessary to revise your assumptions to reconcile the differing values. Then, once you understand why remaining differences exist and feel comfortable with the

magnitude of the differences, select a terminal value based on your assessment of the relative merits of each estimate for the target company.

Five Terminal Value Estimates

Liquidation Value Highly relevant when liquidation at the end of the forecast period is under consideration, liquidation value usually grossly understates a healthy business's terminal value.

Book Value Popular perhaps among accountants, book value usually yields a quite conservative terminal value estimate.

Warranted Price-to-Earnings Multiple To implement this approach, multiply the target firm's estimated earnings to common stock at the end of the forecast horizon by a "warranted" price-to-earnings ratio; then add projected interest-bearing liabilities to estimate the firm's terminal value. As a warranted price-to-earnings ratio, consider the multiples of publicly traded firms that you believe represent what the target will become by the end of the forecast period.[5] If, for example, the target company is a startup but you believe it will be representative of other, mature companies in its industry by the end of the forecast period, the industry's current price-to-earnings multiple may be a suitable ratio. Another strategy is to bracket the value by trying multiples of, say, 10 and 20 times. The approach generalizes easily to other "warranted" ratios, such as market value to book value, price to cash flow, or price to sales.

No-Growth Perpetuity We saw in Chapter 7 that the present value of a no-growth perpetuity is the annual cash flow divided by the discount rate. This suggests the following terminal value estimate:

$$\text{Terminal value of no-growth firm} = \frac{\text{FCF}_{T+1}}{K_W}$$

where FCF_{T+1} is free cash flow in the first year beyond the forecast horizon, and K_W is the target's weighted-average cost of capital. As further refinement, we might note that when a company is not growing, its capital expenditures should about equal its annual depreciation charges, and its net working capital should neither increase nor decrease over time, both of which imply that free cash flow should simplify to EBIT(1 − Tax rate).

Because most businesses expand over time, if due only to inflation, many analysts believe this equation understates the terminal value of a typical business. I am more skeptical. For, as noted repeatedly in earlier chapters, growth creates value only when it generates returns above capital costs, and

[5] For industry price-to-earnings ratios, see **pages.stern.nyu.edu/~adamodar.** Select "Data" and under "Current Data" go to "Multiples."

in competitive product markets over the long run, such performance is more the exception than the rule. Hence, even if many companies are capable of expanding, they may be worth no more than similar no-growth companies. The implication is that the no-growth equation is applicable to more firms than might first be supposed. I am also mindful of economist Kenneth Boulding's observation that "Anyone who believes that exponential growth can go on forever in a finite world is either a madman or an economist."

Perpetual Growth In Chapter 8, we saw that the present value of a perpetually growing stream of cash equals next year's cash flow divided by the difference between the discount rate and the growth rate. Thus, another terminal value estimate is

$$\text{Terminal value of perpetually growth firm} = \frac{\text{FCF}_{T+1}}{K_W - g}$$

where g is the perpetual-growth rate of free cash flow.

A few words of caution are in order about this popular expression. It is a simple arithmetic fact that any business growing faster than the economy *forever* must eventually become the economy. (When I made this point one time at a Microsoft seminar, the immediate response was "Yeah! Yeah! We can do it!") The intended conclusion for mere mortal firms is that the absolute upper limit on g must be the long-run growth rate of the economy, or about 2 to 3 percent a year, plus expected inflation. Moreover, because even inflationary growth invariably requires higher capital expenditures and increases in working capital, free cash flow falls as g rises. This implies that unless this inverse relation is kept in mind, the preceding expression may well overstate a company's terminal value—even when the perpetual growth rate is kept to a low figure.[6]

The Forecast Horizon

Terminal values of growing businesses can easily exceed 60 percent of firm value, so it goes without saying that proper selection of the forecast horizon

[6] Here is a modestly more complex version of the perpetual-growth expression, to which I am partial:

$$\text{Terminal value} = \frac{\text{EBIT}_{T+1}(1 - \text{Tax rate})(1 - g/r)}{K_W - g}$$

where r is the rate of return on new investment. One virtue of this expression is that growth does not add value unless returns exceed capital cost. To confirm this, set $r = K_W$ and note that the expression collapses to the no-growth equation. A second virtue is that growth is not free, for as growth rises, so must capital expenditures and net working capital. In the equation, higher g reduces the numerator, which is equivalent to reducing free cash flow. See page 29 in the Koller, Goedhart, and Wessels book referenced at the end of this chapter for a demonstration that this expression is mathematically equivalent to the earlier perpetual-growth equation.

and terminal value are critical to the successful application of discounted cash flow approaches to business valuation. Because most tractable terminal value estimates implicitly assume the firm is a mature, slow-growth, or no-growth perpetuity from that date forward, it is important to extend the forecast horizon far enough into the future that this assumption plausibly applies. When valuing a rapidly growing business, this perspective suggests estimating how long the company can be expected to sustain its supernormal growth before reaching maturity and setting the forecast horizon at or beyond this date.

POLARIS A Numerical Example

Table 9.1 offers a quick look at a discounted cash flow valuation of our friend from earlier chapters: Polaris Inc. It goes without saying that if I were being

TABLE 9.1 **Discounted Cash Flow Valuation of Polaris Inc. on December 31, 2020* ($ millions except per share)**

	Year					
	2021	**2022**	**2023**	**2024**	**2025**	**2026**
Sales	$8,174	$ 9,400	$10,810	$12,432	$14,297	
EBIT	605	696	800	920	1,058	
Tax at 21%	127	146	168	193	222	
Earnings after tax	478	550	632	727	836	
+ Depreciation	286	329	378	435	500	
– Capital expenditures	278	320	368	423	486	
– Working capital investments	49	56	65	75	86	
Free cash flow	**$ 437**	**$ 503**	**$ 578**	**$ 665**	**$ 764**	**$795**
PV at 12.3% of FCFs ’21–’25	**$2,041**					
Terminal value estimates:			Terminal value in 2025			
Perpetual growth at 4% [FCF ’26/(Kw–g)]			$ 9,577			
Warranted MV firm/Sales in ’25 = 1.5 times			21,445			
Projected book value of debt & equity in ’25			5,773			
Best guess terminal value			**12,000**			
PV of terminal value		**$ 6,719**				
Estimated value of firm		**$ 8,760**				
Value of debt		1,436				
Value of equity		$ 7,324				
Shares outstanding		61.9 million				
Estimated value per share		**$118.32**				

*Totals may not add due to rounding.

paid by the hour to value Polaris and you were being similarly compensated to read about it, we would both proceed much more thoroughly and deliberately. In particular, we would want to know a great deal more about the company's products, markets, and competitors, for a discounted cash flow valuation is only as good as the projections on which it is based. Nonetheless, the table should give you a basic understanding of how to execute a discounted cash flow valuation.

The valuation date is December 31, 2020. The free cash flows appearing in the table are percent-of-sales projections assuming 15 percent annual sales growth for the next five years. Most of the estimates used in the forecast are based on a careful review of the common-size historical financial statements appearing in Chapter 2, Table 2.3. The tax rate is Polaris's average effective rate for the last three years, and the growth rate reflects security analysts' expectations.[7] The present value of these free cash flows discounted at Polaris's 12.3 percent weighted-average cost of capital, estimated in the last chapter, amounts to $2,041 million.

The valuation considers three terminal value estimates. The first relies on the perpetual-growth equation and assumes that beginning in 2025 Polaris's free cash flows will commence growing at 4 percent per year into the indefinite future. Free cash flow in 2026 will thus be $795 million [$\$795 = \$764 \times (1 + 0.04)$]. Plugging these values into the perpetual growth equation, one estimate of Polaris's terminal value at the end of 2025 is

$$\text{Terminal value} = \frac{\text{FCF in 2026}}{K_W - g} = \frac{\$795 \text{ million}}{0.123 - 0.04} = \$9,577 \text{ million}$$

The second terminal value estimate assumes that at the end of the forecast horizon, Polaris will command a price-to-sales multiple of 1.5 times, a figure reflecting current valuations of comparable firms. I will say more about this multiple a little later. Applying this warranted price-to-sales ratio to Polaris's sales in 2025 yields a second terminal value estimate:

$$\text{Terminal value} = 1.5 \times \$14,297 \text{ million} = \$21,445 \text{ million}$$

Finally, I estimate that Polaris's projected book value of interest-bearing debt and equity in 2025 will be $5,773. This constitutes a third estimate of the company's terminal value, although certainly a low one.

Wide variation in terminal value estimates, as seen here, does not produce a warm, comforting feeling, given that the estimated terminal value usually has a large effect on the ultimate valuation of the company. In a more detailed

[7] For example, see the analyst growth expectations provided by Yahoo! Finance.

valuation of Polaris, we would want to consider the sensitivity of our value estimate to different assumptions about the terminal value. Nevertheless, after reflecting on the relative merits of these three estimates, my best guess is that Polaris will be worth about $12,000 million in 2025. Discounting this figure back to 2020 and adding it to the present value of free cash flows in the first five years suggests that Polaris is worth $8,760 million on December 31, 2020:

$$\text{FMV}_{\text{firm}} = \$2{,}041 \text{ million} + \$6{,}719 \text{ million} = \$8{,}760 \text{ million}$$

The rest is just arithmetic. Polaris's equity is worth $8,760 million less $1,436 million in interest-bearing debt presently outstanding, or $7,324 million. With 61.9 million shares outstanding, this equates to an estimated price per share of $118.32.

My discounted cash flow valuation thus indicates that Polaris is worth $118.32 a share, provided the projected free cash flows accurately reflect expected future performance. I take the fact that the company's actual price at the time was $95.28 as evidence that investors were less enthusiastic about the company's prospects than I am.

Problems with Present Value Approaches to Valuation

If you are a little hesitant at this point about your ability to apply these discounted cash flow techniques to anything but simple textbook examples, welcome to the club. While DCF approaches to business valuation are conceptually correct, and even rather elegant, they are devilishly difficult to apply in practice. Valuing a business may be conceptually equivalent to any other capital expenditure decision, but there are several fundamental differences in practice:

1. The typical investment opportunity has a finite—usually brief—life, while the life expectancy of a company is indefinite.
2. The typical investment opportunity promises stable or perhaps declining cash flows over time, while the ability of a company to reinvest earnings customarily produces a growing cash flow.
3. The cash flows from a typical investment belong to the owner, while the cash flows generated by the company go to the owner only when management chooses to distribute them. If management decides to invest in Mexican diamond mines rather than pay dividends, a minority owner can do little other than sell out.

As the problems in the accompanying box illustrate, these practical differences introduce potentially large errors into the valuation process and can make the resulting FMV estimates quite sensitive to small changes in the discount rate and the growth rate employed.

10. Stryker Corporation is a leading medical technology company headquartered in Kalamazoo, Michigan, that trades on the New York Stock Exchange. Use the following information on Stryker and seven other similar companies (indicated by their ticker symbols) to value Stryker's common stock as of December 31, 2013.

Stryker Corp. ($ millions)	
Sales	9,021
Net income	1,616
Number of common shares (millions)	378
Earnings before interest and tax	2,163
Tax rate	22.3%
Book value of equity	9,047
Book value interest-bearing debt	2,764

	Stryker	BAX	BDX	COV	MDT	SNN	STJ	ZMH
Comparison of Stryker with comparable companies:								
5-year growth rate in sales (%)	6.1	4.3	2.4	0.6	4.2	2.7	4.7	2.9
5-year growth rate in EPS (%)	8.4	3.0	0.9	3.5	11.6	7.7	17.8	4.5
Analysts' projected growth (%)	9.1	8.0	9.0	NA	6.6	9.9	10.7	8.4
Interest coverage ratio (X)	15.6	12.2	7.8	10.8	12.0	81.2	10.7	15.0
Total liabilities to assets (%)	42.5	67.2	58.5	53.6	46.4	30.5	57.0	34.2
Total assets ($ millions)	15,743	25,869	12,035	19,619	37,231	5,819	10,248	9,581
Indicators of Value								
Price/earnings (X)	17.8	18.8	23.1	20.3	15.8	23.2	24.6	20.8
MV firm/EBIT(1–Tax rate) (X)	18.5	21.7	23.9	20.5	17.7	23.5	28.1	21.5
MV firm/sales (X)	3.5	3.1	3.1	3.5	4.1	3.0	3.9	3.8
MV equity/BV equity (X)	3.1	4.5	4.2	3.3	2.9	3.2	4.1	2.5
MV firm/BV firm (X)	2.0	1.8	2.1	1.8	1.8	2.3	2.1	1.8

MV = Market value; BV = Book value. Market value is estimated as book value of interest-bearing debt + market value of equity.

11. The following is a four-year forecast for Torino Marine.

Year	2022	2023	2024	2025
Free cash flow ($ millions)	–52	76	92	112

a. Estimate the fair market value of Torino Marine at the end of 2021. Assume that after 2025, earnings before interest and tax will remain

constant at $200 million, depreciation will equal capital expenditures in each year, and working capital will not change. Torino Marine's weighted-average cost of capital is 11 percent and its tax rate is 40 percent.

b. Estimate the fair market value per share of Torino Marine's equity at the end of 2021 if the company has 40 million shares outstanding and the market value of its interest-bearing liabilities on the valuation date equals $250 million.

c. Now let's try a different terminal value. Estimate the fair market value of Torino Marine's equity per share at the end of 2021 under the following assumptions:
   i. Free cash flows in years 2022 through 2025 remain as above.
   ii. EBIT in year 2025 is $200 million, and then grows at 5 percent per year forever.
   iii. To support the perpetual growth in EBIT, capital expenditures in year 2026 exceed depreciation by $30 million, and this difference grows 5 percent per year forever.
   iv. Similarly, working capital investments are $15 million in 2026, and this amount grows 5 percent per year forever.

d. Lastly, let's try a third terminal value. Estimate the fair market value of Torino Marine's equity per share at the end of 2021 under the following assumptions:
   i. Free cash flows in years 2022 through 2025 remain as above. EBIT in year 2025 will be $200 million.
   ii. At year-end 2025, Torino Marine has reached maturity, and its equity sells for a "typical" multiple of year 2025 net income. Use 12 as a typical multiple.
   iii. At year-end 2025, Torino Marine has $250 million of interest-bearing liabilities outstanding at an average interest rate of 10 percent.

The following three problems test your knowledge of this chapter's Appendix.

12. A venture capital company buys 400,000 shares of a startup's stock for $5 million. If the company has 1.6 million shares outstanding prior to the purchase, what is the company's pre-money value? What is its post-money value?`

13. New ventures commonly set aside 10 to 20 percent of company shares at the valuation date for employee bonuses and stock options. Modify the valuation of ZMW Enterprises in Panel B of Table 9A.1 to include an employee set aside equal to 20 percent of the company in year 5. Specifically, calculate Touchstone's required percentage ownership in year 0 under these revised conditions. Assume as before that Touchstone and the second-round investor continue to target returns of 60 percent and 40 percent, respectively.

Ross

ROSS, WESTERFIELD, JORDAN_ LAMB - Fundamentos de Administraç

- Fluxos de caixa incrementais: custos irrecuperáveis, de oportunidade, CCL: Cap. 10, §10.1–10.2 (p. 319–
- Mais sobre fluxo de caixa de projetos: CCL, depreciação, valor residual: Cap. 10, §10.4 (p. 325–338)

Em seguida, concentramos nossa atenção na definição de uma análise por fluxos de caixa descontados. Conforme aprendemos no capítulo anterior, os fluxos de caixa futuros são o principal elemento de tal avaliação. Assim, enfatizamos o trabalho com informações financeiras e contábeis para chegar a esses números.

Ao avaliar um investimento proposto, prestamos atenção especial à determinação de quais informações são relevantes para a decisão em questão e quais não são. Como veremos, é fácil negligenciar peças importantes do quebra-cabeça do orçamento de capital.

Aguardaremos até o próximo capítulo para descrever com detalhes como proceder para avaliar os resultados de nossa análise por fluxos de caixa descontados. Da mesma forma, onde for necessário, assumiremos que sabemos qual é o retorno exigido, isto é, a taxa de desconto. Deixaremos as discussões mais detalhadas para a Parte 5.

10.1 Fluxos de caixa de um projeto: um contato inicial

Assumir um projeto tem como efeito a alteração dos fluxos de caixa da empresa hoje e no futuro. Para avaliar um investimento proposto, devemos considerar essas alterações e, em seguida, resolver se elas agregam ou não valor à empresa. A primeira (e mais importante) etapa, portanto, é resolver quais desses fluxos de caixa são relevantes.

Fluxos de caixa relevantes

O que é um fluxo de caixa relevante para um projeto? O princípio geral é muito simples: um fluxo de caixa relevante para um projeto é uma variação no fluxo de caixa futuro geral da empresa que ocorre como consequência direta da decisão de realizar tal projeto. Como os fluxos de caixa relevantes são definidos em termos das variações ou incrementos no fluxo de caixa da empresa, eles são chamados de **fluxos de caixa incrementais** associados ao projeto.

fluxos de caixa incrementais
a diferença entre os fluxos de caixa futuros de uma empresa com e sem a realização de um projeto.

O conceito de fluxo de caixa incremental é fundamental para nossa análise, de modo que estabeleceremos uma definição geral e voltaremos a ela quando necessário:

> **Os fluxos de caixa incrementais para a avaliação de um projeto consistem em *cada uma e em todas* as mudanças nos fluxos de caixa futuros da empresa que são consequências diretas da aceitação do projeto.**

Essa definição de fluxos de caixa incrementais tem um corolário óbvio e importante: todo fluxo de caixa que existe independentemente da realização de um projeto *não* é relevante.

O princípio da independência

Na prática, seria complicado calcular os fluxos de caixa totais futuros da empresa com e sem um projeto, principalmente no caso de uma empresa grande. Por sorte, não é necessário fazer isso. Uma vez que identifiquemos o efeito da realização do projeto sobre os fluxos de caixa da empresa, precisamos nos concentrar apenas nos fluxos de caixa incrementais resultantes do projeto. Isso é chamado de **princípio da independência**.

princípio da independência
a pressuposição de que a avaliação de um projeto pode se basear nos seus fluxos de caixa incrementais.

O princípio da independência diz que, após determinarmos os fluxos de caixa incrementais de um projeto, podemos encará-lo como um tipo de "miniempresa", com suas próprias receitas e despesas futuras, seu próprio ativo e, obviamente, seus próprios fluxos de caixa. A partir daí, estaremos interessados principalmente em comparar os fluxos de caixa dessa miniempresa com o custo de sua aquisição. Uma consequência importante dessa abordagem é que avaliaremos o projeto proposto simplesmente por seus próprios méritos, sem levar em conta outras atividades ou projetos.

Questões conceituais

10.1a Quais são os fluxos de caixa incrementais relevantes para avaliação de um projeto?

10.1b O que é o princípio da independência?

10.2 Fluxos de caixa incrementais

Estamos preocupados aqui apenas com os fluxos de caixa que são incrementais e que resultam de um projeto. Olhando nossa definição geral, podemos pensar que é fácil decidir se um fluxo de caixa é ou não incremental. Mesmo assim, existem algumas situações nas quais é fácil cometer erros. Nesta seção, descrevemos algumas das armadilhas mais comuns e como evitá-las.

Custos irrecuperáveis

custo irrecuperável
um custo que já foi incorrido, que não pode ser eliminado e, portanto, não deve ser considerado em uma decisão de investimento.

Um **custo irrecuperável**, por definição, é um custo que já pagamos ou cuja obrigação de pagar já foi assumida. Tal custo não pode ser alterado pela decisão de aceitar ou rejeitar um projeto hoje. Em outras palavras, a empresa terá de pagar esse custo de qualquer maneira. Com base em nossa definição geral de fluxo de caixa incremental, tal custo é claramente irrelevante para a decisão em questão. Assim, sempre tomaremos o cuidado de excluir os custos irrecuperáveis de nossa análise.

Dada nossa discussão, parece óbvio que um custo irrecuperável não é relevante. Entretanto, é fácil cair no conto de que um custo irrecuperável deve ser associado a um projeto. Por exemplo, suponhamos que a General Milk Company contrate um consultor financeiro para ajudar a avaliar se uma linha de achocolatados deve ou não ser lançada. Quando o consultor entrega o relatório, a General Milk discorda da análise porque o consultor não incluiu os altos custos de consultoria como parte do custo do projeto de achocolatados.

Quem está correto? Neste momento, sabemos que os custos de consultoria são um custo irrecuperável e devem ser pagos independentemente do lançamento da linha de achocolatados (essa é uma característica atraente do negócio de consultoria).

Custos de oportunidade

custo de oportunidade
a alternativa de maior valor que é abandonada se determinado investimento for aceito.

Quando pensamos em custos, normalmente pensamos nos custos tangíveis, ou seja, aqueles em que realmente precisamos tirar dinheiro do bolso. Um **custo de oportunidade** é ligeiramente diferente, pois exige que desistamos de um benefício. Uma situação comum surge quando uma empresa já possui algum dos ativos que um projeto proposto utilizará. Por exemplo, podemos estar pensando em converter um velho e rústico moinho comprado há anos por $100.000 em um condomínio de luxo.

Se aceitarmos esse projeto, não haverá fluxo de saída de caixa direto associado à compra do velho moinho, porque já o temos. Para fins de avaliação do projeto do condomínio, devemos tratar o moinho como "gratuito"? A resposta é não. O moinho é um recurso valioso utilizado pelo projeto. Se não o usássemos aqui, poderíamos fazer alguma outra coisa com ele. Como o que, por exemplo? A resposta óbvia é que, no mínimo, poderíamos vendê-lo. O uso do moinho para o complexo de condomínios, portanto, tem um custo de oportunidade: desistimos da oportunidade valiosa de fazer outra coisa com o moinho.[1]

Existe outra questão aqui. Depois de concordarmos que o uso do moinho tem um custo de oportunidade, quanto deveremos cobrar do projeto de condomínios por esse uso? Dado que pagamos $100.000, parece que devemos cobrar esse montante do projeto de condomínios. Isso está certo? A resposta é não, e o motivo tem por base nossa discussão sobre os custos irrecuperáveis.

O fato de que pagamos $100.000 há alguns anos é irrelevante. Esse custo é irrecuperável. No mínimo, o custo de oportunidade que cobramos do projeto é o preço pelo qual o moinho seria vendido hoje (preço líquido de qualquer custo de venda), porque esse é o montante do qual desistimos usando o moinho em vez de vendê-lo.[2]

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[1] Os economistas, às vezes, usam a sigla TANSTAAFL, que é uma abreviação de "There ain't no such thing as a free lunch" (não existe almoço grátis), para descrever o fato de que apenas raramente algo é gratuito de verdade.

[2] Se o ativo em questão é exclusivo, então o custo de oportunidade pode ser mais alto, porque pode haver outros projetos valiosos que poderíamos realizar e que o utilizariam. Entretanto, se o ativo em questão é de um tipo rotineiramente comprado e vendido (por exemplo, um carro usado), então o custo de oportunidade sempre é o preço atual do mercado, porque esse é o custo de compra de outro ativo semelhante.

Efeitos colaterais

Lembre-se de que os fluxos de caixa incrementais de um projeto incluem todas as alterações que ocorrem nos fluxos de caixa futuros da *empresa*. É comum um projeto ter efeitos colaterais (bons e ruins). Por exemplo, em 2008, o tempo entre o lançamento de um filme no cinema e em DVD havia diminuído para 98 dias, em comparação com os 200 dias de 1998. Esse tempo mais curto de lançamento foi o culpado por, pelo menos, parte do declínio nas receitas dos filmes. Obviamente, os varejistas comemoraram a mudança, porque a ela foi atribuído o aumento nas vendas de DVDs. Um impacto negativo nos fluxos de caixa de um produto existente causado pela introdução de um produto novo é chamado de **erosão**.[3] Nesse caso, os fluxos de caixa da nova linha de produtos devem ser diminuídos para refletir a perda de lucros em outras linhas.

erosão
a geração de fluxos de caixa de um projeto novo à custa dos projetos existentes de uma empresa.

Ao contabilizar a erosão, é importante reconhecer que quaisquer vendas perdidas por causa do lançamento de um produto também podem se perder de qualquer modo devido à concorrência futura. A erosão só é relevante se as vendas não fossem perdidas de outra maneira.

Existem vários tipos de efeitos colaterais. Por exemplo, uma das preocupações da Walt Disney Company ao criar a Euro Disney era que o novo parque tiraria os visitantes do parque da Flórida, um conhecido destino de férias para os europeus.

É claro que também existem efeitos colaterais benéficos. Por exemplo, você pode achar que a Hewlett-Packard ficou preocupada quando o preço de uma impressora que era vendida entre $500 e $600, em 1994, caiu para menos de $100 em 2009; mas a empresa não se importou. A HP percebeu que o verdadeiro lucro estava nos suprimentos que os proprietários compram para manter as impressoras funcionando, como cartuchos de tinta, *toners* e papel especial. As margens de lucro desses produtos são substanciais.

Capital de giro

Normalmente, um projeto exige que a empresa invista também em ativos circulantes, e não só em ativos imobilizados. Por exemplo, um projeto, em geral, precisa de um determinado montante em dinheiro disponível para pagar todas as despesas que surgirem. Além disso, um projeto precisará de um investimento inicial em estoques e contas a receber (para cobrir as vendas a crédito). Parte do financiamento dessas vendas assumirá a forma de montantes devidos aos fornecedores (contas a pagar), mas a empresa terá de fornecer o saldo restante. Esse saldo representa o investimento em capital circulante.

Antes de continuarmos, são necessárias algumas considerações para os temas: capital circulante, capital circulante líquido e capital de giro.

As contas a receber, os estoques, a parcela a recuperar dos impostos sobre a produção, o caixa mínimo necessário às operações são investimentos em capital circulante, são as contas operacionais do ativo circulante. No balanço patrimonial de uma empresa, outras contas não operacionais integram o ativo circulante como, por exemplo, as aplicações financeiras. Por sua vez, no passivo circulante encontramos os montantes a pagar devidos a fornecedores, salários e impostos sobre a produção. Essas são as contas operacionais do passivo circulante que, no balanço de uma empresa, também apresenta contas não operacionais.

Chamamos de *capital circulante líquido* a diferença de valores entre ativo circulante e passivo circulante, e é essa diferença que exige financiamento de recursos obtidos fora das operações da empresa, no capital de giro.

Enquanto o capital circulante líquido é uso de recursos, o capital de giro é fonte de recursos para as operações circulantes. A origem do capital de giro está no passivo não circulante e no patrimônio líquido (se você tiver dúvidas, esboce um balanço patrimonial e identifique fontes e usos de recursos).

Quando avaliamos a necessidade de capital de giro para as operações continuadas de uma empresa, nós precisamos distinguir entre contas operacionais e contas não operacionais do circulante porque, ao longo do tempo, alguns direitos e obrigações não operacionais se acrescentam. Calculamos então a necessidade de capital de giro pela diferença entre os valores das apli-

[3] De modo mais ilustrativo, a erosão às vezes é chamada de *pirataria* ou *canibalismo*.

cações operacionais e das fontes operacionais. Surge assim o conceito de necessidade de capital de giro, que quase sempre terá valor diferente do capital circulante líquido. Porém, quando avaliamos um projeto, não há sentido em alocar fontes e usos não operacionais no balanço que estimamos para o projeto. Assim, a diferença entre usos e fontes operacionais do circulante pode ser representada pelo capital circulante líquido do balanço projetado.

Ao longo deste capítulo, as referências ao capital de giro dizem respeito à necessidade de aporte de recursos para sustentar as operações, medida pelo capital circulante líquido do balanço projetado. Em outros capítulos, quando fizermos a estimativa do capital de giro para as operações de uma empresa, utilizaremos o conceito de necessidade de capital de giro.

Feitos esses esclarecimentos, tratemos então de outra característica importante do capital de giro na avaliação de projetos – característica essa que é fácil de ser ignorada. À medida que um projeto se desenvolve, os estoques são vendidos, as contas a receber são recebidas, as contas são pagas e os saldos de caixa podem ser resgatados. Ao final do projeto, essas atividades liberam o capital de giro investido originalmente. Assim, o investimento em capital de giro se parece muito com um empréstimo. A empresa fornece capital de giro no início e o recupera no final do projeto.

Custos de financiamento

Ao analisar um investimento proposto, *não* incluiremos os juros ou outros custos de financiamento, como dividendos ou parcelas do principal da dívida, porque estamos interessados no fluxo de caixa gerado pelos ativos do projeto. Como mencionamos no Capítulo 2, os juros, por exemplo, são um componente do fluxo de caixa para os credores, e não do fluxo de caixa dos ativos.

De modo geral, nosso objetivo na avaliação de projetos é comparar o fluxo de caixa de um projeto com o custo de aquisição desse projeto para estimar o VPL. A combinação específica entre capital próprio e capital de terceiros que uma empresa resolve usar ao financiar um projeto é uma variável de gestão e determina principalmente como os fluxos de caixa do projeto se dividem entre proprietários e credores. Isso não quer dizer que as estruturas de financiamento não tenham importância. Mas elas são algo a ser analisado separadamente. Abordaremos esse assunto em capítulos futuros.

Outras questões

Existem outros elementos a serem observados. Em primeiro lugar, estamos interessados apenas em mensurar o fluxo de caixa. Além disso, estamos interessados em mensurá-lo quando realmente ocorre, não quando ele é registrado no sentido contábil. Em segundo lugar, estamos sempre interessados no fluxo de caixa *após* impostos, porque os impostos são, definitivamente, uma saída de caixa. Na verdade, sempre que escrevemos *fluxos de caixa incrementais*, estamos querendo dizer fluxos de caixa incrementais *após* impostos. Lembre-se, porém, de que o fluxo de caixa após impostos e o lucro contábil ou o lucro líquido são coisas totalmente diferentes.

Questões conceituais

10.2a O que é um custo irrecuperável? E um custo de oportunidade?

10.2b Explique o que é erosão e por que ela é relevante.

10.2c Explique por que os juros não são um fluxo de caixa relevante para a avaliação de um projeto.

10.3 Demonstrações financeiras projetadas e fluxos de caixa do projeto

A primeira coisa necessária para começarmos a avaliar uma proposta de investimento é uma série de demonstrações financeiras projetadas. Em seguida, podemos desenvolver os fluxos de caixa esperados do projeto. Após termos os fluxos de caixa, podemos estimar o valor do projeto usando as técnicas descritas no capítulo anterior.

em capital circulante. O fluxo de saída de caixa imediato é, portanto, de $110.000. No final do projeto, o ativo imobilizado nada valerá, mas a empresa recuperará o capital de giro de $20.000 investido no circulante.[6] Isso resultará em uma *entrada de caixa* de $20.000 no último ano.

Em um nível puramente mecânico, observe que todo investimento em capital circulante tem de ser recuperado; ou seja, o mesmo número tem de aparecer em algum momento futuro com sinal invertido.

Fluxos de caixa totais projetados e valor

Dadas as informações que acumulamos, podemos terminar a análise preliminar de fluxos de caixa, como ilustramos na Tabela 10.5.

Agora que temos as projeções de fluxos de caixa, estamos prontos para aplicar os diversos critérios que discutimos no capítulo anterior. Primeiro, o VPL ao retorno exigido de 20% é:

$$\begin{aligned} \text{VPL} &= -\$110.000 + 51.780/1{,}2 + 51.780/1{,}2^2 + 71.780/1{,}2^3 \\ &= \$10.648 \end{aligned}$$

Assim, com base nessas projeções, o projeto cria mais de $10.000 de valor e deve ser aceito. Da mesma forma, o retorno sobre esse investimento obviamente excede aos 20% (porque o VPL é positivo a 20%). Após alguma tentativa e erro, descobrimos que a TIR é de aproximadamente 25,8%.

Além disso, se for necessário, podemos ir em frente e calcular o *payback* e o retorno contábil médio, ou RCM. Um exame dos fluxos de caixa mostra que o *payback* desse projeto é de apenas um pouco mais do que dois anos (verifique que ele é de 2,1 anos).[7]

Do último capítulo, sabemos que o RCM é o lucro líquido médio dividido pelo valor contábil médio. O lucro líquido a cada ano é de $21.780. A média (em milhares) dos quatro valores contábeis (da Tabela 10.2) para o investimento total é ($110 + $80 + $50 + $20)/4 = $65. Assim, o RCM é $21.780/$65.000 = 33,51%.[8] Já vimos que o retorno sobre esse investimento (a TIR) é de cerca de 26%. O fato de que o RCM é maior ilustra novamente o motivo pelo qual ele não pode ser interpretado de modo significativo como o retorno sobre um projeto.

Questões conceituais

10.3a Qual é a definição de fluxo de caixa operacional do projeto? Como isso difere do lucro líquido?

10.3b Para o projeto de iscas para tubarão, por que somamos o investimento em capital circulante líquido no último ano?

10.4 Mais sobre fluxo de caixa de projetos

Nesta seção, veremos mais atentamente alguns aspectos dos fluxos de caixa de projetos. Em particular, discutimos o capital de giro para o projeto com maiores detalhes. Em seguida, examinamos a legislação tributária relativa à depreciação. Por último, criamos um exemplo mais complicado de decisão sobre investimento de capital.

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[6] Na verdade, a empresa provavelmente recuperaria um pouco menos de 100% dessa quantia por causa de dívidas incobráveis, perda de estoques, etc. Se quiséssemos, poderíamos pressupor que, por exemplo, somente 90% seriam recuperados, e continuar a partir desse valor.

[7] Somos culpados de uma pequena inconsistência aqui. Ao calcularmos o VPL e a TIR, assumimos que todos os fluxos de caixa ocorreram *ao final* do ano. Quando calculamos o *payback*, assumimos que os fluxos de caixa ocorreram de modo uniforme *ao longo* do ano.

[8] Observe que o valor contábil total médio não é o total inicial de $110.000 dividido por 2. O motivo disso é que os $20.000 de capital de giro não são "depreciados".

Um exame mais detalhado do capital de giro

Ao calcular o fluxo de caixa operacional, não consideramos explicitamente o fato de que parte de nossas vendas poderia ser a crédito. Além disso, podemos não ter realmente pago algumas das despesas mostradas. Em ambos os casos, os fluxos de caixa em questão ainda não teriam ocorrido. Mostramos aqui que essas possibilidades não são um problema, desde que não nos esqueçamos de incluir as variações do capital circulante em nossa análise. Essa discussão, portanto, enfatiza a importância e o efeito disso.

Suponhamos que, durante determinado ano de um projeto, tenhamos a seguinte demonstração de resultados simplificada:

Vendas	$500
Custos	310
Lucro líquido	$190

A depreciação e os impostos são iguais a zero. Nenhum ativo imobilizado foi comprado durante o ano. Assim, para ilustrar a questão, assumimos que os únicos componentes do capital circulante são as contas a receber e a pagar. Os montantes inicial e final dessas contas são os seguintes:

	Início do ano	Final do ano	Variação
Contas a receber	$880	$910	+$30
Contas a pagar	550	605	+ 55
Capital circulante líquido	$330	$305	−$25

Com base nessas informações, qual é o fluxo de caixa total para o ano? Primeiro podemos simplesmente aplicar de maneira mecânica o que temos discutido para chegar à resposta. O fluxo de caixa operacional, neste caso em particular, é igual ao Lajir, porque não há impostos ou depreciação e, assim, é igual a $190. Da mesma forma, observe que o capital circulante líquido na verdade *diminuiu* em $25. Isso significa simplesmente que $25 foram liberados durante o ano. Não houve gastos de capital e, portanto, o fluxo de caixa total para o ano é:

$$\begin{aligned}\text{Fluxo de caixa total} &= \text{Fluxo de caixa operacional} - \text{Variação no CCL} - \text{Gastos de capital} \\ &= \$190 - (-\$25) - 0 \\ &= \$215\end{aligned}$$

Agora sabemos que esses $215 do fluxo de caixa total têm de ser os "reais que entram" menos os "reais que saem" durante o ano. Portanto, poderíamos fazer uma pergunta diferente: quais foram as receitas realizadas no ano? Além disso, quais foram os custos realizados?

Para determinar as receitas realizadas em caixa, precisamos examinar mais atentamente o capital circulante. Durante o ano, tivemos vendas de $500. Entretanto, as contas a receber aumentaram em $30 no mesmo período. O que isso significa? O aumento de $30 nos diz que as vendas excederam aos recebimentos em $30. Em outras palavras, ainda não recebemos o caixa de $30 dos $500 em vendas. Como resultado, nosso fluxo de entradas de caixa é de $500 − $30 = $470. Em geral, as entradas de caixa são as vendas menos o aumento nas contas a receber.

As saídas de caixa podem ser determinadas do mesmo modo. Mostramos custos de $310 na demonstração de resultados, mas as contas a pagar aumentaram em $55 durante o ano. Isso significa que ainda não pagamos $55 dos $310, de modo que os custos realizados em caixa do período são apenas $310 − $55 = $255. Em outras palavras, nesse caso, os custos realizados são iguais aos custos menos o aumento em contas a pagar.[9]

[9] Se houver outras contas, talvez sejam precisos alguns outros ajustes. Por exemplo, um aumento líquido no estoque seria um fluxo de saída de caixa.

EM SUAS PRÓPRIAS PALAVRAS...

Samuel Weaver sobre orçamento de capital na Hershey Company

O programa de gastos de capital da Hershey Company e da maior parte das empresas da Fortune 500 ou Fortune 1.000 envolve uma abordagem em três etapas: planejamento ou orçamento, avaliação e revisões pós-implantação.

A primeira etapa envolve a identificação de projetos possíveis durante o período de planejamento estratégico. Esses projetos são selecionados para dar suporte aos objetivos estratégicos da corporação. Essa identificação, em geral, é ampla no escopo, com pouquíssima avaliação financeira. Os projetos são classificados como produto novo, redução de custos, expansão de capacidade, etc. À medida que o processo de planejamento se concentra mais nos planos (ou orçamentos) de curto prazo, os principais gastos de capital são examinados com maior rigor. Os custos do projeto são mais bem ajustados, e projetos específicos podem ser reconsiderados.

Em seguida, cada projeto é revisado e autorizado individualmente. O planejamento, o desenvolvimento e o refinamento dos fluxos de caixa são a base da análise de capital na Hershey. Depois que os fluxos de caixa foram determinados, a aplicação de técnicas de avaliação de capital, como aquelas que usam valor presente líquido, taxa interna de retorno e período de *payback*, é rotineira. A apresentação dos resultados é melhorada usando análise de sensibilidade, que tem um papel importante para a administração na avaliação dos pressupostos críticos e do impacto resultante.

A etapa final se relaciona às revisões pós-implantação, nas quais as previsões originais de desempenho do projeto são comparadas com os resultados reais e/ou expectativas revisadas.

A análise de gastos de capital será tão boa quanto os pressupostos subjacentes ao projeto. O velho clichê "entra lixo, sai lixo" se aplica nesse caso. Os fluxos de caixa incrementais resultam primariamente das vendas incrementais ou aperfeiçoamentos de margem (redução de custos). Na maioria das vezes, uma variedade de fluxos de caixa incrementais pode ser identificada por meio de pesquisas de mercado ou de estudos da engenharia. Entretanto, para alguns projetos, a distinção correta das implicações e dos fluxos de caixa relevantes é um desafio em termos de análise. Por exemplo, quando um produto é lançado e espera-se que gere milhões de dólares em vendas, a análise apropriada se concentra nas vendas incrementais depois de se contabilizar a canibalização dos produtos existentes.

Um dos problemas que enfrentamos na Hershey está relacionado à aplicação do valor presente líquido (VPL) *versus* a taxa interna de retorno (TIR). O VPL nos oferece a indicação do investimento correto ao lidar com alternativas mutuamente excludentes. Entretanto, os tomadores de decisão em todos os níveis, às vezes, acham difícil compreender o resultado. Ou seja, um VPL de, digamos, $535.000 precisa ser interpretado. Não basta saber que o VPL é positivo ou mesmo que ele é mais positivo do que outra alternativa. Os tomadores de decisão buscam determinar um nível de "conforto" quanto à rentabilidade do investimento comparando-o com outros padrões.

Embora a TIR possa fornecer uma indicação enganosa sobre qual projeto deva ser selecionado, o resultado é fornecido de uma forma que pode ser interpretada por todos. A TIR resultante pode ser comparada mentalmente à inflação esperada, às taxas de empréstimo atuais, ao custo de capital, ao retorno de uma carteira de ações e assim por diante. Uma TIR de, digamos, 18% pode ser interpretada facilmente pela administração. Talvez essa facilidade de entendimento explique por que pesquisas indicam que muitas empresas da Fortune 500 ou Fortune 1.000 usam o método da TIR (juntamente com o do VPL) como a primeira técnica de avaliação.

Além do problema de VPL *versus* TIR, existe um número limitado de projetos para os quais a análise tradicional dos gastos de capital é difícil de ser aplicada porque os fluxos de caixa não podem ser determinados. Quando um novo computador é comprado, um prédio de escritórios é reformado ou um estacionamento é repavimentado, é praticamente impossível identificar os fluxos de caixa, de modo que o uso das técnicas tradicionais de avaliação é limitado. Decisões de "gastos de capital" desse tipo são tomadas por meio de outras técnicas que se apoiam no julgamento dos administradores.

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Reunindo essas informações, calculamos que os fluxos de entrada menos os fluxos de saída de caixa são de $470 − $255 = $215, exatamente como tínhamos antes. Observe que:

$$
\begin{aligned}
\text{Fluxo de caixa} &= \text{Fluxo de entradas de caixa} - \text{Fluxo de saídas de caixa} \\
&= (\$500 - \$30) - (\$310 - \$55) \\
&= (\$500 - \$310) - (\$30 - \$55) \\
&= \text{Fluxo de caixa operacional} - \text{Variação do CCL} \\
&= \$190 - (-\$25) \\
&= \$215
\end{aligned}
$$

De modo geral, esse exemplo ilustra que incluir as variações no capital circulante líquido em nossos cálculos ajusta a diferença entre as vendas e os custos contábeis e os recebimentos e pagamentos efetivos de caixa.

EXEMPLO 10.1 Recebimentos em caixa e custos

No ano que acaba de terminar, a Combates, Embates & Cia. Ltda. (CECL) reporta vendas de $998 e custos de $734. Você reuniu as seguintes informações do balanço patrimonial inicial e final:

	Inicial	Final
Contas a receber	$100	$110
Estoque	100	80
Contas a pagar	100	70
Capital circulante líquido	$100	$120

Com base nesses números, quais são os fluxos de entrada de caixa? E os fluxos de saída de caixa? O que aconteceu com cada conta? O que é o fluxo de caixa líquido?

As vendas foram de $998, mas as contas a receber aumentaram em $10. Assim, os recebimentos realizados foram de $10 a menos do que as vendas, ou $988. Os custos foram de $734, mas os estoques caíram em $20. Isso significa que não substituímos os $20 em estoque e, portanto, os custos, na verdade, estão registrados por valor maior. Além disso, as contas a pagar caíram em $30. Isso significa que, em base líquida, realmente pagamos aos nossos fornecedores $30 a mais do que recebemos deles, resultando em um registro a menor de $30 nos custos. Fazendo o ajuste desses eventos, calculamos que os custos realizados são $734 − $20 + $30 = $744. O fluxo de caixa líquido é de $988 − $744 = $244.

Por fim, observe que o capital circulante líquido aumentou em $20 no geral. Podemos verificar nossa resposta observando que as vendas originalmente contabilizadas menos os custos (= $998 − $734) são de $264. Além disso, a CECL utilizou $20 de seu capital de giro, de modo que o resultado líquido é um fluxo de caixa de $264 − $20 = $244, como havíamos calculado.

Depreciação

Como já observamos, a depreciação contábil é uma dedução que não afeta o caixa. Como resultado, a depreciação tem consequências para o fluxo de caixa apenas porque influencia a conta dos tributos. Como a depreciação é calculada para finalidades tributárias é, portanto, o método de depreciação que importa para as decisões de investimento de capital. Não é surpresa, por tanto, que as considerações dos efeitos da depreciação sejam regidas pela legislação fiscal. Nas seções a seguir, discutiremos alguns detalhes de sistemas de depreciação, comparando os efeitos do sistema vigente nos Estados Unidos, conforme a Lei da Reforma Fiscal norte-americana de 1986, e do sistema vigente no Brasil, desde 1998. O sistema norte-americano em vigor é uma modificação do **sistema de recuperação acelerada do custo** (*accelerated cost recovery system* – ACRS) que lá havia sido instituído em 1981 e será referido ao longo deste texto como MACRS (*modified accelerated cost recovery system*). Consulte a norma brasileira em http://www.receita.fazenda.gov.br/legislacao/ins/ant2001/1998/in16298.htm.

sistema de recuperação acelerada do custo (ACRS)
um método de depreciação de acordo com a legislação fiscal estadunidense que permite a baixa acelerada de uma propriedade sob diversas classificações.

É necessário alertar que, no Brasil, a consideração da depreciação para fins fiscais e para avaliação de fluxos de caixa somente se aplica para aquelas empresas que declaram seus lucros com base no lucro real. Nos regimes de tributação pelo lucro presumido e pelo regime simples, os tributos são calculados com base em percentuais sobre o faturamento, e quaisquer despesas, inclusive as despesas de depreciação, não influem no valor apurado para o lucro tributável.

Com a adoção dos critérios da contabilidade internacional no Brasil, também é preciso distinguir entre a depreciação para fins fiscais e a depreciação para fins societários. Temos ainda, para alguns setores regulados, a depreciação regulatória, estabelecida pelo órgão que regula o setor.

As taxas de **depreciação para fins fiscais** têm por objetivo o cálculo do lucro real e da base de cálculo da CSLL e são estabelecidas pela Secretaria da Receita Federal (SRF), conforme o chamado RTT – Regime de Transição Tributária que visa assegurar a neutralidade tributária na adoção das normas IFRS. O RTT passou a ser obrigatório para as demonstrações financeiras publicadas a partir do ano de 2010 com efeitos até a entrada em vigor de lei que discipline os efeitos tributários dos métodos e critérios contábeis IFRS. O RTT é aplicado na apuração do imposto

sobre a renda – IRPJ, da contribuição social sobre o lucro líquido – CSLL, da contribuição para o PIS/PASEP e da contribuição para o financiamento da seguridade social – COFINS.

A **depreciação para fins societários** decorre das regras contábeis instituídas pela lei 11.638/07. O Comitê de Pronunciamentos Contábeis, em seu CPC 27, estabelece o tratamento contábil para ativos imobilizados e o tratamento da depreciação.

Conforme o Comitê de Pronunciamentos Contábeis, *valor depreciável* é o custo de um ativo, ou outro valor que substitua o custo, menos o seu valor residual, e *depreciação* é a alocação sistemática do valor depreciável de um ativo ao longo da sua vida útil (ver: http://www.cpc.org.br/pdf/CPC%2027.pdf).

Conforme a SRF, somente são admitidas, para fins de apuração do lucro real, despesas de depreciação de bens móveis ou imóveis que estejam intrinsecamente relacionadas com a produção ou comercialização de bens e serviços objeto da atividade. A SRF também define que a *taxa de depreciação* será fixada em função do prazo durante o qual se possa esperar a utilização econômica do bem, pelo contribuinte, na produção dos seus rendimentos e que os prazos de *vida útil* admissíveis para fins de depreciação são fixados por Instrução Normativa SRF.

A **depreciação para fins regulatórios** é um caso especial que se aplica a algumas empresas reguladas. É o caso das concessionárias de serviços de distribuição de energia elétrica. A tarifa cobrada pelas distribuidoras é estabelecida pela ANEEL – Agência Nacional de Energia Elétrica, com base em diversos fatores, sendo um deles a chamada "base de remuneração" constituída pelos ativos da concessão. O valor desses ativos, para fins regulatórios, é depreciado conforme taxas que refletem a vida útil dos ativos calculada pelo regulador. Assim, a decisão de investimento no caso especial das empresas distribuidoras de energia precisa levar em conta três depreciações:

a) A depreciação regulatória, que afeta a base de remuneração, a receita bruta da concessionária e os fluxos de caixa.
b) A depreciação fiscal, que afeta o lucro tributável e a base de cálculo da CSLL e os fluxos de caixa.
c) A depreciação societária, que afeta o valor dos ativos para cálculo de juros sobre o capital próprio, e o lucro líquido, que afeta a distribuição de dividendos.

A Tabela 10.6 apresenta alguns ativos e suas taxas de depreciação vigentes no contexto brasileiro, conforme estabelecido pela Secretaria da Receita Federal nos Anexos I e II da Instrução Normativa SRF nº 162, de 31 de dezembro de 1998. A tabela apresenta uma visão geral das taxas de depreciação segundo a classe de ativos; para cada caso pode haver exceções. Para maiores informações deve ser consultada a norma, disponível em http://www.receita.fazenda.gov.br/legislacao/ins/ant2001/1998/in16298.htm.

Cálculo de depreciação O cálculo da depreciação é normalmente mecânico. Embora existam várias ressalvas, a ideia básica é que cada ativo é atribuído a uma determinada classe. A classe de um ativo estabelece sua vida útil para fins fiscais. Depois que a vida útil fiscal de um ativo é determinada, a depreciação de cada ano é calculada multiplicando-se o custo do ativo por uma porcentagem fixa.[10] O valor residual esperado (aquilo que achamos que o ativo valerá quando o descartarmos) e a vida econômica esperada (por quanto tempo esperamos que o ativo esteja em serviço) não são considerados explicitamente no cálculo da depreciação fiscal.

Algumas classes de depreciação vigente nos EUA típicas são mostradas na Tabela 10.7. As porcentagens associadas são mostradas nas Tabelas 10.8a[11] para os EUA e 10.8b para o Brasil.

[10] Sob determinadas circunstâncias, o custo do ativo pode ser ajustado antes de calcular a depreciação. O resultado é chamado de *valor depreciável*, e a depreciação é calculada usando esse número em vez do custo real.

[11] Para os curiosos, as porcentagens de depreciação do sistema norte-americano originam-se de um esquema de saldo de duplo declínio com uma mudança para o método linear quando este se torna vantajoso. Além disso, existe uma convenção de meio ano, ou seja, assume-se que todos os ativos estão em funcionamento na metade do ano fiscal. Essa convenção é mantida, a não ser que mais de 40% do custo de um ativo seja incorrido no último trimestre. Nesse caso, uma convenção de meio de trimestre é usada. O arredondamento estranho para as empresas nos Estados Unidos é cortesia da Receita Federal dos Estados Unidos.

TABELA 10.6 Exemplos de taxas de depreciação de ativos conforme a legislação fiscal brasileira

Ativos	Taxa anual	Prazo (anos)
Animais vivos, aves	50,00%	2
Discos, fitas e cartões magnéticos	33.33%	3
Caminhões fora de estrada	25,00%	4
Tratores, máquinas e aparelhos para obras públicas, construção civil ou trabalhos semelhantes	25,00%	4
Veículos automotores para transporte de 10 pessoas ou mais, incluído o motorista	25,00%	4
Veículos automóveis para transporte de mercadoria	25,00%	4
Animais vivos, exceto aves	20,00%	5
Aquisição e desenvolvimento de *software*	20,00%	5
Computadores e periféricos, máquinas para processamento de dados, leitores magnéticos	20,00%	5
Ferramentas em geral	20,00%	5
Gravadores e reprodutores de som e imagem	20,00%	5
Veículos de passageiros e outros veículos automóveis principalmente concebidos para transporte de pessoas, incluídos os veículos de uso misto e os automóveis de corrida	20,00%	5
Instalações	10,00%	10
Motores, máquinas motrizes, máquinas-ferramenta e aparelhos em geral	10,00%	10
Reatores nucleares	10,00%	10
Edificações	4,00%	25
Terrenos, salvo os melhoramentos ou construções. Prédios ou construções não alugados nem utilizados na produção dos seus rendimentos, e os destinados à revenda. Bens que normalmente aumentam de valor com o tempo, como obras de arte e antiguidades. Bens para os quais seja registrada quota de exaustão.	Não depreciáveis	



Os percentuais de depreciação são apresentados por um ano a mais do que o número de anos da respectiva classe. Isso acontece porque um projeto inicia e termina durante o ano. Para que os percentuais depreciáveis coincidissem com o número de anos da classe, os projetos deveriam ser implantados no primeiro dia de um ano fiscal e terminar no último dia de outro ano fiscal.

Na Tabela 10.8b, para o caso brasileiro, consideramos que os projetos iniciam e terminam, em média, no meio do ano, de forma que, para uma classe de ativos com cinco anos de vida contábil e 20% de taxa de depreciação anual, a taxa de depreciação no primeiro e último anos é de 10% (20%/2). O mesmo se aplica às demais classes de ativos nos exemplos com base nas normas tributárias brasileiras.

Nos Estados Unidos, um imóvel não residencial (como um prédio de escritórios) é depreciado ao longo de 31,5 anos, usando-se uma depreciação de base linear, enquanto um imóvel residencial (como um prédio de apartamentos) tem depreciação linear ao longo de 27,5 anos. No Brasil, imóveis classificados como "edificações" que estejam intrinsecamente relacionados com a produção ou comercialização de bens e serviços objeto da atividade empresarial são depreciados em 25 anos. Terrenos não podem ser depreciados nos dois sistemas.[12]

Depreciação acelerada Conforme a Receita Federal do Brasil, há duas espécies de depreciação acelerada:

a. a reconhecida e registrada contabilmente, relativa à depreciação acelerada dos bens móveis, resultante do desgaste pelo uso em regime de operação superior ao normal, calculada com base no número de horas diárias de operação (turnos de trabalho); e

b. a relativa à depreciação acelerada incentivada, considerada como benefício fiscal e reconhecida, apenas, pela legislação tributária, para fins da apuração do lucro real, sendo registrada no Lalur, sem qualquer lançamento contábil.

[12] Existem, porém, provisões de exaustão para empresas que estão nas linhas de negócios do tipo extração (por exemplo, mineração). Essas provisões são semelhantes às provisões de depreciação.

Para bens móveis do Ativo Imobilizado, a depreciação acelerada fiscal, em função do número de horas diárias de operação são os seguintes coeficientes de depreciação acelerada sobre as taxas normalmente utilizáveis:

a. 1,0 – para um turno de oito horas de operação;
b. 1,5 – para dois turnos de oito horas de operação; e
c. 2,0 – para três turnos de oito horas de operação.

Nessas condições, um bem cuja taxa normal de depreciação é de 10% ao ano poderá ser depreciado em 15% ao ano se operar 16 horas por dia, ou 20% ao ano, se em regime de operação de 24 horas por dia.

É permitida a aplicação dos coeficientes de aceleração da depreciação dos bens móveis do ativo imobilizado, em razão dos turnos de operação, conjuntamente com os coeficientes multiplicativos concedidos como incentivo fiscal a determinados setores da atividade econômica. Em qualquer caso, o montante acumulado das quotas de depreciação deduzidas na apuração do lucro real não pode ultrapassar o custo de aquisição do bem integrado contabilmente.

Sistema de recuperação acelerada do custo (ACRS) um método de depreciação de acordo com a legislação fiscal norte-americana que permite a baixa acelerada de um ativo sob diversas classificações.

TABELA 10.7 Classes de ativos com MACRS modificado (Estados Unidos)

Classe	Exemplos
Três anos	Equipamento usado em pesquisa
Cinco anos	Carros, computadores
Sete anos	Maioria dos equipamentos industriais

TABELA 10.8A Provisões de depreciação com MACRS modificado (Estados Unidos)

	Classe de bens		
Ano	Três anos	Cinco anos	Sete anos
1	33,33%	20,00%	14,29%
2	44,45	32,00	24,49
3	14,81	19,20	17,49
4	7,41	11,52	12,49
5		11,52	8,93
6		5,76	8,92
7			8,93
8			4,46

TABELA 10.8B Taxas de depreciação típicas vigentes no Brasil†

Ano	2 anos	3 anos	4 anos	5 anos	10 anos
1	25,00%	16,67%	12,50%	10,00%	5,00%
2	50,00%	33,33%	25,00%	20,00%	10,00%
3	25,00%	33,33%	25,00%	20,00%	10,00%
4		16,67%	25,00%	20,00%	10,00%
5			12,50%	20,00%	10,00%
6				10,00%	10,00%
7					10,00%
8					10,00%
9					10,00%
10					10,00%
11					5,00%

†Projetos iniciando e terminando no meio do ano.

depreciação acelerada contábil depreciação reconhecida e registrada contabilmente, relativa ao desgaste pelo uso em regime de operação superior ao normal, calculada com base no número de horas diárias de operação em turnos de trabalho.

depreciação acelerada incentivada depreciação decorrente de incentivos especiais, considerada como benefício fiscal e reconhecida apenas pela legislação tributária, para fins da apuração do lucro real, sendo registrada no Lalur, sem qualquer lançamento contábil.

Para ilustrar como a depreciação é calculada, consideremos um automóvel que custe $12.000. Normalmente, os automóveis são classificados como bens de cinco anos. Examinando a Tabela 10.8a, vemos que o número relativo ao primeiro ano de um bem de cinco anos é 20%.[13] A depreciação do primeiro ano é, portanto, de $12.000 × 0,20 = $2.400 no sistema norte-americano, enquanto no sistema tributário brasileiro será igual a n/12 × R$2.400,00 onde "n" é o número de meses do primeiro ano em que o projeto tem faturamento. No sistema brasileiro, os mesmos 20% são aplicados até o final da vida contábil do automóvel, mas no sistema norte-americano a depreciação referente ao segundo ano é 32%, $12.000 × 0,32 = $3.840, como mostra a Tabela 10.8a. Esses cálculos estão resumidos a seguir:

Ano	Porcentagem do MACRS	Depreciação	
1	20,00%	0,2000 × $12.000 =	$ 2.400,00
2	32,00%	0,3200 × $12.000 =	3.840,00
3	19,20%	0,1920 × $12.000 =	2.304,00
4	11,52%	0,1152 × $12.000 =	1.382,40
5	11,52%	0,1152 × $12.000 =	1.382,40
6	5,76%	0,0576 × $12.000 =	691,20
	100,00%		$12.000,00

Depreciação pelo sistema vigente no Brasil†			
Ano	Percentagem	Depreciação	
1	10,00%	0,1000 × 12.000 =	1.200,00
2	20,00%	0,2000 × 12.000 =	2.400,00
3	20,00%	0,2000 × 12.000 =	2.400,00
4	20,00%	0,2000 × 12.000 =	2.400,00
5	20,00%	0,2000 × 12.000 =	2.400,00
6	10,00%	0,1000 × 12.000 =	1.200,00
	100,00%		$12.000,00

†Projetos iniciando e terminando no meio do ano (n=6).

Observe que as porcentagens somam 100%. Como resultado, ao final do projeto baixamos 100% do custo do ativo, ou $12.000 neste caso.

Valor contábil *versus* valor de mercado Tradicionalmente, ao calcular a depreciação de acordo com a legislação fiscal, a vida econômica e o valor futuro de mercado do ativo não importam. Como resultado, o valor contábil de um ativo pode ser bem diferente de seu valor de mercado. Por exemplo, após o primeiro ano, o valor contábil de nosso automóvel é de $12.000 menos a depreciação de $2.400 do primeiro ano, ou seja, $9.600. Os valores contábeis restantes estão resumidos nas Tabelas 10.9a e 10.9b. Após seis anos, o valor contábil do automóvel é zero.

Suponha que queiramos vender o automóvel após cinco anos. Com base nas médias históricas, ele valeria, digamos, 25% do preço de compra, ou 0,25 × $12.000 = $3.000. Se realmente o vendermos por isso, teremos de pagar impostos à alíquota geral do imposto de renda sobre a diferença entre o preço de venda de $3.000 e o valor contábil (de $691,20 nos Estados Unidos ou de R$1.200 no Brasil). Para uma empresa, na faixa de 34% (e optante pelo lucro

[13] Pode parecer estranho que um ativo de cinco anos seja depreciado ao longo de seis anos. A explicação da contabilidade fiscal é a pressuposição de que temos o ativo por apenas seis meses no primeiro ano e, consequentemente, seis meses no último ano. Como resultado, existem cinco períodos de 12 meses, mas temos depreciação em cada um dos seis diferentes anos fiscais.

real no Brasil), a carga tributária seria de 0,34 × $2.308,80 = $784,99[14] nos Estados Unidos e de 0,34 × 1.800,00 = R$612,00 no Brasil.

No caso brasileiro, R$612,00 é a conta dos impostos a ser recolhida sobre a venda do automóvel. Entretanto, com a adoção das normas do IFRS, o saldo contabilizado de um ativo deve refletir a sua efetiva capacidade econômica de recuperar o valor nele investido e, se o automóvel tiver uma vida útil diferente da sua vida contábil, isso deverá ser refletido nas demonstrações financeiras e impactará o lucro para fins de distribuição de dividendos. Isso também afeta o processo de tomada de decisão.[†]

O motivo pelo qual os impostos devem ser pagos neste caso é que a diferença entre o valor de mercado e o valor contábil resulta de uma depreciação em "excesso", a qual deve ser "recuperada" quando o ativo é vendido. Isso significa que superdepreciamos o ativo em $3.000 – $691,20 = $2.308,80 (R$3.000 – R$1.200 = R$1.800). Como deduzimos $2.308,80 (R$1.800) a mais em depreciação, pagamos $784,99 (R$612) a menos em impostos e precisamos compensar a diferença.

TABELA 10.9A Valores contábeis com MACRS (Estados Unidos)

Ano	Valor contábil inicial	Depreciação	Valor contábil final
1	$12.000,00	$2.400,00	$9.600,00
2	9.600,00	3.840,00	5.760,00
3	5.760,00	2.304,00	3.456,00
4	3.456,00	1.382,40	2.073,60
5	2.073,60	1.382,40	691,20
6	691,20	691,20	0,00

TABELA 10.9B Valores contábeis conforme legislação brasileira

Taxas de depreciação pelo sistema vigente no Brasil[†]			
Ano	**Valor contábil inicial**	**Depreciação**	**Valor contábil final**
1	12.000,00	1.200,00	10.800,00
2	10.800,00	2.400,00	8.400,00
3	8.400,00	2.400,00	6.000,00
4	6.000,00	2.400,00	3.600,00
5	3.600,00	2.400,00	1.200,00
6	1.200,00	1.200,00	—

[†]Projetos iniciando e terminando no meio do ano.

Observe que isso não é imposto sobre ganhos de capital. Como regra geral (pelo menos aproximadamente), um ganho de capital ocorre apenas quando o preço de mercado excede ao custo original. Entretanto, em última análise, são as autoridades fiscais que determinam o que é um ganho de capital, e as regras específicas podem ser bastante complexas. Na maioria dos casos, iremos ignorar os impostos sobre ganhos de capital.

Por fim, se o valor contábil exceder ao valor de mercado, a diferença é tratada como prejuízo para fins fiscais. Por exemplo, se vendermos o carro após dois anos por $4.000, então o valor contábil excede ao valor de mercado em $1.760, 00 (R$4.400,00). Nesse caso, ocorre uma economia fiscal de 0,34 × $1.760 = $598,40 (0,34 × R$4.400,00 = R$1.496,00) que poderá ser utilizada para economizar impostos sobre outros lucros da empresa.

[14] No caso norte-americano, as regras são diferentes e mais complicadas no caso de um imóvel. Nesse caso, somente seria recuperada a diferença entre o valor contábil real e o valor contábil que teria existido se fosse usada uma depreciação linear. Tudo o que estivesse acima do valor contábil linear seria considerado ganho de capital.

[†] N. de R. T.: Para melhor informação sobre os procedimentos para depreciação sob a norma IFRS, consulte o CPC 27 – Ativo Imobilizado, que trata dos principais pontos a serem considerados na contabilização dos ativos imobilizados: o reconhecimento dos ativos, a determinação dos seus valores contábeis e os valores de depreciação e as perdas por desvalorização a serem reconhecidas em relação aos mesmos.

EXEMPLO 10.2 Depreciação com MACRS (Estados Unidos) e o caso brasileiro

A Companhia Fornecedora de Grampos acaba de comprar um novo sistema de informações computadorizado com um custo de instalação de $160.000. O computador é tratado como um ativo depreciável em cinco anos. Quais são as despesas de depreciação anual? Com base na experiência histórica, cremos que o sistema valerá apenas $10.000 quando a empresa desativá-lo daqui a quatro anos. Quais são as consequências fiscais da venda? Qual é o fluxo de caixa total da venda após impostos?

As despesas de depreciação anual são calculadas multiplicando $160.000 pelas porcentagens de cinco anos encontradas na Tabela 10.8a e 10.8b:

EXEMPLO 10.2A

Ano	Porcentagem do MACRS	Depreciação		Valor contábil final
1	20,00%	0,2000 × $160.000 =	$ 32.000	$128.000
2	32,00	0,3200 × 160.000 =	51.200	76.800
3	19,20	0,1920 × 160.000 =	30.720	46.080
4	11,52	0,1152 × 160.000 =	18.432	27.648
5	11,52	0,1152 × 160.000 =	18.432	9.216
6	5,76	0,0576 × 160.000 =	9.216	0
	100,00%		$160.000	

EXEMPLO 10.2B

Ano	Porcentagem norma brasileira†	Depreciação		Valor contábil final
1	10,00%	0,1000 × 160.000 =	16.000	144.000
2	20,00%	0,2000 × 160.000 =	32.000	112.000
3	20,00%	0,2000 × 160.000 =	32.000	80.000
4	20,00%	0,2000 × 160.000 =	32.000	48.000
5	20,00%	0,2000 × 160.000 =	32.000	16.000
6	10,00%	0,1000 × 160.000 =	16.000	—
	100,00%		$160.000	

† Iniciando e terminando no meio do ano.

Observe que também calculamos o valor contábil do sistema ao final de cada ano. O valor contábil ao final do ano 4 é de $27.648 no sistema tributário norte-americano e de R$48.000 no sistema tributário brasileiro. Se a empresa vender o sistema por $10.000 ao final do quarto ano, a diferença será um prejuízo de $17.648 (R$38.000) para fins fiscais. Esse prejuízo, obviamente, é como depreciação, porque não é despesa realizada.

O que acontece de fato? Duas coisas. Em primeiro lugar, a empresa recebe $10.000 do comprador. Em segundo lugar, ela economiza 0,34 x $17.648 = $6.000 (0,34 x R$38.000 = R$12.920) em impostos. Assim, o fluxo de caixa total da venda, após o efeito tributário, é um fluxo de entrada de caixa de $16.000 (R$22.920).

Um exemplo: A Majestosa – Máquina de Compostagem e Cobertura (MMCC)

Neste ponto, queremos discutir uma análise de orçamento de capital um pouco mais complicada. Ao ler esta análise, lembre-se de que a abordagem básica é exatamente a mesma do exemplo de iscas para tubarão usado anteriormente. Apenas acrescentamos detalhes mais reais (e muito mais números).

A MMCC está investigando a viabilidade de uma nova linha de equipamentos elétricos visando ao número crescente de compostadores domésticos. Com base em conversas exploratórias com compradores de grandes lojas de jardinagem, a MMCC projeta as vendas unitárias da seguinte maneira:

Ano	Vendas unitárias
1	3.000
2	5.000
3	6.000
4	6.500
5	6.000
6	5.000
7	4.000
8	3.000

O novo compostador elétrico será vendido inicialmente a $120 por unidade. Entretanto, após três anos, quando o produto tiver concorrência, a MMCC prevê que o preço cairá para $110.

O projeto do compostador elétrico exigirá $20.000 de capital de giro no início. Posteriormente, o capital de giro total necessário, ao final de cada ano, será de cerca de 15% das vendas daquele ano. O custo variável por unidade é de $60, e os custos fixos totais são de $25.000 por ano.

O custo do equipamento necessário para iniciar a produção será de cerca de $800.000. Esse investimento será feito principalmente em máquinas e equipamentos industriais e se qualifica como equipamento depreciável em 10 anos no sistema brasileiro e de sete anos no sistema MACRS norte-americano. Os equipamentos ainda terão valor de mercado de cerca de 20% de seu custo em oito anos, ou $0,20 \times \$800.000 = \$160.000$. A alíquota de imposto considerada é de 34%, e o retorno exigido é de 15%. Com base nessas informações, a MMCC deve ir adiante com o projeto?

Fluxos de caixa operacionais Existem muitas informações que precisamos organizar. A primeira coisa que podemos fazer é calcular as vendas projetadas. As vendas do primeiro ano são projetadas como 3.000 unidades a $120 a unidade, ou um total de $360.000. Os demais números são mostrados na Tabela 10.10.

A seguir, calculamos a depreciação sobre o investimento de $800.000 nas Tabelas 10.11a e 10.11b. Com essas informações, podemos preparar as demonstrações de resultados projetadas, como mostram as Tabelas 10.12a e 10.12b. Desse ponto em diante, o cálculo dos fluxos de caixa operacionais é simples. Os resultados são ilustrados na primeira parte das Tabelas 10.14a e 10.14b.

Variação do CCL Agora que temos os fluxos de caixa operacionais, precisamos determinar as variações do CCL. Por hipótese, as demandas do capital de giro para financiar o CCL mudam à medida que as vendas mudam. Em cada ano, a MMCC adicionará ou recuperará parte do capital circulante de seu projeto. Lembrando que a necessidade de capital circulante inicia em $20.000 e, em seguida, se eleva até 15% das vendas, podemos calcular o montante das necessidades de capital de giro de cada ano como ilustra a Tabela 10.13.

TABELA 10.10 Receitas projetadas – projeto do compostador elétrico

Ano	Preço unitário	Vendas unitárias	Receitas
1	$120	3.000	$360.000
2	120	5.000	600.000
3	120	6.000	720.000
4	110	6.500	715.000
5	110	6.000	660.000
6	110	5.000	550.000
7	110	4.000	440.000
8	110	3.000	330.000

TABELA 10.11A Depreciação anual – projeto do compostador elétrico (Estados Unidos)

Ano	Porcentagem do MACRS	Depreciação	Valor contábil final
1	14,29%	0,1429 × $800.000 = $114.320	$685.680
2	24,49	0,2449 × $800.000 = 195.920	489.760
3	17,49	0,1749 × $800.000 = 139.920	349.840
4	12,49	0,1249 × $800.000 = 99.920	249.920
5	8,93	0,0893 × $800.000 = 71.440	178.480
6	8,92	0,0892 × $800.000 = 71.360	107.120
7	8,93	0,0893 × $800.000 = 71.440	35.680
8	4,46	0,0446 × $800.000 = 35.680	0
	100,00%	$800.000	

TABELA 10.11B Depreciação anual – projeto do compostador elétrico (BR)

Ano	Percentagem no sistema brasileiro†	Depreciação	Valor contábil final
1	10,00%	0,0500 × 800.000 = 80.000	720.000
2	10,00%	0,1000 × 800.000 = 80.000	640.000
3	10,00%	0,1000 × 800.000 = 80.000	560.000
4	10,00%	0,1000 × 800.000 = 80.000	480.000
5	10,00%	0,1000 × 800.000 = 80.000	400.000
6	10,00%	0,1000 × 800.000 = 80.000	320.000
7	10,00%	0,1000 × 800.000 = 80.000	240.000
8	10,00%	0,0500 × 800.000 = 80.000	160.000

† Agora consideramos fluxos de caixa do projeto coincidentes com o ano fiscal e 12/12 avos de depreciação por período.

Observe que as quantidades vendidas, os preços e os custos nas Tabelas 10.12a e 10.12b são os mesmos, porém a forma de contabilizar a depreciação altera o lucro (e a tributação) de forma substancial. A partir da linha "Depreciação" tudo muda – os impostos cobrados no Brasil se mostram maiores até o quarto ano (a partir do quinto ano, os impostos cobrados no Brasil são menores).

Como foi ilustrado, durante o primeiro ano, a necessidade de financiamento do capital circulante líquido aumenta de $20.000 para 0,15 × $360.000 = $54.000. Portanto, o aumento do capital de giro necessário é de $54.000 − $20.000 = $34.000. Os números restantes são calculados da mesma forma.

Lembre-se de que um aumento na necessidade de capital de giro é um consumo de caixa e, portanto, usamos um sinal negativo nas Tabelas 10.14a e 10.14b para indicar um investimento adicional que a empresa faz no capital circulante líquido. Um sinal positivo representa o capital circulante que retorna para a empresa. Assim, $16.500 de CCL voltam para a empresa no ano 6. Ao longo da vida útil do projeto, a necessidade líquida de capital circulante chega ao valor máximo de $108.000 e diminui à medida que as vendas começam a cair.

TABELA 10.12A Demonstrações de resultados projetadas – projeto do compostador elétrico (Estados Unidos)

	Ano							
	1	2	3	4	5	6	7	8
Preço unitário	$ 120	$ 120	$ 120	$ 110	$ 110	$ 110	$ 110	$ 110
Vendas unitárias	3.000	5.000	6.000	6.500	6.000	5.000	4.000	3.000
Receitas	$360.000	$600.000	$720.000	$715.000	$660.000	$550.000	$440.000	$330.000
Custos variáveis	180.000	300.000	360.000	390.000	360.000	300.000	240.000	180.000
Custos fixos	25.000	25.000	25.000	25.000	25.000	25.000	25.000	25.000
Depreciação	114.320	195.920	139.920	99.920	71.440	71.360	71.440	35.680
Lajir	$ 40.680	$ 79.080	$195.080	$200.080	$203.560	$153.640	$103.560	$ 89.320
Impostos (34%)	13.831	26.887	66.327	68.027	69.210	52.238	35.210	30.369
Lucro líquido	$ 26.849	$ 52.193	$128.753	$132.053	$134.350	$101.402	$ 68.350	$ 58.951

TABELA 10.12B Demonstrações de resultados projetadas – projeto do compostador elétrico (BR)

Ano	1	2	3	4	5	6	7	8
Preço unitário	$ 120	$ 120	$ 120	$ 110	$ 110	$ 110	$ 110	$ 110
Vendas unitárias	3.000	5.000	6.000	6.500	6.000	5.000	4.000	3.000
Receita	$360.000	$600.000	$720.000	$715.000	$660.000	$550.000	$440.000	$330.000
Custos variáveis	180.000	300.000	360.000	390.000	360.000	300.000	240.000	180.000
Custos fixos	25.000	25.000	25.000	25.000	25.000	25.000	25.000	25.000
Depreciação	80.000	80.000	80.000	80.000	80.000	80.000	80.000	80.000
Lajir	$ 75.000	$ 195.000	$255.000	$220.000	$195.000	$145.000	$ 95.000	$ 45.000
Impostos (34%)	25.500	66.300	86.700	74.800	66.300	49.300	32.300	15.300
Lucro líquido	$ 49.500	$128.700	$168.300	$145.200	$128.700	$ 95.700	$ 62.700	$ 29.700

TABELA 10.13 Variações do CCL – projeto do compostador elétrico

Ano	Receitas	CCL	Fluxo de caixa
0		$ 20.000	−$20.000
1	$360.000	54.000	−34.000
2	600.000	90.000	−36.000
3	720.000	108.000	−18.000
4	715.000	107.250	750
5	660.000	99.000	8.250
6	550.000	82.500	16.500
7	440.000	66.000	16.500
8	330.000	49.500	16.500

Os resultados das variações do capital circulante líquido são mostrados na segunda parte das Tabelas 10.14a e 10.14b. Observe que, ao final da vida do projeto, há $49.500 de capital circulante ainda a ser recuperado. Assim, no último ano, o projeto retorna $16.500 de CCL durante o ano e, em seguida, retorna os $49.500 restantes ao final do ano, somando $66.000.

Gastos de capital Por fim, precisamos contabilizar o capital de longo prazo investido no projeto. Neste caso, a MMCC investe $800.000 no ano 0. A hipótese é de que esse equipamento valerá $160.000 ao final do projeto. Terá um valor contábil de zero no sistema tributário norte-americano e um valor contábil de R$160.000 no sistema tributário brasileiro. O excedente do valor de mercado em relação ao valor contábil é tributável, de modo que os resultados após impostos serão de $160.000 × (1 − 0,34) = $105.600 para o sistema norte-americano, enquanto que, no sistema brasileiro, não há excedente e, portanto, não há tributação. Esses números são mostrados na terceira parte das Tabelas 10.14a e 10.14b.

Fluxos de caixa totais e valor Agora temos todas as partes do fluxo de caixa e as reunimos nas Tabelas 10.15a e 10.15b. Além dos fluxos de caixa totais do projeto, calculamos os fluxos de caixa acumulados e os fluxos de caixa descontados. Neste ponto, é só inserir os valores para calcular o valor presente líquido, a taxa interna de retorno e o *payback*.

Se somarmos os fluxos descontados e o investimento inicial, temos que os valores presentes líquidos (a 15%) são de $65.485 e R$34.789. Isso é positivo, de modo que, com base nessas projeções preliminares, o projeto do compostador elétrico é aceitável nos dois sistemas tributários. A taxa interna de retorno é maior do que 15% porque o VPL é positivo. Ela é de 17,24% (Estados Unidos) ou de 16,13% (BR), indicando novamente que o projeto é aceitável.

Ao observar os fluxos de caixa acumulados, vemos que o projeto está quase pago após três anos, porque as Tabelas 10.15a e 10.15b mostram que eles somam quase zero nesse momento. Como foi indicado, o ano fracionário é $17.322/$214.040 = 0,07 e R$95.550/R$225.950 = 0,42. Portanto, o *payback* é de 3,07 anos no sistema norte-americano e 3,42 no sistema brasileiro. Não podemos dizer se isso é bom ou ruim, porque não temos uma referência para a MMCC. Esse é o problema mais comum dos períodos de *payback*.

TABELA 10.14A Fluxos de caixa projetados – projeto do compostador elétrico (Estados Unidos)

	Ano								
	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
I. Fluxo de caixa operacional									
Lajir		$ 40.680	$ 79.080	$195.080	$200.080	$203.560	$153.640	$103.560	$ 89.320
Depreciação		114.320	195.920	139.920	99.920	71.440	71.360	71.440	35.680
Impostos		– 13.831	– 26.887	– 66.327	– 68.027	– 69.210	– 52.238	– 35.210	– 30.369
Fluxo de caixa operacional		$141.169	$248.113	$268.673	$231.973	$205.790	$172.762	$139.790	$ 94.631
II. Capital de giro									
CCL inicial	–$ 20.000								
Variação do CCL		–$ 34.000	–$ 36.000	–$ 18.000	$ 750	$ 8.250	$ 16.500	$ 16.500	$ 16.500
Recuperação do CCL									49.500
Variação total do CCL	–$ 20.000	–$ 34.000	–$ 36.000	–$ 18.000	$ 750	$ 8.250	$ 16.500	$ 16.500	$ 66.000
III. Gastos de capital									
Desembolso inicial	–$800.000								
Valor residual após impostos									$105.600
Gastos de capital	–$800.000								$105.600

TABELA 10.14B Fluxos de caixa projetados – projeto do compostador elétrico (BR)

	Ano							
0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
I. Fluxo de caixa operacional								
Lajir	$ 75.000	$195.000	$255.000	$220.000	$195.000	$145.000	$ 95.000	$ 45.000
Depreciação	80.000	80.000	80.000	80.000	80.000	80.000	80.000	80.000
Impostos	– 25.500	– 66.300	– 86.700	– 74.800	– 66.300	– 49.300	– 32.300	– 15.300
Fluxo de caixa operacional	$129.500	$208.700	$248.300	$225.200	$208.700	$175.700	$142.700	$109.700
II. Capital de giro								
CCL inicial	–$ 20.000							
Variação do CCL	–$ 34.000	–$ 36.000	–$ 18.000	$ 750	$ 8.250	$ 16.500	$ 16.500	$ 16.500
Recuperação do CCL								49.500
Variação total do CCL	–$ 20.000	–$ 34.000	–$ 36.000	–$ 18.000	$ 750	$ 8.250	$ 16.500	$ 16.500	$ 66.000
III. Gastos de capital								
Desembolso inicial	–$800.000							
Valor residual após impostos								$160.000
Gastos de capital	–$800.000							$160.000

TABELA 10.15A Fluxos de caixa totais projetados – projeto do compostador elétrico (Estados Unidos)

	Ano								
	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Fluxo de caixa operacional		$141.169	$248.113	$268.673	$231.973	$205.790	$172.762	$139.790	$ 94.631
Variação do CCL	–$ 20.000	– 34.000	– 36.000	– 18.000	750	8.250	16.500	16.500	66.000
Gastos de capital	– 800.000								105.600
Fluxos de caixa totais do projeto	–$820.000	$107.169	$212.113	$250.673	$232.723	$214.040	$189.262	$156.290	$266.231
Fluxos de caixa acumulados	–$820.000	–$712.831	–$500.718	–$250.045	–$ 17.322	$196.718	$385.980	542.270	$808.501
Fluxos de caixa descontados a 15%	– 820.000	93.190	160.388	164.822	133.060	106.416	81.823	58.755	87.031
Valor presente líquido (15%)	= $65.485								
Taxa interna de retorno	= 17,24%								
Payback	= 3,07 anos								

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4. Dependendo da localização das lâmpadas (ou do número de lâmpadas), pode haver um custo que não pode ser negligenciado para a troca (por exemplo, os custos de mão de obra em uma empresa).
5. A geração de energia elétrica em usinas de carvão é responsável por uma parcela substancial das emissões de mercúrio nos Estados Unidos, mas elas devem ser drasticamente reduzidas em um futuro relativamente próximo. Isso não é um problema no Brasil, ainda, porque nossa matriz energética é predominantemente hidráulica; mas a geração térmica está aumentando.
6. A geração de energia é responsável por uma parcela substancial das emissões de CO_2 nos Estados Unidos. Por outro lado, as emissões de gases com efeito estufa em represas para usinas hidrelétricas, devidas à decomposição de florestas submersas, também tem sido apontadas como um problema ambiental.
7. As LFCs consomem mais energia e matéria-prima para serem produzidas. Além disso, existe o perigo de contaminação de mercúrio no local de produção das lâmpadas e também o perigo para os trabalhadores.
8. Se você instala uma LFC em uma luminária permanente de um prédio, você provavelmente se mudará antes de ela queimar.
9. Existe outra tecnologia baseada em diodos emissores de luz (LEDs) que está evoluindo. Atualmente, os LEDs são muito mais caros que as LFCs, mas os custos estão caindo. Os LEDs duram muito mais que as LFCs e usam muito menos energia. Além disso, os LEDs não contêm mercúrio.
10. Em 2007, a GE anunciou outra tecnologia: lâmpadas incandescentes de alta eficiência (*high efficiency incandescent* — HEI). Essas novas lâmpadas reduzem drasticamente o uso de energia elétrica e, de acordo com a GE, serão comparáveis às LFCs. Elas tinham previsão de lançamento em 2010, mas o produto inicial não seria tão eficiente quanto uma LFC.

Qualitativamente, como essas questões afetam sua posição diante do debate lâmpada incandescente *versus* fluorescente? Desde julho de 2012, lâmpadas incandescentes com potências de 150 W e 200W, e que não tiverem determinada eficiência energética não podem mais ser produzidas ou importadas pelo Brasil. O prazo para sua venda no atacado e varejo foi fixado para junho de 2013. As lâmpadas incandescentes devem ser substituídas de forma gradativa até junho de 2017. Sua análise sugere que isso faz sentido? Existem outras opções de regulamentação além do banimento que fariam sentido?

29. **Decisões de substituição [OA2]** Sua pequena empresa de reformas tem dois veículos de trabalho. Um é um pequeno carro de passeio usado para visitar os locais de trabalho e fazer outras atividades gerais da empresa. O outro é um caminhão usado para transportar equipamentos. O carro faz 10,5 km por litro. O caminhão faz 4,22 km por litro. Você quer reduzir os custos com gasolina e tem dinheiro suficiente para trocar apenas um veículo. Os custos de substituição seriam iguais para ambos os veículos. Um carro novo faria 17 km por litro; um caminhão novo faria 5,5 km por litro. O custo da gasolina é de $2,00 por litro. Pressupondo que uma substituição é uma boa ideia, qual veículo deveria ser substituído? Ambos andam 19 mil quilômetros por ano.

30. **Decisões de substituição [OA2]** No problema anterior suponha que você usa o caminhão por X quilômetros por ano. Qual a quilometragem do caminhão que torna a opção de substituí-lo por um novo uma boa escolha? Dica: Observe as economias relativas de gasolina.

DESAFIO (Questões 31-36)

31. **Calculando o VPL do projeto [OA1]** Você foi contratado como consultor da Cítaras Urbanas De Tecnologia Original Ltda. (CUTO), fabricante de cítaras de qualidade. O mercado de cítaras está crescendo rapidamente. A empresa comprou terras há três anos por $1,4 milhão, prevendo o seu uso como depósito de lixo tóxico, mas contratou outra empresa para lidar com todo o material tóxico. Com base em uma avaliação recente, a empresa acredita que poderia vender o terreno por $1,5 milhão após impostos. Em quatro

anos, o terreno poderia ser vendido por $1,6 milhão após impostos. A empresa também contratou uma empresa de marketing para analisar o mercado de cítaras a um custo de $125.000. Este é um trecho extraído do relatório de marketing:

> A indústria de cítaras terá uma expansão rápida nos próximos quatro anos. Com o reconhecimento de marca que a CUTO tem, cremos que a empresa poderá vender 3.200, 4.300, 3.900 e 2.800 unidades por ano nos próximos quatro anos, respectivamente. Mais uma vez, aproveitando o reconhecimento da CUTO, acreditamos que um preço-prêmio de $780 por cítara possa ser cobrado. Como as cítaras parecem ser uma moda passageira, é provável que, ao final do período de quatro anos, as vendas sejam interrompidas.

A CUTO acredita que os custos fixos do projeto serão de $425.000 por ano, e os custos variáveis serão de 15% das vendas. O equipamento necessário para a produção custará $4,2 milhões e será depreciado de forma linear e acelerada, em três anos, aproveitando uma depreciação incentivada combinada com a depreciação fiscal. Ao final do projeto, o equipamento pode ser sucateado por $400.000. Um capital de giro de $125.000 é necessário imediatamente para financiar o investimento em ativos circulantes. A CUTO tem uma alíquota tributária de 34% e o retorno exigido sobre o projeto é de 13%. Qual é o VPL do projeto? Assuma que a empresa tenha outros projetos lucrativos.

32. Avaliação de projeto com depreciação MACRS [OA1] A Acústica Avançada S/A (AASA), pretende implantar uma unidade de produção no Vale do Silício, na Califórnia. Ela projeta vendas unitárias para um novo implante de emulação de voz de sete oitavas da seguinte maneira:

Ano	Vendas unitárias
1	93.000
2	105.000
3	128.000
4	134.000
5	87.000

A produção dos implantes exigirá inicialmente $1.800.000 em capital de giro e aportes adicionais anuais para financiar o ativo circulante equivalentes a 15% do aumento projetado para as vendas do ano seguinte. Os custos fixos totais são de $1.200.000 por ano, os custos variáveis de produção são de $265 por unidade, e as unidades são vendidas a $380 cada. O equipamento necessário para iniciar a produção tem um custo de instalação de $24.000.000. Como os implantes destinam-se aos cantores profissionais, esse equipamento é considerado maquinário industrial e, portanto, qualifica-se como ativo depreciável pelo método MACRS de sete anos pela regra fiscal norte-americana. Em cinco anos, esse equipamento pode ser vendido por cerca de 20% de seu custo de aquisição. A unidade nos EUA está na faixa de imposto marginal de 35% e tem um retorno exigido de 18% sobre todos os seus projetos. Com base nessas estimativas preliminares de projeto, qual é o VPL do projeto? E qual é a TIR? Refaça os cálculos supondo que o investimento seja realizado no Brasil, e o equipamento seja depreciável em dez anos. A faixa de imposto de renda é de 34% neste caso.

33. Cálculo de economia exigida [OA2] Avalie o seguinte projeto proposto nos EUA: um equipamento proposto para economizar custos tem um custo de instalação de $610.000. O equipamento será usado em um projeto de 5 anos, mas está classificado como ativo com MACRS de 3 anos para fins fiscais. O investimento necessário para financiar o capital circulante líquido inicial é de $55.000, a alíquota tributária é de 35% e a taxa de desconto do projeto é de 12%. O equipamento tem um valor residual estimado em $40.000 no ano 5.

a. Qual nível de economia de custos antes dos impostos necessário para que esse projeto seja lucrativo?

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